Table S1: Summary Table
Diagnostic testing for liver fibrosis (by biopsy)
Categorical outcomes

PMID	Author	Year	Type of article	Study design	Population	Sample size	Description of Diagnostic Test: Diagnostic test	Description of Diagnostic Test: Reference standard (Sensitivity threshold)	Timpoint
20357753	**Liu CH., Liang CC., Liu CJ., Hsu SJ., Lin JW., Chen SI., Hung PH., Tsai HB., Lai MY., Chen PJ., Chen JH., Chen DS., Kao JH.**	2010	Peer-reviewed publication	Cohort study, prospective	HD	279	APRI (≥0.30)	Liver biopsy (F2-F4)	**0 months**
						175			**12-18 months**
					HD	279	APRI (≥0.40)	Liver biopsy (F2-F4)	**0 months**
						175			**12-18 months**
					HD	279	APRI (≥0.60)	Liver biopsy (F2-F4)	**0 months**
						175			**12-18 months**
					HD	279	APRI (≥0.80)	Liver biopsy (F2-F4)	**0 months**
						175			**12-18 months**
					HD	279	APRI (≥0.95)	Liver biopsy (F2-F4)	**0 months**
						175			**12-18 months**
17634962	**Schiavon LL., Schiavon JL., Filho RJ., Sampaio JP., Lanzoni VP., Silva AE., Ferraz ML.**	2007	Peer-reviewed publication	Cross-sectional study	HD	203	APRI (≥0.4)	Liver biopsy (F2-F4)	**nd**
							APRI (≥0.95)	Liver biopsy (F2-F4)	

PMID	Author	Year	Type of article	Study design	Population	Sample size	Description of Diagnostic Test		Timpoint
							Diagnostic test	Reference standard (Sensitivity threshold)	
							APRI (>0.55)	Liver biopsy (F3-F4)	
							APRI (≥1.00)	Liver biopsy (F3-F4)	
19758273	**Schiavon LL., Carvalho-Filho RJ., Narciso-Schiavon JL., Pinheiro SR., Barbosa DV., Lanzoni VP., Ferraz ML., Silva AE.**	2010	Peer-reviewed publication	Cross-sectional study	Transplant	102	APRI (>0.50)	Liver biopsy (F2-F4)	**nd**
							APRI (>1.50)	Liver biopsy (F2-F4)	
ATC-abstract 596	**A. Brar, M. Salifu, M. Rampal, N. Sumrani, D. John, F. Tedla.**	2012	Abstract	Cross-sectional study	HD	26	APRI	Liver biopsy (F2-F4)	**nd**
24353319	**Jiang Y., Huang E., Mehrnia A., Kamgar M., Pham PT., Ogunorunyinka O., Brown I., Danovitch GM., Bunnapradist S.**	2014	Peer-reviewed publication	Cross-sectional study	HD? (Transplant candidates)	210	APRI (≥0.4)	Liver biopsy (F3-F4)	**nd**
							APRI (>0.95)	Liver biopsy (F3-F4)	
21393486	**Liu CH., Liang CC., Huang KW., Liu CJ., Chen SI., Lin JW., Hung PH., Tsai HB., Lai MY., Chen PJ., Chen JH., Chen DS., Kao JH.**	2011	Peer-reviewed publication	Cross-sectional study	HD	304	APRI (1.5)	Liver biopsy (≥F2)	**nd**
							APRI (0.55)	Liver biopsy (≥F2)	**nd**
							APRI (0.5)	Liver biopsy (≥F2)	**nd**
							APRI (1.75)	Liver biopsy (F3)	**nd**
							APRI (0.75)	Liver biopsy (F3)	**nd**

PMID	Author	Year	Type of article	Study design	Population	Sample size	Description of Diagnostic Test		Timpoint
							Diagnostic test	Reference standard (Sensitivity threshold)	
16371924	**Varaut A., Fontaine H., Serpaggi J., Verkarre V., Vallet-Pichard A., Nalpas B., Imbertbismuth F., Lebray P., Pol S.**	2005	Peer-reviewed publication	Cross-sectional study	HD & Transplant	110	Fibrotest (>0.6)	Liver biopsy (F2-F4)	**nd**
							Fibrotest (≥0.2)	Liver biopsy (F2-F4)	
							Fibrotest (>0.6)	Liver biopsy (F3-F4)	
							Fibrotest (≥0.2)	Liver biopsy (F3-F4)	
					HD	50	Fibrotest (>0.6)	Liver biopsy (F2-F4)	
							Fibrotest (≥0.2)	Liver biopsy (F2-F4)	
							Fibrotest (>0.6)	Liver biopsy (F3-F4)	
							Fibrotest (≥0.2)	Liver biopsy (F3-F4)	
					Transplant	60	Fibrotest (>0.6)	Liver biopsy (F2-F4)	
							Fibrotest (≥0.2)	Liver biopsy (F2-F4)	
							Fibrotest (>0.6)	Liver biopsy (F3-F4)	
							Fibrotest (≥0.2)	Liver biopsy (F3-F4)	

PMID	Author	Year	Type of article	Study design	Population	Sample size	Description of Diagnostic Test		
							Diagnostic test	Reference standard (Sensitivity threshold)	Timpoint
20847572	**Canbakan M., Senturk H., Canbakan B., Toptas T., Tabak O., Ozaras R., Tabak F., Balcı H., Sut N., Ozbay G.**	2011	Peer-reviewed publication	Cross-sectional study	HD	33	Fibrotest (>0.19)	Liver biopsy (F2-F4)	**nd**
							Fibrotest (>0.6)	Liver biopsy (F2-F4)	
21393486	**Liu CH., Liang CC., Huang KW., Liu CJ., Chen SI., Lin JW., Hung PH., Tsai HB., Lai MY., Chen PJ., Chen JH., Chen DS., Kao JH.**	2011	Peer-reviewed publication	Cross-sectional study	HD	304	Fibroscan (8.8)	Liver biopsy (F2-F4)	**nd**
							Fibroscan (7.1)	Liver biopsy (F2-F4)	**nd**
							Fibroscan (5.3)	Liver biopsy (F2-F4)	**nd**
							Fibroscan (9.5)	Liver biopsy (F3-4)	**nd**
							Fibroscan (9.6)	Liver biopsy (F3-4)	**nd**
							Fibroscan (8.3)	Liver biopsy (F3-4)	**nd**
ATC-abstract 596	**A. Brar, M. Salifu, M. Rampal, N. Sumrani, D. John, F. Tedla.**	2012	Abstract	Cross-sectional study	HD	26	FIB-4	Liver biopsy (F2-F4)	
20847572	**Canbakan M., Senturk H., Canbakan B., Toptas T., Tabak O., Ozaras R., Tabak F., Balcı H., Sut N., Ozbay G.**	2011	Peer-reviewed publication	Cross-sectional study	HD	33	ActiTest (>0.11)	Liver biopsy (≥A2)	

Table S1: Summary Table
Diagnostic testing for liver fibrosis (by biopsy)
Categorical outcomes

PMID	Author	Year	Outcome	Sensitivity (%)	Specificity (%)	ROC-AUC	Quality
20357753	**Liu CH., Liang CC., Liu CJ., Hsu SJ., Lin JW., Chen SI., Hung PH., Tsai HB., Lai MY., Chen PJ., Chen JH., Chen DS., Kao JH.**	2010	**Fibrosis**	**97**	**23**	**0.83**	**Good**
				76	**55**	**0.71**	
				84	**51**		
				50	**73**		
				61	**90**		
				45	**89**		
				45	**95**		
				33	**95**		
				32	**96**		
				14	**98**		
17634962	**Schiavon LL., Schiavon JL., Filho RJ., Sampaio JP., Lanzoni VP., Silva AE., Ferraz ML.**	2007	**Fibrosis**	**88**	**54**	**0.80**	**Good**
				44	**93**		

PMID	Author	Year	Outcome	Sensitivity (%)	Specificity (%)	ROC-AUC	Quality
				95	69	0.84	
				42	89		
19758273	Schiavon LL., Carvalho-Filho RJ., Narciso-Schiavon JL., Pinheiro SR., Barbosa DV., Lanzoni VP., Ferraz ML., Silva AE.	2010	Fibrosis	85	51	0.73	Good
				25	89		
ATC-abstract 596	A. Brar, M. Salifu, M. Rampal, N. Sumrani, D. John, F. Tedla.	2012	Fibrosis	nd	nd	0.81	Fair
24353319	Jiang Y., Huang E., Mehrnia A., Kamgar M., Pham PT., Ogunorunyinka O., Brown I., Danovitch GM., Bunnapradist S.	2014	Fibrosis	67	56	nd	Good
				27	92		
21393486	Liu CH., Liang CC., Huang KW., Liu CJ., Chen SI., Lin JW., Hung PH., Tsai HB., Lai MY., Chen PJ., Chen JH., Chen DS., Kao JH.	2011	Fibrosis	3	100	nd	Good
			Fibrosis	74	82	0.84 (0.79, 0.88)	
			Fibrosis	79	70	nd	
			Fibrosis	2	99	nd	
			Fibrosis	93	90	0.93 (0.90, 0.97)	

PMID	Author	Year	Outcome	Sensitivity (%)	Specificity (%)	ROC-AUC	Quality
16371924	**Varaut A., Fontaine H., Serpaggi J., Verkarre V., Vallet-Pichard A., Nalpas B., Imbertbismuth F., Lebray P., Pol S.**	2005	**Fibrosis**	**24**	**92**	**0.66**	**Fair**
				84	**45**		
				nd	**nd**	**0.70**	
				nd	**nd**		
				14	**97**	**0.47**	
				71	**52**		
				30	**98**	**0.66**	
				80	**48**		
				69	**88**	**0.71**	
				93	**39**		
				nd	**nd**	**0.72**	
				100	**29**		

PMID	Author	Year	Outcome	Sensitivity (%)	Specificity (%)	ROC-AUC	Quality
20847572	**Canbakan M., Senturk H., Canbakan B., Toptas T., Tabak O., Ozaras R., Tabak F., Balcı H., Sut N., Ozbay G.**	2011	**Fibrosis**	**64**	**32**	**0.46 (0.25, 0.67)**	**Poor**
				13	**84**		
21393486	**Liu CH., Liang CC., Huang KW., Liu CJ., Chen SI., Lin JW., Hung PH., Tsai HB., Lai MY., Chen PJ., Chen JH., Chen DS., Kao JH.**	2011	**Fibrosis**	**30**	**100**	**nd**	**Good**
			Fibrosis	**55**	**96**	**nd**	
			Fibrosis	**93**	**88**	**0.96 (0.94, 0.98)**	
			Fibrosis	**45**	**99**	**nd**	
			Fibrosis	**45**	**99**	**nd**	
			Fibrosis	**95**	**99**	**0.98 (0.97, 1.00)**	
ATC-abstract 596	**A. Brar, M. Salifu, M. Rampal, N. Sumrani, D. John, F. Tedla.**	2012	**Fibrosis**	**nd**	**nd**	**0.54**	**Fair**
20847572	**Canbakan M., Senturk H., Canbakan B., Toptas T., Tabak O., Ozaras R., Tabak F., Balcı H., Sut N., Ozbay G.**	2011	**Activity (Necroinflammation)**	**36**	**40**	**0.36 (0.15, 0.57)**	**Poor**

Table S2: Evidence profile: Diagnostic testing for liver fibrosis (by biopsy)

Diagnostic Test	# of Studies	Total N of Patients	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence, including Applicability	Other Considerations	Summary of Findings		
							Quality of Evidence for Outcome	Description of Findings	Importance of Outcome
APRI*	5 studies	995	No limitations (0)	No important inconsistencies (0)	Direct (0)	None (0)	High	F2-F4: AUC 0.73-0.83 (4 studies, transplant/HD) APRI >0.4 Sn 84-88%, Sp 51-54% (2 studies, HD) APRI >0.95 Sn 32-44%, Sp 93-96% (2 studies, HD) † F3-F4: AUC 0.84 (1 study, HD) APRI >0.4 Sn 67%, Sp 56% (1 of 2 studies, HD) ‡ APRI >0.95 Sn 27%, Sp 92% (1 of 2 studies, HD) §	High
Fibroscan ‖	1 study	304	No limitations (0)	NA	Direct (0)	Sparse data (-1)	Moderate	F2-F4: AUC 0.96 (1 study, HD) F3-F4: AUC 0.98 (1 study, HD)	High
Fibrotest ¶	2 studies	143	Some limitations (-1)	No important inconsistencies (0)	Direct (0)	None (0)	Moderate	F2-F4: AUC 0.46-0.47 (2 studies, HD) Fibrotest >0.2 Sn 64-71%, Sp 32-52% (2 studies)** Fibrotest >0.6 Sn 13-14%, Sp 84-97% (2 studies) †† F3-F4: AUC 0.66 (1 study, HD), AUC 0.72 (1 study, Txp)	High
FIB-4 ‡‡	1 study	26	Some limitations (-1)	NA	Direct (0)	Sparse data (-1)	Low	AUC 0.54 (1 study, HD)	High
ActiTest §§	1 study	33	Serious limitations (-2)	NA	Indirect (-1)	Sparse data (-1)	Very Low	Predicting "Activity" (necroinflammation): ActiTest >0.11 Sn 36%, Sp 40%, AUC 0.36	High

Overall summary (and Quality of Overall Evidence):

High quality evidence that APRI accurately diagnoses METAVIR F2-F4, sparse evidence that APRI accurately diagnoses F3-F4.
Sparse, moderate quality evidence that Fibroscan very accurately diagnoses F2-F4 and F3-F4.
Very Low to Moderate evidence that other tests (Fibrotest, FIB-4, ActiTest) do not accurately diagnose liver fibrosis (or necroinflammation).

Studies included in EP: 16371924, 20847572, 21393486, ATC-abstract 596, 20847572

Abbreviations: ALT = alanine aminotransferase, APRI = AST to platelet ratio index, AST = aspartate aminotransferase, AUC = area under the (receiver operating characteristics) curve, F2/F3/F4 = METAVIR scores, FIB4 = Fibrosis-4 test, GGT = γ-glutamyltranspeptidase, HD = hemodialysis, NA = not applicable (single study), Sn = sensitivity, Sp = specificity.

Annotations:

* APRI = (AST level [U/L]/upper limit of normal)/platelet counts (10^9 /liter) x 100
† Transplant population (1 study): APRI >0.50 Sn 85%, Sp 51%; APRI >1.5 Sn 25%, Sp 89%.
‡ 2nd HD study: APRI >0.55 Sn 95%, Sp 69%.
§ 2nd HD study: APRI >1.0 Sn 42%, Sp 89%.
‖ Also known as transient elastography or liver stiffness measurement, based on proprietary ultrasonography.
¶ Proprietary test based on α2-macroglobulin, haptoglobin, apoliprotein A1, GGT, total bilirubin, age and sex.
** In transplant population (1 study): Fibrotest >0.2 Sn 93%, Sp 39%.
†† In transplant population (1 study): Fibrotest >0.6 Sn 69%, Sp 88%; AUC 0.71.
‡‡ FIB4 = age (years) × AST [U/L]/platelet count[10^9/L] × (ALT [U/L])$^{1/2}$
§§ Proprietary test based on α2-macroglobulin, haptoglobin, apoliprotein A1, GGT, total bilirubin, ALT, age and sex.

Table S3: Summary Table
HCV infection as independent predictor of CKD progression

PMID	Author	Year	Type of article	Country	Study design	Registry Name	Population	Sample Size, Total	CKD stage, baseline	Kidney function, baseline	HCV genotypes	Follow-up Time	Predictor	Predictor Definition	Predictor units	# w/Predictor or mean (95%)
													HCV+	nd	nd	100,518
25529816	**Molnar**	2015	Peer-reviewed publication	US	Retrospective cohort study	nd	US veterans with baseline GFR ≥60 mL.min.1.73m2 with and without HCV	1,021,049	CKD 1-2	HCV+: 93 ± 16	nd	7.8 years (IQR 6.8, 8.4)				
										HCV-: 87 ± 16			HCV-			920,531
													HCV+	nd	nd	100,518
													HCV-			920,531
													HCV+	nd	nd	100,518
												8.0 years (IQR 7.1, 8.5)				
													HCV-			920,531
														detectable RNA		
23904290	**Lucas**	2013	Peer-reviewed publication	US	Retrospective cohort study	NA-ACCORD	HIV+ adults	63,023	CKD 0-5	GFR >90 70%	nd	nd	HCV viremic			57,854 pt-yr
										GFR 60-89 26%			HCV aviremic	seropositiitve, undetectable RNA		4769 pt-yr
										GFR 30-59 3.4%			HCV-			269,805 pt-yr
										GFR 15-29 0.4%				detectable RNA		
													HCV viremic			61 833 pt-yr
										GFR <15 0.7%			HCV aviremic	seropositiitve, undetectable RNA		5139 pt-yr
													HCV-			287,386 pt-yr
														detectable RNA		
													HCV viremic			59 892 pt-yr
													HCV aviremic	seropositiitve, undetectable RNA		4952 pt-yr
													HCV-			279 096 pt-yr
						Taiwanese National Health								seropositive	**nd**	5,189
20400217	**Lee**	2010	Peer-reviewed publication	Taiwan	Cross-sectional study	Insurance Program	Adults	54,966	CKD 1-5	GFR ≥90: 27.7%	nd	nd	HCV alone			
														seropositive	**nd**	5,189
										GFR 60-89: 57.9%			HCV alone			
														seropositive	**nd**	5,189
										GFR 30-59: 13.7%			HCV alone			
														seropositive	**nd**	5,189
										GFR <0.8%			HCV alone			
	Hofmann	2011	Peer-reviewed publication	Sweden	Retrospective cohort study	Swedish Institute for Infectious Diseae Control	HCV and matched controls	43,000	nd	nd	nd	nd				
21386707													**MALES**			
													HCV-infected	nd	**nd**	25,412
													HCV-infected	nd	**nd**	25,412
													HCV-infected	nd	**nd**	25,412
													HCV-infected	nd	**nd**	25,412
													FEMALES			
													HCV-infected	nd	**nd**	25,412
													HCV-infected	nd	**nd**	25,412

PMID	Author	Year	Type of article	Country	Study design	Registry Name	Population	Sample Size, Total	CKD stage, baseline	Kidney function, baseline	HCV genotypes	Follow-up Time	Predictor	Predictor Definition	Predictor units	# w/Predictor or mean (95%)
													HCV-infected	nd	**nd**	25,412
													HCV-infected	nd	**nd**	25,412
18371161	**Ishizaka**	2008	Peer-reviewed publication	Japan	Cross-sectional study	General Health Screen at Mitsui Memorial Hospital	Working adults	12,535	nd	Hep B - and Hep C -: 70 ± 10	nd	nd	HCV +	HCV Ag +	**nd**	72
										Hep B +: 70 ± 9			HCV - and HBV -	HCV Ag - and HBV Ag -	**nd**	12,333
										Hep C +: 67 ± 13			HCV +	HCV Ag +	**nd**	72
													HCV - and HBV -	HCV Ag - and HBV Ag -	**nd**	12,333
													HCV +	HCV Ag +	**nd**	72
													HCV - and HBV -	HCV Ag - and HBV Ag -	**nd**	12,333
23311442	**Cao**	2013	Peer-reviewed publication	China	Cross-sectional Study	nd	HIV infected antiretroviral therapy-naïve	538	nd	Non-CKD: 113.54 ± 24.06	nd	nd	HCV +	HCV Ab +	**nd**	80
										CKD: 98.98 ± 34.48			HCV -	HCV Ab -	**nd**	444
														seropositive	nd	178,447 pt-yr
17592100	**Tsui**	2007	Peer-reviewed	US	Retrospective Cohort Study	Medicare, the Department of Veterans Affairs, and the United States Renal Data System	US Veterans	474,369	HCV+: 9.4% with any stage	HCV+	nd	3.4y (median 3.6y)	HCV+			
									HCV-: 16.6% with any stage	GFR>=60: 83%			HCV-	seronegative	nd	1,442,826 pt-yr
										GFR 30-59: 15%			HCV+, eGFR >= 60	seropositive	nd	163,846 pt-yr
										GFR<30: 2%			HCV-, eGFR >= 61	seronegative	nd	1,219,523 pt-yr
														seropositive	nd	13,295 pt-yr
										HCV-			HCV+, eGFR 30-59			
										GFR>=60: 91%			HCV-, eGFR 30-59	seronegative	nd	207,484 pt-yr
														seropositive	nd	1,307 pt-yr
										GFR 30-59: 8%			HCV+, eGFR <30			
														seronegative	nd	15,819 pt-yr
										GFR<30: 2%			HCV-, eGFR <30			
26526451	**Rogal**	2016	Peer-reviewed	US	Retrospective Cohort Study	HCV Infected Veterans (ERCHIVES)	US Veterans	71,528			1: 33.14%, 2: 4.75%, 3: 5.25%, 4: 0.35%, Mixed: 0.08%, Missing: 56.43%	HCV+: 5.99 (SD 2.84)y	HCV+	nd		nd
												HCV-: 4.91 (SD 2.16)y	HCV-			
													HCV+	nd		nd
													HCV-			
22554802	**Su**	2012	Peer-reviewed	Taiwan	Retrospective Case Control	Taiwan National Health Insurance Research Database	Taiwanese adults	37,746			nd	5.58 (SD 2.04)	HCV+			32,648 pt-yr
													HCV-			176,921 pt-yr
26817874	**Hwang**	2016	Peer-reviewed	Taiwan	Retrospective Case Control	Longitudinal Cohort of Diabetes Patients	DM	19,574			nd	12 y				nd
													HCV+			
													HCV-			
16524948	**Tsui**	2006	Peer-reviewed publication	US	Cross-sectional	NHANES III	US general population	15029 (366 HCV+)	nd	HCV+: 110 (106-114)	nd	nd	HCV+	nd	nd	366
										HCV-: 100 (99-101)			HCV-			14,663
													HCV+	nd	nd	366
													HCV-			14,663
												nd	HCV+	nd	nd	100,518
													HCV-			920,531
19747988	**Asrani**	2010	Peer-reviewed publication	US	Retrospective cohort study	Conglomerate of health insurance carriers database, NOS	US adults	167,569	CKD 0-5	GFR >60 95%	nd	nd	HCV+	EIA, NAT, gentoype test		13,384
										GFR 30-59 5.0%			HCV -			154,185
										GFR 15-29 0.2%			HCV+	EIA, NAT, genotype test		8,063
										GFR <15 0.1%			HCV -			80,759
24257691	**Chen**	2014	Peer-reviewed publication	Taiwan	Retrospective cohort study	Taiwan National Health Insurance Research Database	Taiwanese adults (no renal Txp)	9430 HCV+ and 3772	Non-CKD (per ICD-9)	nd	nd	~7 years	HCV +	ICD-9 codes		9,430

PMID	Author	Year	Type of article	Country	Study design	Registry Name	Population	Sample Size, Total	CKD stage, baseline	Kidney function, baseline	HCV genotypes	Follow-up Time	Predictor	Predictor Definition	Predictor units	# w/Predictor or mean (95%)
													HCV-			37,720 (matched)
25826420	**Mocroft**	2015	Peer-reviewed publication	Europe, U	Prospective cohort study	nd	HIV+	17,594	Healthy (not CKD)	median: 104	nd	103,185 person-years	HCV +	nd		2,262
													HCV -	nd		11,386
21185632	**Butt**	2011	Peer-reviewed publication	US	Retrospective cohort study	ERCHIVES	HCV infected patients, HCV-uninfected controls	43,139	Healthy (not CKD 3-5)	HCV+: 99 (22.6)	nd	HCV+: 3.15 (1.4) y	HCV +	anti-HCV Ab + or HCV RNA +		18,002
										HCV-: 92 (22.2)		HCV-: 3.00 (1.3) y	HCV -	anti-HCV Ab - or HCV RNA -		25,137
22994610	**Bickel**	2015	Peer-reviewed publication	Germany	Prospective cohort study	nd	HIV+	9198	Not ESRD	nd	nd	5 years max	HCV +			1008
													HCV -			4330 (+3860 unknown)

Table S3: Summary Table
HCV infection as independent predictor of CKD progression

PMID	Author	Year	Outcome	Outcome Definition	Cases (# w/Outcome)	Rate	Adjustments (reject if none)	Metric (HR, OR, RR, beta, change)	Estimate (95% CI)	Comparison	P_value	Risk of bias
25529816	**Molnar**	2015	GFR<60 incidence	the presence of two consecutive eGFR values of <60 mL/min/1.73 m2 at least 90 days apart with the added stipulation that the decrease in eGFR should be at least 25% from the baseline eGFR	11,271	16.7 (16.4,17.0)/1000 pt/yr	age, gender, race/ethnicity, baseline GFR, ,comorbidities, SBP, DBP, BMI, socio-demographic parameters	HR	1.15 (1.12, 1.17)	vs. HCV-	nd	**LOW**
					95,837	14.9 (14.8,15.0)/1000 pt-yr						
			GFR loss >5 mL/min/1.73m2/year	Rapid deterioration was defined as slopes of < −5 mL/min/1.73 m2/year (i.e., loss of eGFR of >5 mL/min/1.73 m2) /year).	nd	nd	age, gender, race/ethnicity, baseline GFR, ,comorbidities, SBP, DBP, BMI, socio-demographic parameters	OR	1.22 (1.19, 1.26)	vs. HCV-	nd	
					nd	nd						
			ESRD	initiation of renal replacement therapy or preemptive transplant	904	1.2 (1.1, 1.3)/1000 pt-yr	age, gender, race/ethnicity, baseline GFR, ,comorbidities, SBP, DBP, BMI, socio-demographic parameters	HR	1.98 (1.81, 2.16)	vs. HCV-	nd	
					2479	0.36 (0.34, 0.37)/1000 pt-yr						
23904290	**Lucas**	2013	CKD 3	GFR<60 for >=90 d	1666	28.8 (27.5, 30.2)/1000 pt-yr	Age, sex, race, history of injection drug use, hepatitis B surface antigen positivity, baseline glomerular filtration rate, and time-updated covariates including calendar year, CD4 cell count, HIV RNA, antiretroviral therapy use, tenofovir use, indinavir use, lopinavir/ritonavir use, atazanavir use, hypertension, and diabetes.	HR	1.36 (1.26, 1.46)	vs. HCV-	nd	**LOW**
					122	25.6 (21.4, 30.6)/1000 pt-yr			1.19 (0.98, 1.45)	vs. HCV-	nd	
					5090	18.9 (18.4, 19.4)/1000 pt-yr						
			CKD 5	GFR<15 for >=90 d	376	6.1 (5.5, 6.7)/1000 pt-yr	Age, sex, race, history of injection drug use, hepatitis B surface antigen positivity, baseline glomerular filtration rate, and time-updated covariates including calendar year, CD4 cell count, HIV RNA, antiretroviral therapy use, tenofovir use, indinavir use, lopinavir/ritonavir use, atazanavir use, hypertension, and diabetes.	HR	1.95 (1.64, 2.31)	vs. HCV-	nd	
					23	4.4 (2.9, 6.7)/1000 pt-yr			1.69 (1.07, 2.65)	vs. HCV-	nd	
					699	2.4 (2.3, 2.6)/1000 pt-yr						
			CKD decline	≥25% GFR decline from baseline to a GFR < 60 mL/min/1.73 m2 that persisted for at least 90 days among subjects with baseline GFR ≥ 30 mL/min/1.73 m2.	984	16.4 (15.4, 17.5)/1000 pt-yr	Age, sex, race, history of injection drug use, hepatitis B surface antigen positivity, baseline glomerular filtration rate, and time-updated covariates including calendar year, CD4 cell count, HIV RNA, antiretroviral therapy use, tenofovir use, indinavir use, lopinavir/ritonavir use, atazanavir use, hypertension, and diabetes.	HR	1.31 (1.19, 1.44)	vs. HCV-	nd	
					76	15.4 (12.3, 19.2)/1000 pt-yr			1.31 (1.02, 1.68)	vs. HCV-	nd	
					2885	10.3 (9.9, 10.7)/1000 pt-yr						
20400217	**Lee**	2010	CKD	determined by eGFR and proteinuria based on the K/DOQI definition	nd	nd	age, sex, educational status, BMA, albumin, Hb, cholesterol, uric acid, HTN, and DM	OR	1.26 (1.17, 1.38)	vs. HBV or HBV/HCV	<0.001	**HIGH**
			CKD	defined using GFR estimated using the 6-variable MDRD Study equation	nd	nd	age, sex, educational status, BMA, albumin, Hb, cholesterol, uric acid, HTN, and DM	OR	1.23 (1.14, 1.34)	vs. HBV or HBV/HCV	<0.001	
			proteinuria	presence of urine protein of at least grade 1+ in repeated measures	nd	nd	age, sex, educational status, BMA, albumin, Hb, cholesterol, uric acid, HTN, and DM	OR	1.14 (1.003, 1.300)	vs. HBV or HBV/HCV	0.04	
			low GFR	eGFR <60 mL/min/1.73 m2	nd	nd	age, sex, educational status, BMA, albumin, Hb, cholesterol, uric acid, HTN, and DM	OR	1.30 (1.20, 1.42)	vs. HBV or HBV/HCV	<0.001	
21386707	**Hofmann**	2011										**HIGH**
			glomerular diseases	risk of hospitalization for glomerular disease; ICD-9 codes: 580–583, ICD-10 codes: N00–N08	104	nd	age, sex, and interaction between HCV status and sex	HR	2.9 (2.0, 4.3)	vs. no HCV infection	nd	
			renal failure	risk of hospitalization for renal failure; ICD-9 codes: 584–586, ICD-10 codes: N17–N19	245	nd	age, sex, and interaction between HCV status and sex	HR	4.6 (3.7, 5.8)	vs. no HCV infection	nd	
			other kidney diseases	risk of hospitalization for other kidney diseases; ICD-9 codes: 587–589 and 593, ICD-10 codes:N25–N28	30	nd	age, sex, and interaction between HCV status and sex	HR	2.9 (1.4, 6.0)	vs. no HCV infection	nd	
			all noncancer kidney diseases combined	risk of hospitalization for combination of above	349	nd	age, sex, and interaction between HCV status and sex	HR	3.9 (3.2, 4.8)	vs. no HCV infection	nd	
			glomerular diseases	risk of hospitalization for glomerular disease; ICD-9 codes: 580–583, ICD-10 codes: N00–N08	34	nd	age, sex, and interaction between HCV status and sex	HR	5.1 (3.0, 8.8)	vs. no HCV infection	nd	
			renal failure	risk of hospitalization for renal failure; ICD-9 codes: 584–586, ICD-10 codes: N17–N19	59	nd	age, sex, and interaction between HCV status and sex	HR	7.2 (4.9, 10.6)	vs. no HCV infection	nd	

PMID	Author	Year	Outcome	Outcome Definition	Cases (# w/Outcome)	Rate	Adjustments (reject if none)	Metric (HR, OR, RR, beta, change)	Estimate (95% CI)	Comparison	P_value	Risk of bias
			other kidney diseases	risk of hospitalization for other kidney diseases; ICD-9 codes: 587–589 and 593, ICD-10 codes:N25–N28	4	nd	age, sex, and interaction between HCV status and sex	HR	10.0 (2.7, 37.2)	vs. no HCV infection	nd	
			all noncancer kidney diseases combined	risk of hospitalization for combination of above	94	nd	age, sex, and interaction between HCV status and sex	HR	5.8 (4.2, 7.9)	vs. no HCV infection	nd	
18371161	**Ishizaka**	2008	Low eGFR	eGFR<60 mL/min/1.73 m2	22	31%	age, sex, SBP, and FPG	OR	1.63 (0.95, 2.80)	vs. HCV - and HBV -	0.077	**HIGH**
					1887	15%						
			Albuminuria	UAER of 330 mg/g	14	19%	age, sex, SBP, and FPG	OR	2.00 (1.06, 3.76)	vs. HCV - and HBV -	0.034	
					1157	9%						
			CKD	eGFR<60 mL/min/1.73 m2, or UAER of 330 mg/g	nd	nd	age, sex, SBP, and FPG	OR	1.83 (1.10, 3.05)	vs. HCV - and HBV -	0.02	
					nd	nd						
23311442	**Cao**	2013	CKD	GFR < 60 mL/min per 1.73 m2 and/or isolated proteinuria (>= 1 + on urine dipstick) for 3 months	20	25%	age, HIV viral load	OR	2.074 (1.036, 4.149)	vs. HCV -	0.011	**HIGH**
					64	14%						
17592100	**Tsui**	2007	ESRD	onset of chronic dialysis or renal transplantation	760	4.26 (3.97-4.57)/1000 pt-yr	age, sex, race/ethnicity, diabetes mellitus, hypertension, human immunodeficiency virus infection, congestive heart failure, coronary heart disease, peripheral vascular disease, chronic obstructive pulmonary disease, cerebrovascular disease, substance abuse, and baseline eGFR, also stratified by age	HR	1.68 (1.54-1.82)	vs. HCV-	nd	**LOW**
					4393	3.05 (2.96-3.14)/1000 pt-yr						
					106	0.65 (0.54-0.78)/1000 pt-yr		HR	2.74 (2.13-3.51)	vs. HCV-	nd	
					308	0.25 (0.23-0.28)/1000/ pt-yr						
					228	17.15 (15.06-19.53)/1000 pt-yr		HR	2.57 (2.20-3.01)	vs. HCV-	nd	
					1039	5.01 (4.71-5.32)/1000 pt-yr						
					426	326.03 (296.49-358.50)/1000 pt-yr		HR	1.28 (1.15-1.44)	vs. HCV-	nd	
					3046	192.55 (185.83-199.51)/1000 pt-yr						
26526451	**Rogal**	2016	CKD 3-5		nd	nd	baseline GFR with FIB-4 included as a time-varying covariate	HR	0.86 (0.79, 0.92)	HCV+ vs. HCV-	<0.0001	**LOW**
			Progressive CKD	decline in eGFR of >= 30 from baseline	nd	nd	baseline GFR with FIB-4 included as a time-varying covariate	HR	0.93 (0.86, 1.00)	HCV+ vs. HCV-	0.06	
22554802	**Su**	2012	ESRD		77		age, sex, occupation, urbanization, and Charlson Comorbidity Index score	HR	1.53 (1.17, 2.01)	vs. HCV-	0.002	**LOW**
					196		results also stratified by age, sex, and followup interval					
26817874	**Hwang**	2016	ESRD		nd	nd	sex, age at index date and various comorbidities including hypertension, CAD, hyperlipidemia, HBV infection, gout and liver cirrhosis	HR	1.47 (1.11-1.93)	vs. HCV-	<0.05	**LOW**
16524948	**Tsui**	2006	GFR<60 prevalence	eGFR <60	366	2.0 (1.1,3.6) %	age, gender, race/ethnicity, educational status, smoking status, DM,and HTN	OR	0.89 (0.49, 1.62)	vs. HCV-	0.7	**LOW**
					14,663	4.3 (3.7,4.9) %						
			Albuminuria prevalence	Albuminuria	366	nd	age, gender, race/ethnicity, educational status, smoking status, DM,and HTN	OR	1.22 (1.19, 1.26)	vs. HCV-	nd	
					14,663	nd						
19747988	**Asrani**	2010	CKD prevalence (baseline)	GFR<60 (single measurement)	13,384	5.3%	Age, sex, cirrhosis, DM, HTN, CHF, PVD, CAD, COPD, HIV, drug/alcohol abuse, depression	OR	0.90 (0.36, 2.27)	vs. HCV-	0.7	**LOW**
					154,185	5.1%						
			Progression to CKD	GFR <60 without baseline CKD	8,063	3.8%	Age, sex, cirrhosis, DM, HTN, CHF, PVD, CAD, COPD, HIV, drug/alcohol abuse, depression	OR	0.92 (0.79, 1.08)	vs. HCV-		
					80,759	3.5%						
24257691	**Chen**	2014	CKD, incident	ICD-9 code	367	3.9% (5.46/1000 person-yr)	age per year, sex, comorbidities, geographic region, urbanization level, enrollee category, and number of medical visits in 1 year before study entry	HR	1.28 (1.12, 1.46)	vs HCV-	<0.001	**LOW**

PMID	Author	Year	Outcome	Outcome Definition	Cases (# w/Outcome)	Rate	Adjustments (reject if none)	Metric (HR, OR, RR, beta, change)	Estimate (95% CI)	Comparison	P_value	Risk of bias
					960	2.5% (3.43/1000 person-yr)						
25826420	**Mocroft**	2015	CKD	eGFR <= 60 mL/min per 1.73 m2 (>3mo)	89	13.9%	age, sex, HIV risk, eGFR, nadir CD4 <200, HTN, prior CVD, diabetes	OR [exp(beta)]	1.40 (nd)	vs. HCV -	nd (at least <0.10)	**HIGH**
					391	61.0%						
21185632	**Butt**	2011	CKD 3-5	eGFR < 60 mL/min per 1.73 m2 for >=2 times	3,140	17.4%	age, sex, race, eGFR, HTN, smoking, COPD, diabetes, dyslipidemia, anemia, alcohol abuse, drug abuse, decompensated liver disease, ACEi/ARB use	HR	1.30 (1.23, 1.37)	vs. HCV -	< 0.0001	**LOW**
					3,738	14.9%						
22994610	**Bickel**	2015	ESRD	Initiation of RRT	~13	nd	race (black), intravenous drug use (sex, AIDS, HBV coinfection were NS)		0.84 (0.24, 2.86)	vs. HCV -	0.79	**UNCLEAR**
					~22	nd						

Table S4: Evidence profile: HCV infection as independent predictor of CKD progression

Outcome	# of Studies and Study Design*	Total N of Participants	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence, including Applicability	Other Considerations	Summary of Findings: Quality of Evidence for Outcome	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
ESRD	6 retrospective, 1 prospective cohorts	Cohorts: 9198-1,021,049	No limitations (0)	No important inconsistencies (0)†	Direct (0)	None	High (Moderate for HIV coinfection)	No HIV coinfection: All studies (6 of 6 analyses in 5 studies) found significantly higher rates of ESRD in HCV+ (vs. HCV-). ES range 1.47-7.2 (women)/4.6 (men). HIV coinfection: Inconsistent: One study found no significant association between HCV coinfection and RRT (ES 0.84) One found significant association with CKD 5 (ES 1.95)	Critical
CKD	10 retrospective, 1 prospective cohorts	Cohorts: 538-1,021,049	No limitations (0)	Some inconsistencies (-1)	Direct (0)	None	Moderate	No HIV coinfection: Most studies(5 of 7) found significantly higher CKD incidence in HCV+ (vs. HCV-). ES range 0.86-1.83. One study found significantly *lower* risk of CKD in patients with *newly acquired* HCV. HIV coinfection: All studies (3 of 3) found significantly higher CKD incidence in HCV+ coinfection (vs. HCV-). ES range 1.36-2.07.	High
Decline in Kidney Function	3 retrospective cohorts	Cohorts: 63,023-1,021,049	No limitations (0)	Some inconsistencies (-1)	Direct (0)	None	Moderate	No HIV coinfection: 2 studies had conflicting findings with significant association between HCV+ and GFR loss >5 mL/min (OR 1.22) in 1 study, and no association with GFR loss ≥30 mL/min (HR 0.93) in 1 study. HIV coinfection: 1 study significant association between HCV+ and GFR decline ≥25% to <60 mL/min (HR 1.31).	High
Total N	15 retrospective, 1 prospective cohorts	Cohorts: 538-1,021,049							

Association of HCV+ status and CKD:
People with HCV infection are at increased risk of CKD and ESRD compared to people without HCV infection. Among people with HIV, coinfection with HCV increases the risk for CKD, but it remains unclear whether it also increases the risk for ESRD.

Quality of Overall Evidence:
Moderate

Studies included in EP: 25529816, 23904290, 20400217, 21386707, 18371161, 23311442, 17592100, 26526451, 22554802, 26817874, 16524948, 19747988, 24257691, 25826420, 21185632, 22994610

Abbreviations: CKD: chronic kidney disease; DAA: direct-acting antivirals; GFR: glomerular filtration rate; RCT: randomized controlled trial; SVR: sustained virological response

Annotations:
* Limited to largest studies with the addition of smaller studies of HIV population and from otherwise unrepresented countries.
† For general population. Two studies of HIV population were inconsistent with each other.

Table S5. Evidence Profile: Treatment with direct-acting antiviral regimens in CKD G4-G5 non-dialysis patients

Outcome	Regimen[1]	# of Studies[2]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings			
								Certainty of Evidence for Outcome	Description of Findings		Importance of Outcome
Death, ~6-12 mo	Any	0							ND		Critical
SVR12	DCV/ASV	1	10	No limitations	N/A	Direct	Sparse	Very low	100% (52, 100)	Very high SVR12 for all treatments. Mostly ~97%. No direct evidence of differences among regimens.	High
	GLE/PIB	3	132	No limitations	Consistent	Direct	None	High	98.5% (94.1, 99.6)		
	GZR/EBR	5	857	No limitations	Consistent	Direct	None	High	96.7% (95.4, 97.8)		
	PrO±D	3	103	No limitations	Inconsistent	Direct	Imprecise	Low	89.4% (75.7, 97.8)		
	SOF	2	41	No limitations	Inconsistent	Direct	Imprecise	Very Low	71.7% (29.1, 98.6)		
	SOF/DCV	4	571	No limitations	Consistent	Direct	None	High	97.1% (95.7, 98.3)		
	SOF/LDV	2	43	No limitations	Consistent	Direct	Imprecise	Very Low	100% (84, 100)		
	SOF/SIM	1	41	No limitations	N/A	Direct	Sparse	Very Low	92.7% (79.6, 97.6)		
	SOF/VEL	2	99	No limitations	Consistent	Direct	Imprecise	Low	96.0% (89.7, 98.5)		
Serious AE due to DAA	DCV/ASV	0							ND	Rare, but insufficient evidence. No evidence of differences among regimens.	High
	GLE/PIB	2	67	Some limitations[3]	Consistent	Direct	Imprecise	Very Low	0% (0, 11)		
	GZR/EBR	1	14	Serious limitations[4]	N/A	Direct	Sparse	Very Low	0% (0, 39)		
	PrO±D	2	53	Serious limitations[5]	Consistent	Direct	Imprecise	Very Low	0% (0, 13)		
	SOF	3	41	No limitations	Consistent	Direct	Imprecise	Very Low	0% (0, 17)		
	SOF/DCV	2	52	Serious limitations[6]	Consistent	Direct	Imprecise	Very Low	0% (0, 14)		
	SOF/LDV	2	43	No limitations	Consistent	Direct	Imprecise	Very Low	0% (0, 16)		
	SOF/SIM	0							ND		
	SOF/VEL	1	74	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 10)		
Discontinue due to AE	DCV/ASV	0							ND	Rare, but insufficient evidence. No evidence of differences among regimens.	High
	GLE/PIB	2	67	Some limitations[7]	Consistent	Direct	Imprecise	Very Low	3.0% (0.7, 11)		
	GZR/EBR	1	14	Serious limitations[8]	N/A	Direct	Sparse	Very Low	0% (0, 39)		
	PrO±D	2	53	Serious limitations[9]	Consistent	Direct	Imprecise	Very Low	0% (0, 13)		
	SOF	2	41	No limitations	Inconsistent	Direct	Imprecise	Very Low	11% (1.2, 28)		
	SOF/DCV	2	25	Serious limitations[10]	Consistent	Direct	Imprecise	Very Low	0% (0, 25)		
	SOF/LDV	2	43	No limitations	Consistent	Direct	Imprecise	Very Low	0% (0, 16)		
	SOF/SIM	0							ND		
	SOF/VEL	1	74	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 10)		

Outcome	Regimen [11]	# of Studies[12]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings		
								Certainty of Evidence for Outcome	Description of Findings	Importance of Outcome
Change in CKD category	DCV/ASV	0							ND	
	GLE/PIB	1	17	No limitations	N/A	Direct	Sparse	Very Low	Variable	
	GZR/EBR	0							ND	
	PrO±D	1	12	No limitations	N/A	Direct	Sparse	Very Low	Incident dialysis 0% (0, 43)	
	SOF	0							ND	
	SOF/DCV	0							ND	
	SOF/LDV	0							ND	
	SOF/SIM	0							ND	
	SOF/VEL	0							ND	
									Insufficient evidence	High
eGFR	DCV/ASV	0							ND	
	GLE/PIB	0							ND	
	GZR/EBR	0							ND	
	PrO±D	2	48	No limitations	Consistent	Direct	Imprecise	Very Low	Change NS	
	SOF	1	17	No limitations	N/A	Direct	Imprecise	Very Low	Change NS	
	SOF/DCV	1	11	No limitations	N/A	Direct	Imprecise	Very Low	Change NS	
	SOF/LDV	1	18	No limitations	N/A	Direct	Imprecise	Very Low	Change NS	
	SOF/SIM	0							ND	
	SOF/VEL	1	67	No limitations	N/A	Direct	Sparse	Very Low	Change NS	
									Insufficient evidence	Moderate
Balance of Potential Benefits and Harms: DAAs yield very high rates of SVR12 with rare adverse events (although evidence on adverse events is sparse). Some regimens may have poorer SVR12, but there are no randomized or other comparisons of DAA regimens in comparable patients.								**Certainty of Overall Evidence:** High (for DAAs in general)		

Abbreviations: AE: adverse events, ASV: asunaprevir, CKD: chronic kidney disease (category), DAA: direct-acting antivirals, DCV: daclatasvir, EBR: elbasvir, GFR: glomerular filtration rate, GLE: glecaprevir, GZR: grazoprevir, LDV: ledipasvir, N/A: not applicable, ND: no data, NS: nonsignificant, PIB: pibrentasvir, PrO±D: paritaprevir/ritonavir/ombitasvir ± dasabuvir, SIM: simeprevir, SOF: sofosbuvir, SVR12: sustained virologic response at 12 weeks post-treatment, VEL: velpatasvir.

1 Notation of inclusion of ribavirin omitted from this table.

2 Single groups, mostly retrospective.

3 Reporting bias (larger study did not report outcome).

4 Reporting bias (many studies did not report outcome).

5 Reporting bias (study with half available patients did not report outcome).

6 Reporting bias (study with 90% of patients did not report outcome).

7 Reporting bias (larger study did not report outcome).

[8] Reporting bias (many studies did not report outcome).
[9] Reporting bias (study with half available patients did not report outcome).
[10] Reporting bias (study with 90% of patients did not report outcome).
[11] Notation of inclusion of ribavirin omitted from this table.
[12] Single groups, mostly retrospective.

Table S6. Evidence Profile: Treatment with direct-acting antiviral regimens in CKD G5D (dialysis) patients

Outcome	Regimen[1]	# of Studies[2]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings: Certainty of Evidence for Outcome	Summary of Findings: Description of Findings		Summary of Findings: Importance of Outcome
Death, ~6-12 mo	DCV/ASV	0								Reported death rates low (0.9-6.4%) but very sparse, imprecise estimates. Reporting bias likely inflating estimates. No reported deaths related to DAA or HCV. No evidence of differences among regimens.	Critical
	GLE/PIB	2	109	Serious limitations[3]	Consistent	Direct	Imprecise	Very low	0.9% (0.1, 6.2)		
	GZR/EBR	0									
	PrO±D	2	60	Serious limitations[4]	N/A	Direct	Sparse	Very low	1.7% (0.2, 11)		
	SOF	0									
	SOF/DCV	1	31	Serious limitations[5]	N/A	Direct	Sparse	Very low	6.4% (1.6, 22)		
	SOF/LDV	0									
	SOF/SIM	0									
	SOF/VEL	1	59	Serious limitations[6]	N/A	Direct	Sparse	Very low	3.4% (0.8, 13)		
SVR12	DCV/ASV	9	341	Some limitations[7]	Some inconsistency	Direct	None	Low	93.6% (89.5, 96.8)	Very high SVR12 for all treatments. Mostly ≥94%. No direct evidence of differences among regimens.	High
	GLE/PIB	11	529	Some limitations[8]	Consistent	Direct	None	Moderate	96.9% (95.1, 98.3)		
	GZR/EBR	11	962	Some limitations[9]	Consistent	Direct	None	Moderate	96.5% (94.9, 97.8)		
	PrO±D	16	582	Some limitations[10]	Consistent	Direct	None	Moderate	96.8% (95.2, 98.1)		
	SOF	3	123	No limitations	Inconsistent	Direct	None	Low	91.9% (74.5, 99.8)		
	SOF/DCV	8	278	No limitations	Some inconsistency	Direct	None	Moderate	93.7% (88.9, 97.2)		
	SOF/LDV	7	220	Some limitations[11]	Consistent	Direct	None	Moderate	95.9% (92.8, 98.1)		
	SOF/SIM	1	12	No limitations	N/A	Direct	Sparse	Very Low	83.3% (52.3, 95.8)		
	SOF/VEL	8	629	No limitations	Consistent	Direct	None	High	93.0% (93.0, 97.3)		

Outcome	Regimen[12]	# of Studies[13]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings			
								Certainty of Evidence for Outcome	Description of Findings		Importance of Outcome
Serious AE due to DAA	DCV/ASV	8	274	Some limitations[14]	Consistent	Direct	None	Moderate	0.4% (0.1 2.5)	Rare. No evidence of differences among regimens.	High
	GLE/PIB	9	435	Some limitations[15]	Consistent	Direct	None	Moderate	0.5% (0.1, 1.8)		
	GZR/EBR	6	163	Serious limitations[16]	Consistent	Direct	Incomplete reporting	Low	0.6% (0.1, 4.2)		
	PrO±D	13	406	Some limitations[17]	Consistent	Direct	None	Moderate	0.2% (0.03, 1.7)		
	SOF	2	63	Serious limitations[18]	Consistent	Direct	Imprecise	Very Low	0% (0, 11.5)		
	SOF/DCV	4	112	Serious limitations[19]	Consistent	Direct	Imprecise	Very Low	0% (0, 6.8)		
	SOF/LDV	6	208	No limitations	Consistent	Direct	Some imprecision	Moderate	0% (0, 3.7)		
	SOF/SIM	1	12	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 43)		
	SOF/VEL	6	408	No limitations	Consistent	Direct	None	High	0% (0, 1.9)		
Discontinue due to AE	DCV/ASV	9	341	No limitations	Consistent	Direct	None	High	3.8% (2.0, 6.0)	Rare. Mostly due to minor AE. No evidence of differences among regimens.	High
	GLE/PIB	8	352	Some limitations[20]	Consistent	Direct	None	Moderate	1.6% (0.6, 3.1)		
	GZR/EBR	6	166	Serious limitations[21]	Consistent	Direct	Incomplete reporting	Low	2.5% (0.7, 5.4)		
	PrO±D	14	446	Some limitations[22]	Consistent	Direct	None	Moderate	1.8% (0.8, 3.3)		
	SOF	3	123	No limitations	Consistent	Direct	Imprecise	Very Low	0% (0, 6.2)		
	SOF/DCV	8	247	No limitations	Consistent	Direct	None	High	0.4% (0.1, 2.8)		
	SOF/LDV	7	220	No limitations	Consistent	Direct	Some imprecision	Low	0% (0, 3.5)		
	SOF/SIM	1	12	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 43)		
	SOF/VEL	6	407	Serious limitations[23]	Consistent	Direct	Some imprecision	Low	0% (0, 3.2)		
Quality of life	GLE/PIB	1	77	Serious limitations[24]	N/A	Indirect[25]	Sparse	Very Low	1/3 with clinical improvement in SF-12 PCS and MCS	Insufficient evidence	High
	SOF/DCV	1	41	Serious limitations[26]	N/A	Indirect[27]	Sparse	Very Low	SF-12 improvement 3-4 points		
Balance of Potential Benefits and Harms: DAAs yield very high rates of SVR12 with low rates of discontinuation due to adverse events or serious adverse events attributable to DAAs. Some regimens may have poorer SVR12, but there are no randomized or other comparisons of DAA regimens in comparable patients.								**Certainty of Overall Evidence:** High (for DAAs in general)			

Abbreviations. AE: adverse events, ASV: asunaprevir, DAA: direct acting antivirals, DCV: daclatasvir, EBR: elbasvir, GLE: glecaprevir, Gt: genotype, GZR: grazoprevir, HCV: hepatitis C virus, LDV: ledipasvir, MCS: Mental Component Summary, N/A: not applicable, PCS: Physical Component Summary, PIB:

pibrentasvir, PrO±D: paritaprevir/ritonavir/ombitasvir ± dasabuvir, SF-12: Short Form 12 item questionnaire, SIM: simeprevir, SOF: sofosbuvir, SVR12: sustained virologic response at 12 weeks after end of treatment, VEL: velpatasvir

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[1] Notation of inclusion of ribavirin omitted from this table.
[2] Single groups, mostly retrospective.
[3] Reporting bias (many studies did not report outcome).
[4] Reporting bias (many studies did not report outcome).
[5] Reporting bias (many studies did not report outcome).
[6] Reporting bias (many studies did not report outcome).
[7] Some per protocol and analyses with missing data.
[8] Some per protocol and unclear analyses.
[9] Some per protocol and unclear analyses.
[10] Some per protocol analyses.
[11] Some per protocol analyses.
[12] Notation of inclusion of ribavirin omitted from this table.
[13] Single groups, mostly retrospective.
[14] Reporting bias (large study did not report outcome).
[15] Reporting bias (large study did not report outcome).
[16] Reporting bias (many studies did not report outcome).
[17] Reporting bias (large studies did not report outcome).
[18] Reporting bias (large study with half of available patients did not report outcome).
[19] Reporting bias (many studies did not report outcome).
[20] Reporting bias (large studies did not report outcome).
[21] Reporting bias (many studies did not report outcome).
[22] Reporting bias (large studies did not report outcome).
[23] Reporting bias (large study with half of available patients did not report outcome).
[24] Inadequate reporting of quality of life outcome.
[25] No comparison with untreated group.
[26] Inadequate reporting from conference abstract; unclear overlap with prior study.
[27] No comparison with untreated group.

Table S7. Evidence Profile: Treatment with direct-acting antiviral regimens in kidney transplant recipients

Outcome	Regimen[1]	# of Studies[2]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings			
								Certainty of Evidence for Outcome	Description of Findings		Importance of Outcome
Death, Long-term	Any specific regimen	0							ND	Death rates may be much lower with DAA treatment than without.	Critical
	DAA vs. no DAA	3	86 vs. 63	Serious limitations[3]	Consistent	Direct	None	Low	Unadj OR 0.16 (0.04, 0.61)		
Allograft loss	DCV/ASV	0							ND	Allograft loss rates low (≤1%) but very sparse, imprecise estimates. Reporting bias may be inflating estimates, but follow-up duration mostly short (SVR12 or end of treatment). No evidence of differences among regimens. Allograft loss rate with DAA vs. no DAA unclear.	Critical
	GLE/PIB	1	20	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 30)		
	GZR/EBR	1	11	No limitations	N/A	Direct	Sparse	Very Low	0.1% (1.3, 44)		
	PrO±D	0							ND		
	SOF	1	10	Serious limitations[4]	N/A	Direct	Sparse	Very Low	0% (0, 48)		
	SOF/DCV	2	141	Serious limitations[5]	Consistent	Direct	Imprecise	Very Low	0% (0, 5.4)		
	SOF/LDV	3	84	Serious limitations[6]	Consistent	Direct	None	Low	1.2% (0.2, 8.0)		
	SOF/SIM	0							ND		
	SOF/VEL	0							ND		
	DAA vs. no DAA	3	86 vs. 63	Serious limitations[7]	Consistent	Direct	Imprecise	Very Low	Unadj OR 0.50 (0.09, 2.73)		
SVR12	DCV/ASV	0							ND	Very high SVR12 for evaluated treatments. Mostly 100%. No evidence of differences among regimens.	High
	GLE/PIB	1	20	No limitations	N/A	Direct	Sparse	Very Low	100% (70, 100)		
	GZR/EBR	2	21	No limitations	N/A	Direct	Imprecise	Very Low	100% (71, 100)		
	PrO±D	2	33	No limitations	Consistent	Direct	Imprecise	Very Low	100% (80, 100)		
	SOF	6	117	No limitations	Consistent	Direct	Imprecise	Moderate	94.8% (88.2, 98.8)		
	SOF/DCV	6	290	No limitations	Consistent	Direct	None	High	99.7% (97.6, 100)		
	SOF/LDV	10	300	No limitations	Consistent	Direct	None	High	97.3 (94.9, 99.0)		
	SOF/SIM	0							ND		
	SOF/VEL	1	10	No limitations	Consistent	Direct	Sparse	Very Low	100% (52, 100)		
Serious AE due to DAA	DCV/ASV	0							ND	Rare. No evidence of differences among regimens.	High
	GLE/PIB	0							ND		
	GZR/EBR	2	21	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 29)		
	PrO±D	2	33	No limitations	Consistent	Direct	Imprecise	Very Low	0% (0, 20)		
	SOF	4	91	Serious limitations[8]	Consistent	Direct	Imprecise	Very Low	2.9% (0.5, 7.2)		
	SOF/DCV	5	166	Some limitations[9]	Consistent	Direct	Imprecise	Very Low	0% (0, 4.6)		
	SOF/LDV	5	170	Serious limitations[10]	Consistent	Direct	None	Low	2.6% (0.7, 5.7)		
	SOF/SIM	0							ND		
	SOF/VEL	1	10	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 48)		

Outcome	Regimen [11]	# of Studies[12]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings			
								Certainty of Evidence for Outcome	Description of Findings		Importance of Outcome
Discontinue due to AE	DCV/ASV	0							ND	Rare. No evidence of differences among regimens.	High
	GLE/PIB	0							ND		
	GZR/EBR	2	21	No limitations	N/A	Direct	Sparse	Very Low	4.8% (0.7, 27)		
	PrO±D	2	33	No limitations	Consistent	Direct	Imprecise	Very Low	0% (0, 20)		
	SOF	4	91	Serious limitations[13]	Consistent	Direct	Imprecise	Very Low	2.9% (0.5, 7.2)		
	SOF/DCV	5	186	Some limitations[14]	Consistent	Direct	Imprecise	Very Low	0% (0, 4.2)		
	SOF/LDV	7	224	Some limitations[15]	Consistent	Direct	None	Moderate	1.7% (0.4, 3.7)		
	SOF/SIM	0							ND		
	SOF/VEL	1	10	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 48)		
Acute rejection	DCV/ASV	0							ND	Reported acute rejection rates, treated with SOF and DCV or LDV may be about 3% to 6%. No evidence of differences among regimens.	Moderate
	GLE/PIB	1	20	No limitations	N/A	Direct	Sparse	Very Low	0% (0, 30)		
	GZR/EBR	0							ND		
	PrO±D	0							ND		
	SOF	1	10	Serious limitations[16]	N/A	Direct	Sparse	Very Low	0% (0, 48)		
	SOF/DCV	3	246	Some limitations[17]	Consistent	Direct	None	Moderate	3.4% (1.5, 6.1)		
	SOF/LDV	4	109	Some limitations[18]	Consistent	Direct	None	Moderate	6.2% (2.3, 12)		
	SOF/SIM	0							ND		
	SOF/VEL	0							ND		
Balance of Potential Benefits and Harms: DAAs yield near 100% rates of SVR12 with rarely reported discontinuation due to adverse events or serious adverse events attributable to DAAs. Long-term death rates may be lower with DAA treatment than without.								**Certainty of Overall Evidence:** High (for DAAs in general)			

Abbreviations: ASV: asunaprevir, DAA: direct-acting antiviral, DCV: daclatasvir, EBR: elbasvir, GLE: glecaprevir, Gt: genotype, GZR: grazoprevir, LDV: ledipasvir, N/A: not applicable, ND: no data, PIB: pibrentasvir, PrO±D: paritaprevir/ritonavir/ombitasvir±dasabuvir, SIM: simeprevir, SOF: sofosbuvir, SVR12: sustained virologic response at 12 weeks post-treatment, Unadj OR: unadjusted odds ratio, VEL: velpatasvir.

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[1] Notation of inclusion of ribavirin omitted from this table.
[2] Single groups, mostly retrospective.
[3] No adjustment for potential confounders.

[4] Reporting bias (many studies did not report outcome).
[5] Reporting bias (study with half available patients did not report outcome).
[6] Reporting bias (many studies did not report outcome).
[7] No adjustment for potential confounders.
[8] Reporting bias (study with half available patients did not report outcome).
[9] Reporting bias (larger study did not report outcome).
[10] Reporting bias (many studies did not report outcome).
[11] Notation of inclusion of ribavirin omitted from this table.
[12] Single groups, mostly retrospective.
[13] Reporting bias (study with half available patients did not report outcome).
[14] Reporting bias (larger study did not report outcome).
[15] Reporting bias (larger study did not report outcome).
[16] Reporting bias (many studies did not report outcome).
[17] Reporting bias (larger study did not report outcome).
[18] Reporting bias (larger study did not report outcome).

Table S8. Summary Table: DAAs in CKD G4-G5ND (non-dialysis) patients, part 1 (SVR12 and adverse events)

Regimen (SOF Dose, mg/d)	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12 Analysis Type	DAA-Related Serious AE[1]	D/C DAA Due to AE[2]
DCV/ASV	Mawatari 2017 28078469 Japan (1)	24 wk	NR	NR	10/10 (100%)[3] ITT	NR	NR
DCV/ASV Summary					**100% (52, 100) all ITT**		
GLE/PIB	Atsukawa 2019A 30873651 Japan (2)	8 wk (65%)[4]	Gt 1b 20% Gt 2 46% Gt 3 19%[5]	NR	32/32 (100%) ITT	0/32 (0%)	2/32 (6.3%)
GLE/PIB	Liu C-H 2020 31981264 Taiwan (3)	8 wk (most)[6]	Gt 1a 3% Gt 1b 44% Gt 2 47% Gt 3 3% Gt 6 3%[7]	31%[8] (multiple[9])	35/35 (100%) ITT	0/35 (0%)	0/35 (0%)
GLE/PIB	Nozaki 2020 32128704 Japan (4)	8 wk (most)[10]	NR	NR	63/65 (96.9%) ITT	NR	NR
GLE/PIB Summary					**98.5% (94.1, 99.6)[11] 130/132 ITT**	**0% (0, 11)[12] 0/67**	**3.0% (0.7, 11)[13] 2/67**

[1] Pooled.

[2] Pooled, unless otherwise noted.

[3] Not fully clear that all were non-dialysis.

[4] 12 weeks if cirrhosis, genotype 3, or prior DAA treatment failure.

[5] Mixed genotypes 1/141; 20/141 not tested.

[6] 12 weeks if cirrhosis or 16 weeks if genotype 3 and prior treatment (numbers unclear).

[7] Excludes n=3 with CKD 5ND.

[8] Excludes n=3 with CKD 5ND.

[9] TE, biopsy, clinical, imaging, FIB-4.

[10] 12 weeks if cirrhosis, genotype 3, or prior DAA treatment failure (numbers not reported)

Regimen (SOF Dose, mg/d)	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12 Analysis Type	DAA-Related Serious AE[1]	D/C DAA Due to AE[2]
GZR/EBR	Atsukawa 2019B 30144366 Japan (5)	12 wk	Gt 1b 100%	NR	14/14 (100%) ITT	0/14 (0%)	0/14 (0%)
GZR/EBR	Bruchfeld 2017 28576451 US (6)	12 wk	Gt 1a 52%[14] Gt 1b 48%	6% (TE)	50/51 (98.0%)[15] Per protocol[16]	NR	NR
GZR/EBR	Choi 2020 31862503 US (7)	12 wk	Gt 1a 63% Gt 1b 33%	42% (ICD 9/10 code)	714/740 (96.5%) ITT	NR	NR
GZR/EBR	Cheng P-N 2021 32499107 Taiwan (8)	12 wk	Gt 1 100%	NR	31/31 (100%) mITT[17]	NR	NR
GZR/EBR	Jang 2021 32539296 S Korea (9)	12 wk	Gt 1 100%	NR	21/21 (100%) ITT	NR	NR
GZR/EBR Summary					**96.7% (95.4, 97.8)[18] 830/857 mostly ITT**	**0% (0, 39) 0/14**	**0% (0, 39) 0/14**

[11] 3 studies. Pooled.

[12] 2 studies. Pooled.

[13] 2 studies. Pooled.

[14] Includes dialysis patients.

[15] Combined immediate and deferred treatment groups from randomized controlled trial. One patient in delayed treatment cohort had SVR4 on placebo; the patient's CKD category was not reported.

[16] Excluding patients who died, were lost to follow-up, withdrew, or were noncompliant (reported as modified intention-to-treat).

[17] Patients who received at least 4 weeks of treatment.

[18] 5 studies. I^2 0% (no heterogeneity)

Regimen (SOF Dose, mg/d)	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12 Analysis Type	DAA-Related Serious AE[1]	D/C DAA Due to AE[2]
PrO±RBV	Elmowafy 2021 33111161 Egypt (10)	12 wk	NR	0% (TE)	SVR24 39/50 (78.0%) ITT	NR	NR
PrOD	Iliescu 2020 31948406 Romania (11)	12 wk	Gt 1b 100%	NR	18/18 (100%) ITT	0/18 (0%)	0/18 (0%)
PrO/RBV	Mekky 2019 30166253 Egypt (12)	12 wk	Gt 4 100%	23% (TE)	32/35 (91.4%) ITT	0/35 (0%)	0/35 (0%)
PrO±D Summary					**89.4% (75.7, 97.8)[19] 89/103 all ITT**	**0% (0, 13) 0/53**	**0% (0, 13) 0/53**
SOF/RBV (200)	Lawitz 2020A 32531259 US, New Zealand (13)	24 wk	Gt 1a 70% Gt 1b 20% Gt 3 3a 10%	0% (not defined)	4/10 (40%) ITT	0/10 (0%)	2/10 (20%)
SOF/RBV (400)		24 wk	Gt 1a 60% Gt 1b 20% Gt 3a 20%	40% (not defined)	6/10 (60%) ITT	0/10 (0%)	2/10 (20%)
SOF/RBV (400)	Manoj 2018 29676846 India (14)	24 wk	Gt 1a 65%[20] Gt 1b 4% Gt 3a 27% Gt 3b 4%	15%[21] (clinical definition)	21/21 (100%) ITT	0/21 (0%)	0/21 (0%)
SOF Summary					**71.7% (29.1, 98.6)[22] 31/41 all ITT**	**0% (0, 17) 0/41**	**11% (1.2, 28)[23] 4/41**

[19] 3 studies. REML. I^2 70% (large heterogeneity). Combination of SVR12 and SVR24.

[20] Includes 5 on HD.

[21] Includes 5 on HD.

[22] 2 studies (3 cohorts). REML. I^2 87% (large heterogeneity).

[23] REML. I^2 51% (moderate heterogeneity).

Regimen (SOF Dose, mg/d)	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12 Analysis Type	DAA-Related Serious AE[1]	D/C DAA Due to AE[2]
SOF/DCV (200)	Taneja 2018 29484572 India (15)	12 wk (most)[24]	Gt 1 65% Gt 2 1% Gt 3 34%	32% (TE), decompensated 9%	11/11 (100%) ITT	0/11 (0%)	0/11 (0%)
SOF/DCV (200)	Goel 2021 33097949 India (16)	8 wk	Gt 1 22% Gt 3 37% Gt 4 7%[25]	NR	25/27 (92.6%) ITT	0/27 (0%)	NR
SOF/DCV (400)	Manoj 2018 29676846 India (14)	12 wk	Gt 3 100%	37%[26] (clinical definition)	14/14 (100%) ITT	0/14 (0%)	0/14 (0%)
SOF/DCV ±RBV (200)	Eletreby 2020 31858694 Egypt (17)	NR	NR	18.5% (Child B, implied)[27]	506/519 (97.5%) ITT	NR	NR
SOF/DCV Summary					**97.1% (95.7, 98.3)[28] 556/571 all ITT**	**0% (0, 14) 0/52**	**0% (0, 25) 0/25**
SOF/LDV (400)	Manoj 2018 29676846 India (14)	12 wk	Gt 1 100%	23%[29] (clinical definition)	25/25 (100%) ITT	0/25 (0%)	0/25 (0%)
SOF/LDV (400)	Lawitz 2020A 32531259 US, New Zealand (13)	12 wk	Gt 1a 78% Gt 1b 22%	11% (not defined)	18/18 (100%) ITT	0/18 (0%)	0/18 (0%)
SOF/LDV Summary					**100% (84, 100)[30] 43/43 all ITT**	**0% (0, 16)[31] 0/43**	**0% (0, 16)[32] 0/43**

[24] 24 weeks if genotype 3 and cirrhosis (number not reported).

[25] Not tested 33%.

[26] Includes 5 on HD.

[27] Across all DAAs.

[28] 4 studies. REML. I^2 0% (no heterogeneity).

[29] Includes 1 on HD.

[30] 2 studies. Pooled.

Regimen (SOF Dose, mg/d)	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12 Analysis Type	DAA-Related Serious AE[1]	D/C DAA Due to AE[2]
SOF/SIM (200)	Eletreby 2020 31858694 Egypt (17)	NR	NR	18.5% (Child B, implied)[33]	38/41 (92.7%) ITT	NR	NR
SOF/SIM Summary					**92.7% (79.6, 97.6) 38/41 all ITT**		
SOF/VEL (400)	Cheng P-N 2022 34967920 Taiwan (18)	12 wk	NR	NR	25/25 (100%) ITT	NR	NR
SOF/VEL (400)	Liu C-H 2022 33408122 Taiwan (19)	12 wk	Gt 1a 3%[34] Gt 1b 55% Gt 2 33% Gt 6 4%	14% (TE), 5% (Child B,C)[35]	70/74 (94.6%) ITT	0/74 (0%)	0/74 (0%)
SOF/VEL Summary					**96.0% (89.7, 98.5)[36] 95/99 all ITT**	**0% (0, 9.9)**	**0% (0, 9.9)**

Abbreviations: AE: adverse events, ASV: asunaprevir, CKD: chronic kidney disease (category), D/C: discontinued, DAA: direct-acting antivirals, DCV: daclatasvir, EBR: elbasvir, GLE: glecaprevir, Gt: genotype, GZR: grazoprevir, ICD: International Classification of Diseases, ITT: intention-to-treat (or all fully analyzed), LDV: ledipasvir, mITT: modified ITT, NR: not reported, PIB: pibrentasvir, PMID: PubMed identifier, PrO/D: paritaprevir/ritonavir/ombitasvir/dasabuvir, RBV: ribavirin, SIM: simeprevir, SOF: sofosbuvir, SVR12: sustained virologic response at 12 weeks post-treatment, TE: transient elastography, VEL: velpatasvir, wk: weeks.

[31] 2 studies. Pooled.

[32] 2 studies. Pooled.

[33] Across all DAAs.

[34] Across both dialysis and non-dialysis populations (n=191)

[35] Across both dialysis and non-dialysis populations (n=191)

[36] 2 studies. Pooled.

Table S9. Summary Table: DAAs in CKD G4-G5 (non-dialysis) patients, part 2 (kidney outcomes)

Regimen (SOF Dose, mg/qD)	Study, PMID Country (Reference)	Duration	Genotype	Liver Cirrhosis (Definition)	CKD Category	eGFR Chg (95% CI), P value [Baseline]
GLE/PIB	Lawitz 2020B[37] 31821716 Multiple[38] (20)	8 wk (83%)[39]	Gt 1 55%[40] Gt 2 27% Gt 3 15% Gt 4 4%	14%[41] (TE)	At EOT: CKD 3b (improve) 2/17 (12%) CKD 4 (no change) 13/17 (76%) CKD 5ND (worsen) 2/17 (12%) CKD 5D (worsen) 0%	NR
PrO+RBV (75/50/12.5[42])	Said 2019[43] 30791838 Egypt (21)	12 wk	Gt 4 100%	NR	NR	At SVR4: Chg median 3, P=0.48 [22]
PrO±D±RBV	Muñoz-Gómez 2017[44] 27976490 Spain (22)	12 wk (most)[45]	NR	NR	At SVR12: Incident dialysis 0/12	At EOT: Chg median −1, NS [21]

[37] Study did not report SVR12 or adverse event data specifically for CKD 4-5ND population.
[38] Canada, Germany, Greece, Italy, Poland, South Korea, Spain, Sweden, US.
[39] 8 weeks (65%) if noncirrhotic except both prior treatment and genotype 3, 12 weeks (13%) if cirrhotic except genotype 3, 16 weeks (4%) if genotype 3 except treatment-naive/noncirrhotic. Percentages include n=7 with CKD 3b.
[40] Including CKD 3b (n=7).
[41] Including CKD 3b (n=7).
[42] PrO dosages.
[43] Study did not report SVR12 or adverse event data specifically for CKD 4-5ND population.
[44] Study did not report SVR12 or adverse event data specifically for CKD 4-5ND population.
[45] 24 weeks if genotype 1a w/compensated cirrhosis (number not reported).

Regimen (SOF Dose, mg/qD)	Study, PMID Country (Reference)	Duration	Genotype	Liver Cirrhosis (Definition)	CKD Category	eGFR Chg (95% CI), P value [Baseline]
SOF/RBV (200)	Lawitz 2020A 32531259 US, New Zealand (13)	24 wk	Gt 1a 70% Gt 1b 20% Gt 3a 10%	0% (not defined)	NR	At SVR12: −3.0 (−7.1, 1.1), P=0.15[46] [21.8]
SOF/RBV (400)		24 wk	Gt 1a 60% Gt 1b 20% Gt 3a 20%	40% (not defined)	NR	At SVR12: −3.0 (−8.9, 2.9), P=0.32[47] [26.2]
SOF/DCV (200)	Taneja 2018 29484572 India (15)	12 wk (most)[48]	Gt 1 65% Gt 2 1% Gt 3 34%	32% (TE), decompensated 9%	NR	At EOT: −0.4 (−4.3, 3.4), P=0.82 [24.8]
SOF/LDV (400)	Lawitz 2020A 32531259 US, New Zealand (13)	12 wk	Gt 1a 78% Gt 1b 22%	11% (not defined)	NR	At SVR12: −1.0 (−3.5, 1.5), P=0.43[49] [24.9]
SOF/VEL (400)	Liu C-H 2022 33408122 Taiwan (19)	12 wk	Gt 1a 3%[50] Gt 1b 55% Gt 2 33% Gt 6 4%	14% (TE), 5% (Child B,C)[51]	NR	CKD G4: 1.6 (−0.1, 3.3), P=0.06 CKD G5ND: −0.3 (−1.6, 1.0), P=0.65

46 Estimated.

47 Estimated

48 24 weeks if genotype 3 and cirrhosis (number not reported).

49 Estimated.

50 Across both dialysis and non-dialysis populations (n=191)

51 Across both dialysis and non-dialysis populations (n=191)

Regimen (SOF Dose, mg/qD)	Study, PMID Country (Reference)	Duration	Genotype	Liver Cirrhosis (Definition)	CKD Category	eGFR Chg (95% CI), P value [Baseline]
DAAs, various[52]	D'Ambrosio 2020 31813755 Italy (23)	12 wk (52%)[53]	Gt 1 61% Gt 2 17% Gt 3 9% Gt 4 13%	43% (TE)	At SVR12: CKD category improvement: 12/22 (55%) CKD category no change: 7/22 (32%) CKD category worsening: 3/22 (13%) CKD 1: 0% CKD 2: 1/22 (5%) CKD 3a: 2/22 (9%) CKD 3b: 9/22 (41%) CKD 4: 7/22 (32%) CKD 5: 3/22 (13%)	At SVR12: Chg median 6, P=0.016 [24]

Abbreviations: Chg: change, CI: confidence interval, CKD: chronic kidney disease (category), DAA: direct-acting antiviral, D/C: discontinued, DAA: direct-acting antivirals, DCV: daclatasvir, eGFR: estimated glomerular filtration rate, EOT: end of treatment, GLE: glecaprevir, Gt: genotype, LDV: ledipasvir, NR: not reported, NS: nonsignificant, PIB: pibrentasvir, PMID: PubMed identifier, PrO/D: paritaprevir/ritonavir/ombitasvir/dasabuvir, qD: per day, RBV: ribavirin, SOF: sofosbuvir, SVR4/12: sustained virologic response at 4/12 weeks post-treatment, TE: transient elastography, VEL: velpatasvir, wk: weeks.

[52] SOF/RBV ± PegIFN (n=7, 30%), SOF/SIM ± RBV (n=2, 9%), SOF/DCV ± RBV (n=4, 17%), SOF/LDV ± RBV (n=5, 22%), ProD ± RBV (n=4, 17%), PrO/RBV (n=1, 5%).

[53] Implied that the remainder had <12 weeks treatment

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Table S10. Summary Table: DAAs in CKD G5D (dialysis[1]) population, part 1 (SVR12 and adverse events)

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
DCV/ASV	Fujii 2018 29480939 Japan (1)	24 wk	Gt 1 100%	NR	64/67 (95.5%) Per protocol	NR	1/67 (1.5%)
DCV/ASV	Kawakami 2016 27346670 Japan (2)	12 wk	Gt 1 100%	17% (NR)	18/18 (100%) ITT	0/18 (0%)	0/18 (0%)
DCV/ASV	Lee 2019[7] 30400729 S Korea (3)	24 wk	Gt 1b 100%	19% (NR)	16/21 (76.1%) ITT	0/21 (0%)	3/21 (14%)
DCV/ASV	Miyazaki 2016 27098678 Japan (4)	24 wk	Gt 1b 100%	NR	10/10 (100%) ITT	0/10 (0%)	0/10 (0%)
DCV/ASV	Otsuka 2017 30483552 Japan (5)	24 wk	Gt 1b 100%	22% or 26% (serum markers)[8]	21/23 (91.3%) ITT	0/23 (0%)	0/23 (0%)
DCV/ASV	Suda 2016 26768604 Japan (6)	24 wk	Gt 1a 5% Gt 1b 91% Unknown 5%	19% (TE or imaging/serum markers)	20/21 (95.2%) missing data[9]	1/21 (4.5%)	1/21 (4.5%)

[1] 100% hemodialysis populations, unless otherwise noted (for peritoneal dialysis).

[2] Sofosbuvir doses were 400 mg daily unless otherwise noted. All other regimens used standard doses.

[3] REML meta-analysis of double arcsine transformed proportion, unless otherwise noted.

[4] Pooled.

[5] Unreliable estimates. Rarely explicitly reported, possibly suggesting no transfusions in most studies. Mostly unclearly reported whether need for transfusion related to ribavirin treatment.

[6] REML meta-analysis of double arcsine transformed proportion, unless otherwise noted.

[7] n=4 on peritoneal dialysis.

[8] 5 or 6/23, unclear if 1 patient with elevated APRI also had abnormal FIB-4.

[9] Missing data on 13 patients.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
DCV/ASV	Suda 2018 28560477 Japan (7)	24 wk	Gt 1a 1% Gt 1b 99%	32% (TE)	118/123 (95.9%) ITT	0/123 (0%)	4/123 (3.3%)
DCV/ASV	Toyoda 2016 26872889 Japan (8)	24 wk	Gt 1b 100%	39% (NR)	28/28 (100%) ITT	0/28 (0%)	1/28 (3.6%)
DCV/ASV	Uojima 2017 Ren Replace Ther Japan (9)	24 wk	Gt 1b 100%	30% (NR)	25/30 (83.3%) ITT	0/30 (0%)	3/30 (10%)
DCV/ASV Summary					**93.6% (89.5, 96.8)[10] 320/341 mostly ITT**	**0.4% (0.1, 2.5) 1/274[11]**	**3.8% (2.0, 6.0) 13/341[12]**
GLE/PIB	Atsukawa 2019 30873651 Japan (10)	8 wk (65%)[13]	Gt 1b 20%[14] Gt 2 46% Gt 3 19% Mixed 1% Unknown 14%	29% (serum markers ± imaging)[15]	99/100 (100%) ITT	0/100 (0%)	1/100 (1%)
GLE/PIB	Lawitz 2020 31821716 Multiple[16] (11)	8 wk (~83%)[17]	Gt 1 55%[18] Gt 2 27% Gt 3 15% Gt 4 4%	14% (TE)[19]	NR	0/77 (0%)	NR

[10] 9 studies. I^2 40% (moderate heterogeneity).

[11] 8 studies.

[12] 9 studies. I^2 0% (no heterogeneity).

[13] 12 weeks (n=35) if cirrhosis, genotype 3, or prior DAA treatment failure.

[14] Including n=9 CKD 5ND.

[15] Including n=9 CKD 5ND.

[16] Canada, Germany, Greece, Italy, Poland, South Korea, Spain, Sweden, US.

[17] 12 weeks (~13%) if cirrhotic except genotype 3, 16 weeks (~4%) if genotype 3 except treatment-naive/noncirrhotic. Percentages include n=7 with CKD 3b.

[18] Including CKD 4-5ND.

[19] Including CKD 4-5ND.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
GLE/PIB	Liu C-H 2020 31981264 Taiwan (12)	8 wk (most)[20]	Gt 1a 5% Gt 1b 32% Gt 2 58% Gt 6 4% % Gt 1b & 2 1%[21]	33%[22] (multiple[23])	72/73 (98.6%) ITT	0/73 (0%)	0/73 (0%)
GLE/PIB	Miyasaka-2022-35024454 Japan (13)	8 (33%) or 12 wk (67%)[24]	Gt 1 62% Gt 2 38%)[25]	67% (NR)	18/18 (100%) ITT	NR	NR
GLE/PIB	Morishita 2020 31883211 Japan (14)	8 wk (62%)[26]	Gt 1b 33% Gt 2 67%	14% (imaging, serum markers)	24/24 (100%) Per protocol	0/24 (0%)	0/24 (0%)
GLE/PIB	Nozaki 2020 32128704 Japan (15)	8 wk (most)[27]	NR	NR	125/132 (94.7%) ITT	NR	NR
GLE/PIB	Ogawa 2019 30849206 Japan (16)	8 or 12 wk[28]	Gt 1 or 2 100%	NR	23/23 (100%) ITT	NR	0/23 (0%)
GLE/PIB	Persico 2019 Gastro Hep Italy (17)	8 (58%) or 12 wk (42%)[29]	Gt 1a 11% Gt 1b 31% Gt 2 36% Gt 3 19% Gt 4 3%	28% (TE)	35/36 (97.2%) ITT	1/36 (2.8%)	0/36 (0%)

[20] 12 weeks if cirrhosis or 16 weeks if genotype 3 and prior treatment (numbers unclear).

[21] Includes n=3 with CKD 5ND.

[22] Includes n=3 with CKD 5ND.

[23] TE, biopsy, clinical, imaging, FIB-4.

[24] 8 weeks if noncirrhotic, treatment naïve. 12 weeks if cirrhosis (or prior treatment failure)

[25] For all (n=230) with any stage of CKD.

[26] 12 weeks if cirrhosis or prior DAA treatment failure (n=9)

[27] 12 weeks if cirrhosis, genotype 3, or prior DAA treatment failure (numbers not reported)

[28] No further details.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
GLE/PIB	Stein 2020[30] 32956186 Germany (18)	8 wk (85%)[31]	Gt 1a 15%[32] Gt 1b 27% Gt 1 undefined 3% Gt 2 9% Gt 3 36% Gt 4 6%	12% (TE, imaging, or clinical)[33]	21/23 (91.3%) Unclear[34]	0/25 (0%)	0/25 (0%)
GLE/PIB	Suda 2019 30778716 Japan (19)	8 (52%) or 12 wk (48%)[35]	Gt 2 100%	30% (serum markers)	26/27 (96.3%) ITT	1/27 (3.7%)	2/27 (7.4%)
GLE/PIB	Yen 2020 32790715 Taiwan (20)	8 wk (68%)[36]	Gt 1 55% Gt 2 39% Gt 3 2% Misc 5%	32% (biopsy, imaging, endoscopy)	42/44 (95.5%) ITT	0/44 (0%)	1/44 (2.2%)
GLE/PIB	Yu M-L 2021 33303567 Taiwan (21)	NR	NR	NR	27/29 (93.1%) ITT	0/29 (0%)	NR
GLE/PIB Summary					**96.9% (95.1, 98.3)[37] 512/529 mostly ITT**	**0.5% (0.1, 1.8) 2/435[38]**	**1.6% (0.6, 3.1) 4/352[39]**

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29 Reasons for different durations not reported.

30 Includes patients on peritoneal dialysis (number not reported).

31 12 weeks (n=2), 16 weeks (n=1), reasons not reported.

32 Include CKD 4-ND (n=8).

33 Include CKD 4-ND (n=8).

34 3/70 omitted from intention-to-treat analysis, but not acknowledged or accounted for.

35 12 week if cirrhosis or failed to respond to previous DAA treatment

36 12 weeks (n=14) if cirrhosis.

37 11 studies. I^2 10% (small heterogeneity).

38 Pooled. 9 studies.

39 8 studies. I^2 0% (no heterogeneity).

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
GZR/EBR	Atsukawa 2017 28457003 Japan (22)	12 wk	Gt 1b 100%	32% (imaging)	30/31 (96.8%) ITT	1/31 (3.2%)	1/31 (3.2%)
GZR/EBR	Bayu 2020 APASL 360 Indonesia (23)	12 wk	NR	NR	37/38 (97%) ITT	NR	NR
GZR/EBR	Bruchfeld 2017 28576451 US (24)	12 wk	NR	NR	162/164 (98.8%)[40] Per protocol[41]	NR	NR
GZR/EBR	Choi 2020 31862503 US (25)	12 wk	Gt 1a 63%[42] Gt 1b 33%	42% (ICD-9/10 code)[43]	541/563 (96.1%) ITT	NR	NR
GZR/EBR	Ji Q 2022 34599755 China (26)	12 wk	Gt 1b 100%	0% (NR)	11/11 (100%) ITT	0/11 (0%)	0/11 (0%)
GZR/EBR	Li C 2020 33235113 China (27)	12 wk	NR	NR	15/15 Per protocol[44]	0/16 (0%)	1/16 (6.3%)
GZR/EBR	Liu C-H 2020 32513953 Taiwan (28)	12 wk	Gt 1b 100%	10% (TE)	38/40 (95%) ITT	0/40 (0%)	1/40 (2.5%)
GZR/EBR	Ogawa 2018 30300717 Japan (29)	12 wk	Gt 1 100%	NR	19/21 (90.5%) ITT	NR	NR

[40] Combined immediate and deferred treatment groups from randomized controlled trial. One patient in delayed treatment cohort had SVR4 on placebo; the patient's CKD category was not reported.

[41] Excluding patients who died, were lost to follow-up, withdrew, or were noncompliant (reported as modified intention-to-treat).

[42] Including CKD 4-5ND population.

[43] Including CKD 4-5ND population.

[44] 1 patient who discontinued due to adverse event did not have SVR12 evaluated.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion[5]*	D/C DAA Due to AE[6]
GZR/EBR±RBV	Stein 2020[45] 32956186 Germany (18)	12 wk (98%)[46]	Gt 1a 23%[47] Gt 1b 72% Gt 4 5%	14% (TE, imaging, or clinical) [48]	39/42 (92.9%) Unclear[49]	0/45 (0%)	0/45 (0%)
GZR/EBR	Suda 2019 30019127 Japan (30)	12 wk	Gt 1b 100%	31% (TE)	22/23 (95.7%) ITT	0/23 (0%)	0/23 (0%)
GZR/EBR	Suryamin 2020 APASL 1182 Indonesia (31)	12 wk	NR	NR	14/14 (100%) ITT	NR	NR
GZR/EBR Summary					**96.5% (94.9, 97.8)[50] 928/962 mostly ITT**	**0.6% (0.1, 4.2) 1/163[51]**	**2.5% (0.7, 5.4) 3/166[52]**
PrOD±RBV	Abad 2017 28166520 Spain (32)	12 wk (most)[53]	Gt 1a 29% Gt 1b 71%	21% (NR)[54]	31/31 (100%) ITT	0/31 (0%) *Transfusion 0/31 (0%), no RBV*	0/31 (0%)
PrO±RBV	Elmowafy 2021 33111161 Egypt (33)	12 wk	Gt 1a 8% Gt 4 92%	2% (imaging), 0 (TE)	40/40[55] (100%) ITT	NR	0/40 (0%)

45 Includes patients on peritoneal dialysis (number not reported).

46 16 weeks (n=2), reason not reported

47 Include CKD 4-5ND (n=15).

48 Include CKD 4-5ND (n=15).

49 3/70 omitted from intention-to-treat analysis, but not acknowledged or accounted for.

50 11 studies. I^2 12% (small heterogeneity).

51 Pooled. 6 studies.

52 6 studies. I^2 0% (no heterogeneity).

53 24 wk if genotype 1a or cirrhosis (numbers not reported).

54 Including n=4 on other treatment (genotype 4).

55 SVR24 = 38/40 (95%)

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
PrOD±RBV[56]	Etik 2019 30719954 Turkey (34)	12 wk	Gt 1a 28% Gt 1b 72%	NR	17/18 (94.4%) ITT	0/18 (0%)	0/18 (0%)
PrOD	Iliescu 2020 31948406 Romania (35)	12 wk	Gt 1b 100%	17% (TE)	48/48 (100%) ITT	0/48 (0%)	0/48 (0%)
PrOD	Liu C-H 2019 30931537 Taiwan (36)	12 wk	Gt 1b 100%	0% (biopsy or TE)	46/46 (100%) ITT	0/46 (0%)	0/46 (0%)
PrO±D±RBV	Londoño 2019[57] 31550267 Spain (37)	12 or 24 wk[58]	Gt 1a 16% Gt 1b 76% Gt 1 unknown 1% Gt 4 7%	30% (TE)	93/100 (93%) ITT	NR	NR
PrO/RBV	El-Raey 2019 SP138 Egypt (38)	12 wk	NR	NR	SVR36 36/36 (100%) Per protocol	NR Transfusion 6/53 (11%), on RBV	NR
PrO/RBV	Mekky 2019 30166253 Egypt (39)	12 wk	Gt 4 100%	11% (TE)	72/75 (96%) ITT	0/75 (0%)	0/75 (0%)
PrO	Morisawa 2017 28621007 Japan (40)	12 wk	Gt 1b 100%	NR	8/10 (80%) ITT	0/10 (0%)	2/10 (20%)
PrOD±RBV	Pockros 2016 26976799 US (41)	12 wk	Gt 1 65%[59] Gt 1b 35%	0% (biopsy, TE, serum markers)	12/14 (85.7%)[60] ITT	0/14 (0%) *Transfusion 0/14 (0%), ~10 on RBV*	0/14 (0%)

56 N=1 on SOF/LDV.

57 Reported as n=12 (12%) on peritoneal dialysis and 92% were on hemodialysis. Discrepancy not explained.

58 Per "SmPC" indications (not defined). Numbers not reported.

59 Includes 6 with CKD 4.

60 1 SVR12 failure had relatively low adherence, 1 died unrelated to HCV. Unclear if all truly on dialysis.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
PrOD	Ponziani 2017 28258770 Italy (42)	12 wk	Gt 1a 10% Gt 1b 80% Gt 4 10%	70% (TE)[61]	10/10 (100%) ITT	1/10 (10%)	0/10 (0%)
PrO±D±RBV	Sanai 2018 29288514 Saudi Arabia, Canada (43)	12 wk (96%)[62]	Gt 1a 37% Gt 1b 13% Gt 1 unknown 3% Gt 4 48%	21% (TE)	65/67 (97.0%)[63] ITT	0/67 (0%) *Transfusion* 3/54 (5.6%), on RBV[64]	2/67 (2.9%)
PrOD±RBV	Sperl 2018 29669332 Czech Republic (44)	12 wk	Gt 1a 9%[65] Gt 1b 87% Unknown 4%	26% (NR)[66]	19/19 (100%) ITT	0/19 (0%) *Transfusion 0/19 (0%), ~6 on RBV*	0/19 (0%)
PrOD±RBV	Tatar 2019 31994628 Turkey (45)	12 wk	Gt 1a 45% Gt 1b 39% Gt 1 undefined 15%	NR	33/33 (100%) Per protocol[67]	0/33 (0%)	0/33 (0%)
PrOD±RBV	Torun 2019 30762283 Turkey (46)	12 wk	Gt 1a 30% Gt 1b 70%	0% (NR)	10/10 (100%) ITT	0/10 (0%)	0/10 (0%)
PrOD±RBV	Yaraş 2019 30666967 Turkey (47)	12 wk	Gt 1a 12% Gt 1b 64% Gt 1 undefined 24%	0% (NR)	25/25 (100%) ITT	0/25 (0%)	0/25 (0%)
PrO±D Summary					**96.8% (95.2, 98.1)[68] 565/582 mostly ITT**	**0.2% (0.03, 1.7) 1/406[69]**	**1.8% (0.8, 3.3) 4/446[70]**

[61] 30% with F3.

[62] 24 weeks (4%), reasons not reported.

[63] Includes n=2 with CKD 4-5ND

[64] 0/13 without RBV.

[65] Includes n=4 with CKD 4.

[66] Includes n=4 with CKD 4.

[67] 3 patients excluded due to death, missed visits, and abandoning treatment.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
SOF(400 QOD)[71]/RBV	Agarwal 2017 29270489 India (48)	12 wk	Gt 1 65%[72] Gt 2 2% Gt 3 29% Gt 4 3% Gt 6 2%	0% (endoscopy)	39/41 (95.1%) ITT	0/41 (0%) *Transfusion* 3/41 (7.3%) *on RBV*	0/41 (0%)
SOF/RBV	Akhil 2018 28339162 India (49)	12 wk	Gt 1 64% Gt 3 27% Gt 4 9%	NR	16/22 (72.7%)[73] ITT	0/22 (0%) *Transfusion* 0/22 (0%), *on RBV*	0/22 (0%)
SOF/RBV/IFN	Mandhwani 2020 32308935 Pakistan (50)	12 wk	Gt 1 50%[74] Gt 2 1% Gt 3 43% Gt 4 1% Mixed Gt 5%	0% (clinical, imaging)	59/60 (98.3%) ITT	NR *Transfusion* 20/60 (33%) *on RBV*	0/60 (0%)
SOF/RBV Summary					**91.9% (74.5, 99.8)[75] 114/123 all ITT**	**0% (0, 11.5) 0/63[76]**	**0% (0, 6.2) 0/123[77]**
SOF (QOD[78])/DCV	Agarwal 2017 29270489 India (48)	12 wk	Gt 1 65%[79] Gt 2 2% Gt 3 29% Gt 4 3% Gt 6 2%	0% (endoscopy)	20/21 (95.2%) ITT	0/21 (0%)	0/21 (0%) *Transfusion 0/21 (0%),* *no RBV*

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[68] 16 studies. I^2 4% (small heterogeneity).

[69] 13 studies.

[70] 14 studies. I^2 0% (no heterogeneity).

[71] n=39/41.

[72] Across interventions.

[73] Includes 2 lost to follow-up pre-SVR12

[74] Across interventions.

[75] 3 studies. I^2 85% (large heterogeneity).

[76] 2 studies.

[77] Pooled. 2 studies.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
SOF/DCV	Butt 2019 31720170 Pakistan (51)	12 wk	Gt 1 32% Gt 3 68%	0% (clinical, history, and serum markers)	SVR40 27/31 (87.1%) ITT	NR	NR
SOF (400 qD)/DCV	Cheema 2019 31779583 Pakistan (52)	12 wk (most)[80]	Gt 1b 100%	22% (Various[81])	15/18 (83.3%)[82] ITT	NR	0/18 (0%)
SOF (400 TIW)/DCV					17/18 (94.4%)[83] ITT	NR	1/18 (5.6%)
SOF/DCV/RBV	Mandhwani 2020 32308935 Pakistan (50)	12 wk	Gt 1 50%[84] Gt 2 1% Gt 3 43% Gt 4 1% Mixed Gt 5%	0% (clinical, imaging)	70/73 (95.9%) ITT	NR *Transfusion* 11/73 (15%), on RBV	0/73 (0%)
SOF/DCV	Mehta 2018 29430587 India (53)	12 wk	Gt 1a 54% Gt 1b 46%	NR	12/13 (92.3%) ITT	NR	0/13 (0%)
SOF (400 QOD)/DCV					9/13 (69.2%) ITT	NR	0/13 (0%)
SOF (200 qD)/DCV	Mostafi 2020 Ukr JND Bangladesh (54)	12 wk	NR	~23% (NR)	26/26 (100%) ITT	0/26 (0%)	0/26 (0%)
SOF/DCV	Singh 2018 30113115 India (55)	12 wk (most)[85]	Gt 3 100%	26%[86] (TE)[87]	~11/~11 (100%)[88] ITT estimated	0/~11 (0%)[89]	0/~11 (0%)[90]

[78] 400 mg every other day in n=6/21

[79] Across interventions.

[80] 24 weeks if cirrhosis (numbers not reported).

[81] Based on Child-Pugh score and abdominal imaging, TE, and esophagogastroduodenoscopy.

[82] SVR24 15/18 (83.3%)

[83] SVR24 14/18 (77.%)

[84] Across interventions.

[85] 24 weeks if cirrhosis (number not reported).

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
SOF (200 qD)/DCV	Taneja 2018 29484572 India (56)	12 wk (most)[91]	Gt 1 65% Gt 2 1% Gt 3 34%	32% (TE)[92]	54/54 (100%) ITT	0/54 (0%)	0/54 (0%)
SOF/DCV Summary					**93.7% (88.9, 97.2)[93] 261/278 all ITT**	**0% (0, 6.8) 0/112[94]**	**0.4% (0.1, 2.8) 1/247[95]**

[86] Includes n=8 CKD 4-5ND.

[87] 15% decompensated.

[88] N's estimated due to incomplete reporting for hemodialysis population specifically; 2 patients across both treatments were reinfected with a new genotype prior to SVR12; thus, viremic. However, counted as achieving SVR12 since original genotype was cleared.

[89] N's estimated due to incomplete reporting for hemodialysis population specifically; 2 patients across both treatments were reinfected with a new genotype prior to SVR12; thus, viremic. However, counted as achieving SVR12 since original genotype was cleared.

[90] N's estimated due to incomplete reporting for hemodialysis population specifically; 2 patients across both treatments were reinfected with a new genotype prior to SVR12; thus, viremic. However, counted as achieving SVR12 since original genotype was cleared.

[91] 24 weeks if genotype 3 and cirrhosis (number not reported).

[92] 9% decompensated.

[93] 8 studies (10 cohorts). I^2 47% (moderate heterogeneity).

[94] Pooled. 4 studies.

[95] Pooled. 7 studies (9 cohorts).

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion[5]*	D/C DAA Due to AE[6]
SOF (NR)/LDV	Asanbek kyzy 2019 SP641 Kyrgyzstan (57)	12 wk	NR	NR	SVR24: 11/11 (100%) ITT	0/11 (0%)	0/11 (0%)
SOF/LDV	Chuang 2021[96] 34465694 Multiple[97] (58)	8 wk	Gt 1 100% Gt 1a 16%	0% (NR)	42/45 (93.3%) ITT	0/45 (0%)	0/45 (0%)
		12 wk	Gt 1 26% Gt 1a 3% Gt 2 61% Gt 5 3% Gt 6 6% Unknown 3%	0% (NR)	31/31 (100%) ITT	0/31 (0%)	0/31 (0%)
		24 wk	Gt 1 79% Gt 1a 21% Gt 2 11% Gt 4 11%	100% (NR)	16/19 (84.2%) ITT	0/19 (0%)	0/19 (0%)
SOF/LDV	Debnath 2020 32294734 India (59)	12 wk	Gt 1 92%[98] Gt 5 8%	0% (TE)	13/13 (100%) Per protocol[99]	0/13 (0%) *Transfusion* 0/13 (0%), *No RBV*	0/13 (0%)
SOF/LDV	Gohel 2020 32507712 India (60)	12 wk	Gt 1 93%[100] Gt 3 7%	0% (NR)	40/40 (100%) ITT	0/40 (0%)	0/40 (0%)
SOF/LDV	Mehta 2018 29430587 India (53)	12 wk	Gt 1a 17% Gt 1b 83%	NR	12/12 (100%) ITT	NR	0/12 (0%)
SOF/LDV	Singh 2018 30113115 India (55)	12 wk (most)[101]	Gt 1 94% Gt 4 6%	26%[102] (TE)[103]	~28/~28 (100%)[104] ITT estimated	0/~28 (0%)[105]	0/~28 (0%)[106]

96 8/95 on peritoneal dialysis (n=1 8 weeks, n=3 12 weeks, n=4 24 weeks).

97 Belgium, Germany, Italy, Taiwan, United States

98 Including n=5 on other regimens.

99 Excludes n=2 patients who died within 4 weeks of treatment initiation, but not reported which treatment they were on.

100 Includes n=3 on different regimen

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
SOF (QOD)/LDV	Surendra 2018 28972699 India (61)	12 wk	Gt 1a 63% Gt 1b 37%	0% (TE)	19/21 (90.5%)[107] ITT	0/21 (0%) *Transfusion 0/21 (0%), no RBV*	0/21 (0%)
SOF/LDV Summary					**95.9% (92.8, 98.1)[108] 212/220 mostly ITT**	**0% (0, 3.7) 0/208[109]**	**0% (0, 3.5) 0/220[110]**
SOF (200 qD)/SIM	Bhamidimarri 2015[111] 26095179 US (62)	12 wk[112]	Gt 1a 67% Gt 1b 33%	60% (biopsy)[113]	10/12 (83.3%) ITT	0/12 (0%)	0/12 (0%)
SOF/SIM Summary					**83.3% (52.3, 95.8) all ITT**	**0% (0, 43)**	**0% (0, 43)**

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101 24 weeks if cirrhosis (number not reported).

102 Includes n=8 CKD 4-5ND.

103 15% decompensated.

104 N's estimated due to incomplete reporting for hemodialysis population specifically; 2 patients across both treatments were reinfected with a new genotype prior to SVR12; thus, viremic. However, counted as achieving SVR12 since original genotype was cleared.

105 N's estimated due to incomplete reporting for hemodialysis population specifically; 2 patients across both treatments were reinfected with a new genotype prior to SVR12; thus, viremic. However, counted as achieving SVR12 since original genotype was cleared.

106 N's estimated due to incomplete reporting for hemodialysis population specifically; 2 patients across both treatments were reinfected with a new genotype prior to SVR12; thus, viremic. However, counted as achieving SVR12 since original genotype was cleared.

107 2 died pre-SVR12 due to inadequate hemodialysis

108 7 studies (1 with 3 arms). I^2 5% (small heterogeneity).

109 6 studies (1 with 3 arms).

110 Pooled. 7 studies (1 with 3 arms).

111 N=1 on peritoneal dialysis.

112 24 wk if cirrhosis (n=1)

113 Including n=3 CKD 5ND.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] ***Transfusion***[5]	D/C DAA Due to AE[6]
SOF/VEL	Borgia 2019[114] 31195062 Multiple[115] (63)	12 wk	Gt 1a 25% Gt 1b 19% Gt 1 undefined1% Gt 2 12% Gt 3 32% Gt 4 7% Gt 6 3%	29% (TE, biopsy, serum markers)	56/59 (94.9%) ITT	0/59 (0%)	0/59 (0%)
SOF/VEL	Cheng P-N 2022 34967920 Taiwan (64)	12 wk	NR	NR	122/126 (96.8%) ITT	NR	NR
SOF/VEL	Gaur 2020 32405174 India (65)	NR	Gt 1 68% Gt 3 32%	10% (TE)	30/31 (96.8%) ITT	0/31 (0%) *Transfusion* 2/31 (6.5%) *no RBV*	0/31 (0%)
SOF/VEL	Hussain 2021 APASL 2021 Pakistan (66)	12 wk	NR	NR	25/28 (89.3%) ITT	NR	NR
SOF/VEL	Liu C-H 2022[116] 33408122 Taiwan (67)	12 wk	Gt 1a 3%[117] Gt 1b 55% Gt 2 33% Gt 6 4%	14% (TE), 5% (Child B,C)[118]	111/117 (94.9%) ITT	0/117 (0%)	0/117 (0%)
SOF/VEL	Mostafi 2020 Ukr JND Bangladesh (54)	12 wk	NR	~23% (NR)	44/44 (100%) ITT	0/44 (0%)	0/44 (0%)
SOF/VEL	Taneja 2020 33025685 India (68)	12 wk	Gt 1 79% Gt 3 15% Gt 4 5%	20% (TE)[119]	49/51 (96.1%) ITT	0/51 (0%)	0/51 (0%)

[114] N=5 (8%) on peritoneal dialysis.
[115] Canada, UK, Spain, Israel, New Zealand, Australia.
[116] 3/117 on peritoneal dialysis.
[117] Across both D and ND populations (n=191)
[118] Across both D and ND populations (n=191)
[119] n=1 decompensated.

Regimen[2]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[3]	DAA-Related Serious AE[4] *Transfusion*[5]	D/C DAA Due to AE[6]
SOF (NR)/VEL	Yu M-L 2021 33303567 Taiwan (21)	12 wk	Gt 1 44% Gt 2 50%% Gt 6 1%	35% (TE and clinical)	95/106 (89.6%) ITT	0/106 (0%)	5/105 (4.8%)[120]
SOF/VEL Summary					**95.4% (93.0, 97.3)** [121] **598/629 all ITT**	**0% (0, 1.9) 0/408**[122]	**0% (0, 3.2) 5/407**[123]

Abbreviations. AE: adverse events, ASV: asunaprevir, CKD 4-5ND: chronic kidney disease category 4 or 5-nondialysis, D/C: discontinued, DAA: direct acting antivirals, DCV: daclatasvir, EBR: elbasvir, GLE: glecaprevir, Gt: genotype, GZR: grazoprevir, ICD: International Classification of Diseases, IFN: interferon, LDV: ledipasvir, NR: not reported, PIB: pibrentasvir, PMID: PubMed identifier, PrOD: combination paritaprevir/ritonavir/ombitasvir/dasabuvir, qD: every day, QOD: every other day, RBV: ribavirin, REML: random effects maximum likelihood (meta-analysis), SIM: simeprevir, SOF: sofosbuvir, SVR12/24/36/52: sustained virologic response at 12/24/36/52 weeks after end of treatment, TE: transient elastography, TIW: three times a week, VEL: velpatasvir, wk: weeks.

[120] One patient was categorized and reported in a separate group, without clear reporting about adverse events.

[121] 8 studies. I^2 35% (small to moderate heterogeneity).

[122] Pooled. 6 studies.

[123] 6 studies. I^2 29% (small to moderate heterogeneity).

Table S11. Summary Table: DAAs in CKD G5D (dialysis[124]) population, part 2 (Death)[125]

Regimen [126]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	Follow-Up	Death[127] % (95% CI)
GLE/PIB	Liu C-H 2020 31981264 Taiwan (12)	8 wk (most)[128]	Gt 1a 5% Gt 1b 32% Gt 2 58% Gt 6 4% % Gt 1b & 2 1%[129]	33%[130] (multiple[131])	40 wk	0/73 0% (0, 10.1)
GLE/PIB	Persico 2019 Gastro Hep Italy (17)	8 (58%) or 12 wk (42%)[132]	Gt 1a 11% Gt 1b 31% Gt 2 36% Gt 3 19% Gt 4 3%	28% (TE)	1 yr	1/36[133] 2.8% (0.4, 17)
PrOD	Liu C-H 2019 30931537 Taiwan (36)	12 wk	Gt 1b 100%	0% (biopsy or TE)	SVR24	0/46 0% (0, 15)
PrOD±RBV	Pockros 2016 26976799 US (41)	12 wk	Gt 1 65%[134] Gt 1b 35%	0% (biopsy, TE, serum markers)	SVR24	1/14[135] 7.1% (1.0, 37)

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[124] 100% hemodialysis populations, unless otherwise noted (for peritoneal dialysis).

[125] The studies not included in this table either did not follow patients beyond SVR12 or did not explicitly report deaths. Only explicitly reported deaths (or survival) included. Therefore, estimates are likely to be inflated (if all studies were included).

[126] Sofosbuvir doses all were 400 mg daily. All other regimens used standard doses.

[127] Restricted to follow-up of about ≥6 month (omitting end of treatment and SVR12 timepoint data)

[128] 12 weeks if cirrhosis or 16 weeks if genotype 3 and prior treatment (numbers unclear).

[129] Includes n=3 with CKD 5ND.

[130] Includes n=3 with CKD 5ND.

[131] TE, biopsy, clinical, imaging, FIB-4.

[132] Reasons for different durations not reported.

[133] Car accident.

[134] Includes 6 with CKD 4.

[135] At SVR2 due to congestive heart failure and lower gastrointestinal bleeding. Not deemed related to treatment.

Regimen [126]	**Study, PMID Country (Reference)**	**Treatment Duration**	**Genotype**	**Liver Cirrhosis (Definition)**	**Follow-Up**	**Death**[127] **% (95% CI)**
SOF/DCV	Butt 2019 31720170 Pakistan (51)	12 wk	Gt 1 32% Gt 3 68%	0% (clinical, history, and serum markers)	1 yr	2/31[136] 6.4% (1.6, 22)
SOF/VEL	Borgia 2019[137] 31195062 Multiple[138] (63)	12 wk	Gt 1a 25% Gt 1b 19% Gt 1 undefined1% Gt 2 12% Gt 3 32% Gt 4 7% Gt 6 3%	29% (TE, biopsy, serum markers)	≥SVR16	2/59[139] 3.4% (0.8, 13)

Abbreviations. CI: confidence interval, CKD 4: chronic kidney disease category 4, DAA: direct acting antivirals, DCV: daclatasvir, ESRD: end-stage renal disease, GLE: glecaprevir, Gt: genotype, GZR: grazoprevir, HCV = hepatitis C virus, PIB: pibrentasvir, PMID: PubMed identifier, PrOD: combination paritaprevir/ritonavir/ombitasvir/dasabuvir, qD: every day, RBV: ribavirin, SOF: sofosbuvir, SVR2/12/24: (time of) sustained virologic response at 2/12/24 weeks after end of treatment, TE: transient elastography, wk: weeks, yr: year.

[136] Both due to "natural progression of ESRD."

[137] N=5 (8%) on peritoneal dialysis.

[138] Canada, UK, Spain, Israel, New Zealand, Australia.

[139] Neither related to HCV, DAA, or CKD (anxiety/suicide, lung cancer).

Table S12. Summary Table: DAAs in CKD G5D (dialysis) population, part 3 (Quality of Life)

Regimen [140]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	Follow-Up	Quality of Life
GLE/PIB	Lawitz 2020 31821716 Multiple[141] (11)	8 wk (~83%)[142]	Gt 1 55%[143] Gt 2 27% Gt 3 15% Gt 4 4%	14% (TE)[144]	1 yr	"Clinical meaningful improvement": SF-12 PCS: 27/77 (35.1%) SF-12 MCS: 29/77 (37.8%)
SOF (200) / DCV	Singhal 2019[145] AASLD 1618 India (69)	12 wk	NR	NR	EOT	SF-12 PCS: Mean Δ 3.4 (1.1, 5.7) SF-12 MCS: Mean Δ 4.0 (1.6, 6.4)

Abbreviations. CKD 4-5ND: chronic kidney disease category 4 or 5-nondialysis, DAA: direct acting antivirals, EOT: end of treatment, GLE: glecaprevir, Gt: genotype, MCS: Mental Component Summary, PCS: Physical Component Summary, PIB: pibrentasvir, PMID: PubMed identifier, SF-12: Short Form 12 item questionnaire, E: transient elastography, wk: weeks, yr: year.

140 Regimens used standard doses.

141 Canada, Germany, Greece, Italy, Poland, South Korea, Spain, Sweden, US.

142 12 weeks (~13%) if cirrhotic except genotype 3, 16 weeks (~4%) if genotype 3 except treatment-naive/noncirrhotic. Percentages include n=7 with CKD 3b.

143 Including CKD 4-5ND.

144 Including CKD 4-5ND.

145 Probably substudy of Taneja 2018 PMID 29484572.

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Table S13. Summary Table: DAAs in kidney transplant recipients, part 1 (SVR12 and adverse events)

Regimen[1]	**Study, PMID Country (Reference)**	**Treatment Duration**	**Genotype**	**Liver Cirrhosis (Definition)**	**SVR12**[2]	**DAA-Related Serious AE**[3] ***Transfusion***[4]	**D/C DAA Due to AE**[5]
GLE/PIB	Reau 2018 29672891 US (1)	12 wk	Gt 1a 30% Gt 1b 55% Gt 3 10% Gt 4 5%	0% (TE)	20/20 (100%) ITT	NR	NR
GLE/PIB Summary					**100% (70, 100) all ITT**		
GZR/EBR	Eisenberger 2019 30656217 Germany (2)	12 wk (most)[6]	Gt 1a 18% Gt 1b 73% Gt 4 9%	NR	11/11 (100%) ITT	0/11 (0%)	0/11 (0%)
GZR/EBR	Lai 2022 34902265 Taiwan (3)	12 wk	Gt 1b 100%	40% (F2)	10/10 (100%) ITT	0/10 (0%)	1/10 (10%)
GZR/EBR Summary					**100% (71, 100)[7]) 21/21 all ITT**	**0% (0, 29)[8] 0/21**	**4.8% (0.7, 27)[9] 1/21**
PrO±D±RBV[10]	Daniş 2019 31418413 Turkey (4)	12 wk (most)[11]	Gt 1a 48% Gt 1b 48%	10% (not defined)	21/21 (100%) ITT	0/21 (0%) *Transfusion 3/21 (14.3%)*[12]	0/21 (0%)

1 Sofosbuvir doses are noted. All other regimens used standard doses.
2 Pooled, unless otherwise noted.
3 Pooled.
4 Unreliable estimates. Rarely explicitly reported, possibly suggesting no transfusions in most studies. Mostly unclearly reported whether need for transfusion related to ribavirin treatment.
5 Pooled, unless otherwise noted.
6 16 weeks (n=1 patient with high HCV mRNA at baseline).
7 Pooled. 2 studies.
8 Pooled. 2 studies.
9 Pooled. 2 studies.
10 Dasabuvir for genotype 1 (n=20), ribavirin for genotypes 1a and 4 (n=10).
11 24 weeks (n=1 patient with cirrhosis).
12 Not reported whether on RBV or not.

Regimen[1]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[2]	DAA-Related Serious AE[3] *Transfusion*[4]	D/C DAA Due to AE[5]
PrOD	Iliescu 2020 31948406 Romania (5)	12 wk	Gt 1b 100%	0% (TE)	12/12 (100%) ITT	0/12 (0%)	0/12 (0%)
PrO±D Summary					**100% (80, 100)**[13] **33/33** **all ITT**	**0% (0, 20)** **0/33** ***Transfusion 14% (5, 36)***	**0% (0, 20)** **0/33**
SOF/RBV (400)	Hanif 2017 28260436 Pakistan (6)	24 wk	Gt 1 57% Gt 3 30% Gt 4 3% Mixed 11%	11% (Decompensated)	37/37 (100%) ITT	0/37	0/37
SOF/RBV (400)	Kirushnan 2016 Indian J Transpl India (7)	12 wk	Gt 1 65% Gt 2 5% Gt 3 30% Gt 4 5%	0% (Clinical)	10/13 (76.9%) Available data[14]	NR	NR
SOF/RBV (400)[15]	Taneja 2018 28703905 India (8)	12 wk	Gt 1 79% Gt 3 14% Gt 4 7%	14% (TE, LSM)	12/14 (85.7%)[16] ITT	2/14 (14.3%) *Transfusion 0/12 (0%)*	2/14 (14.3%)
SOF/RBV (400)	Prasad 2018 29962673 India (9)	12 wk	Gt 1 27%[17] Gt 2 4.5% Gt 3 64% Gt 4 4.5%	45% (TE)[18]	14/14 (100%) ITT	NR	NR
SOF/RBV (400)	Reddy 2018 29942507 India (10)	12 wk	Gt 1 70% Gt 2 10% Gt 3 10% Gt 4 10%	NR	10/10 (100%) ITT	0/10 (0%)	0/10 (0%)

[13] Pooled. 2 studies.
[14] Implied that all 20 completed treatment, but SVR12 data available for only 13.
[15] In n=1, SOF dose 200 mg daily.
[16] 1 patient included in both SOF/RBV and (after rapid virologic response failure) SOF/LDV arms
[17] Including n=8 on other regimens.
[18] Including n=8 on other regimens.

Regimen[1]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[2]	DAA-Related Serious AE[3] *Transfusion*[4]	D/C DAA Due to AE[5]
SOF/RBV (400)	Sharma 2018 29796311 India (11)	24 wk	Gt 1 33%[19] Gt 3 67%	10% (TE)[20]	29/29 (100%) ITT	0/30 (0%) *Transfusion or EPO* *0/30 (0%)*	0/30 (0%)
SOF Summary					**94.8% (88.2, 98.8)**[21] **112/117** **mostly ITT**	**2.9% (0.5, 7.2)**[22] **2/91** ***Transfusion 0% (0, 17)***	**2.9% (0.5, 7.2)**[23] **2/91**
SOF/DCV (400)	D'Ambrosio 2019 30904073 Italy (12)	12 wk (86%)[24]	Gt 2 71% Gt 3 29%	14% (not defined)	14/14 (100%) ITT	0/14 (0%)	0/14 (0%)
SOF/DCV (400)	Duerr 2019 30717681 Germany (13)	12 wk	Gt 1a 6% Gt 1b 94%	0% (biopsy)	15/16 (93.8%) ITT	0/16 (0%)	0/16 (0%)
SOF/DCV±RBV (400)	El Maghrabi 2019 31531807 Egypt (14)	24 wk	Gt 1 13%[25] Gt 4 87%	3% (U/S) [26], 0% (TE)	105/105 (100%) ITT	0/105 (0%)	NR
SOF/DCV±RBV (400)	Elmowafy 2020 33111161 Egypt (15)	24 wk	Gt 1 10% Gt 4 90%	2% (U/S), 0% (TE)	125/125 (100%) ITT	NR	0/125 (0%)
SOF/DCV (400)	Huang 2019 30369001 China (16)	12 wk (89%)[27]	Gt 1 84% Others 16%	16% (FIB4 >3.25)	19/19 (100%) ITT	0/19 (0%)	0/19 (0%)

[19] Including n=15 on other regimens.
[20] Including n=15 on other regimens.
[21] 6 studies. REML MA, I^2 41% (moderate heterogeneity)
[22] 4 studies. REML MA, I^2 2% (no heterogeneity)
[23] 4 studies. REML MA, I^2 2% (no heterogeneity)
[24] 24 weeks (n=2), reason not reported.
[25] Including n=9 who received other regimens.
[26] Including n=9 who received other regimens.
[27] 24 weeks for genotypes 1a or 4, cirrhosis, or prior treatment (n=2).

Regimen[1]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[2]	DAA-Related Serious AE[3] *Transfusion*[4]	D/C DAA Due to AE[5]
SOF/DCV±RBV (400)[28]	Taneja 2018 28703905 India (8)	12 wk (most)[29]	Gt 1 25% Gt 2 8% Gt 3 67% Gt 4 0%	17% (TE, LSM)	12/12 (100%) ITT	0/12 (0%)	0/12 (0%)
SOF/DCV Summary					**99.7% (97.6, 100)[30] 290/291 all ITT**	**0% (0, 4.6)[31] 0/166**	**0% (0, 4.2)[32] 0/186**
SOF/LDV (400)	Colombo 2017 27842383[33] Italy, France, Austria, Germany (17)	12 wk	Gt 1a 12% Gt 1b 74% Gt 1, no subtype 4% Gt 4 11%	14% (multiple)[34]	57/57 (100%) ITT	1/57 (1.8%)	0/57 (0%)
		24 wk	Gt 1a 18% Gt 1b 75% Gt 4 7%	16% (multiple)[35]	57/57 (100%) ITT	2/57 (3.5%)	1/57 (1.8%)
SOF/LDV (400)	Eisenberger 2017 27495770 Germany (18)	12 wk (80%)[36]	Gt 1a 27% Gt 1b 67% Gt 4 7%	NR	15/15 (100%) ITT	NR	NR
SOF/LDV (400)	Weigert 2018 29661427 Portugal (19)	12 wk (most)[37]	Gt 1a 22%)[38] Gt 1b 56% Gt 3 13% Gt 4 9%	13% (TE)[39]	14/14 (100%) ITT	NR	NR

[28] In n=3, SOF dose 200 mg daily.
[29] 24 weeks if cirrhosis or detectable RNA at week 4 (n not reported).
[30] Pooled. 6 studies.
[31] Pooled. 5 studies.
[32] Pooled. 5 studies.
[33] RCT of 12 vs. 24 wk.
[34] TE, Ishak score >= 5 by biopsy, FibroScan >12.5 kPa, or FibroTest score >0.75 plus APRI >2.
[35] TE, Ishak score >= 5 by biopsy, FibroScan >12.5 kPa, or FibroTest score >0.75 plus APRI >2.
[36] 8 weeks (n=3), unclear reason.
[37] 24 weeks for F4 fibrosis (number not reported).

Regimen[1]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[2]	DAA-Related Serious AE[3] *Transfusion*[4]	D/C DAA Due to AE[5]
SOF/LDV +RBV (400)	Musialik 2019 30972854 Poland (20)	12 wk (74%)[40]	Gt 1b 90% Gt 4 10%	5% (TE)[41]	31/31 (100%) Per protocol[42]	NR	0/31 (0%)
SOF/LDV +RBV (400)	Akin 2018 29899985 Turkey (21)	12 or 24 wk[43]	Gt 1a 25% Gt 1b 75%	17% (not defined)	12/12 (100%) ITT	0/12 (0%) Transfusion 0/12 (0%)	0/12 (0%)
SOF/LDV +RBV (400)	Bhamidimarri 2017 28332729 US (22)	12 wk (91%)[44]	Gt 1a 74% Gt 1b 26%	0% (TE)	22/23 (95.7%) ITT	NR	0/23 (0%)
SOF/LDV +RBV (400)	Chascsa 2018[45] 29758123 US (23)	12 wk (77%)[46]	Gt 1 82%[47] Gt 1a 54% Gt 1b 29% Gt 2 4% Gt 3 11%	36% (not defined)[48]	22/22 (100%) ITT	0/22 (0%)	0/22 (0%)
SOF/LDV +RBV (400)	Saxena 2017 28504842 US (24)	12 or 24 wk[49]	Gt 1 90%[50] Gt 1a 48% Gt 1b 37% Gt 2 2% Gt 3 3% Gt 4 3% Gt 6 2%	0% (TE)	42/47 (89.4%) mITT[51]	NR	NR

[38] Including n=9 who received other regimens.
[39] Including n=9 who received other regimens.
[40] 24 weeks for first 8 patients treated.
[41] Including n=9 who received other regimens.
[42] 1 dropout excluded from analysis.
[43] 24 weeks if cirrhosis, previous treatment failure and for specific genotypes (implied) (number not reported).
[44] 8 weeks (n=1), 24 weeks (n=1); reasons not reported.
[45] Study specific to D+/R+ patients.
[46] 24 weeks (n=5), reasons not reported.
[47] Including n=6 who received other regimens.
[48] Including n=6 who received other regimens.

Regimen[1]	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	SVR12[2]	DAA-Related Serious AE[3] *Transfusion*[4]	D/C DAA Due to AE[5]
SOF/LDV +RBV (400)	Taneja 2018 28703905 India (8)	12 wk (most)[52]	Gt 1 86% Gt 4 14%	5% (TE, LSM)	22/22 (100%)[53] ITT	0/22 (0%)	0/22 (0%)
SOF/LDV Summary					**97.3% (94.9, 99.0)**[54] **294/300** **almost all ITT**	**2.6% (0.7, 5.7)**[55] **3/170**	**1.7% (0.4, 3.7)**[56] **1/224**
SOF/VEL (400)	Greco 2019 SP786[57] Italy (25)	12 wk	Gt 1 60% Gt 2 40%	0% (not defined)	10/10 (100%) ITT	0/10 (0%)	0/10 (0%)
SOF/VEL Summary					**100% (52, 100)** **all ITT**	**0% (0, 48)**	**0% (0, 48)**

Abbreviations: AE: adverse event, APRI: aspartate aminotransferase to platelet ratio index, D/C = discontinuation, DAA: direct-acting antiviral, DCV: daclatasvir, EBR: elbasvir, EPO: erythropoietin, FIB4: fibrosis 4 (score), GLE: glecaprevir, Gt: genotype, GZR: grazoprevir, ITT: intention-to-treat analysis (or all participants analyzed), LDV: ledipasvir, LSM: liver stiffness measurement, mITT: modified ITT, NR: not reported, PIB: pibrentasvir, PMID: PubMed identifier, PrOD: paritaprevir/ritonavir/ombitasvir/dasabuvir, RBV: ribavirin, REML MA: restricted maximum likelihood meta-analysis, SOF: sofosbuvir, SVR12: sustained virologic response at 12 weeks post-treatment, TE: transient elastography, U/S: ultrasound, VEL: velpatasvir, wk: weeks.

[49] Numbers and reasons not reported.
[50] Including n=11 on other regimens.
[51] Omitted 1 who discontinued DAA for administrative reasons.
[52] 24 weeks if cirrhosis or detectable RNA at week 4 (n not reported).
[53] 1 patient included in both SOF/RBV and (after rapid virologic response failure) SOF/LDV arms
[54] 10 study arms. REML MA, I^2 18% (small heterogeneity)
[55] 5 study arms. REML MA, I^2 0% (no heterogeneity)
[56] 7 study arms. REML MA, I^2 0% (no heterogeneity)
[57] European Renal Association – European Dialysis and Transplant Association (ERA-EDTA) 2019 conference abstract.

Table S14. Summary Table: DAAs in kidney transplant recipients, part 2 (allograft outcomes and death)

Regimen	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	Timing	Acute Rejection	Allograft Loss	Patient Death (≥12 Months)
GLE/PIB	Reau 2018 29672891 US (1)	12 wk	Gt 1a 30% Gt 1b 55% Gt 3 10% Gt 4 5%	0% (TE)	SVR12	0/20 (0%)	0/20 (0%)	NR
GLE/PIB Summary						**0% (0, 30)**	**0% (0, 30)**	
GZR/EBR	Eisenberger 2019 30656217 Germany (2)	12 wk (91%)[58]	Gt 1a 18% Gt 1b 73% Gt 4 9%	NR	SVR12	NR	1/11 (9.1%)[59]	NR
GZR/EBR Summary							**9.1% (1.3, 44)**	
SOF/RBV	Reddy 2018 29942507 India (10)	12 wk	Gt 1 70% Gt 2 10% Gt 3 10% Gt 4 10%	NR	1 yr	0/10 (0%)	0/10 (0%)	NR
SOF Summary						**0% (0, 48)**	**0% (0, 48)**	
SOF/DCV (400)	Duerr 2019 30717681 Germany (13)	12 wk	Gt 1a 6% Gt 1b 94%	0% (biopsy)	24 wk	0/16 (0%)	0/16 (0%)	NR
SOF/DCV±RBV (400)	El Maghrabi 2019 31531807 Egypt (14)	24 wk	Gt 1 13%[60] Gt 4 87%	3% (U/S) [61], 0% (TE)	EOT	4/105 (3.8%)	NR	NR

[58] 16 weeks (n=1 patient with high HCV mRNA at baseline).
[59] Not associated with DAA, but with progressive diabetic nephropathy.
[60] Including n=9 who received other regimens.
[61] Including n=9 who received other regimens.

Regimen	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	Timing	Acute Rejection	Allograft Loss	Patient Death (≥12 Months)
SOF/DCV±RBV (400)	Elmowafy 2020 33111161 Egypt (15)	24 wk	Gt 1 10% Gt 4 90%	2% (U/S), 0 (TE)	EOT	4/125 (3.2%)	0/125[62]	NR
SOF/DCV Summary						**3.4% (1.5, 6.1)[63] 8/246**	**0% (0, 5.4)[64] 0/141**	
SOF/LDV (400)	Colombo 2017 27842383[65] Italy, France, Austria, Germany (17)	12 wk	Gt 1 4% Gt 1a 12% Gt 1b 74% Gt 4 11%	14% (multiple)[66]	SVR12	NR	NR	NR
SOF/LDV (400)		24 wk	Gt 1a 18% Gt 1b 75% Gt 4 7%	16% (multiple)[67]	SVR12	NR	NR	NR
SOF/LDV (400)	Eisenberger 2017 27495770 Germany (18)	12 wk (80%)[68]	Gt 1a 27% Gt 1b 67% Gt 4 7%	NR	SVR12	0/15 (0%)	0/15 (0%)	NR
SOF/LDV±RBV (400)	Chascsa 2018 29758123 US (23)	12 wk (77%)[69]	Gt 1 82%[70] Gt 1a 54% Gt 1b 29% Gt 2 4% Gt 3 11%	36% (not defined)[71]	Median 413 d	1/22 (4.5%)[72]	0/22 (0%)	NR

[62] Incident dialysis.
[63] 3 studies. REML MA. I^2 0% (no heterogeneity)
[64] Pooled. 2 studies.
[65] RCT of 12 vs. 24 weeks.
[66] TE, Ishak score >= 5 by biopsy, FibroScan >12.5 kPa, or FibroTest score >0.75 plus APRI >2.
[67] TE, Ishak score >= 5 by biopsy, FibroScan >12.5 kPa, or FibroTest score >0.75 plus APRI >2.
[68] 8 weeks (n=3), unclear reason.
[69] 24 weeks (n=5), reasons not reported.
[70] Including n=6 who received other regimens.

Regimen	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	Timing	Acute Rejection	Allograft Loss	Patient Death (≥12 Months)
SOF/LDV±RBV (400)	Saxena 2017 28504842 US (24)	12 or 24 wk[73]	Gt 1 90%[74] Gt 1a 48% Gt 1b 37% Gt 2 2% Gt 3 3% Gt 4 3% Gt 6 2%	0% (TE)	EOT	2/47 (4.3%)	1/47 (2.1%)	NR
SOF/LDV±RBV[75] (400)	Bhamidimarri 2017 28332729 US (22)	12 wk (91%)[76]	Gt 1a 72% Gt 1b 24% Gt 2b 4%	0% (TE)	EOT	4/25 (16%)[77]	NR	NR
SOF/LDV Summary						**6.2% (2.3, 12.0)[78] 7/109**	**1.2% (0.2, 8.0)[79] 1/84**	

[71] Including n=6 who received other regimens.
[72] During first week of treatment.
[73] Numbers and reasons not reported.
[74] Including n=11 on other regimens.
[75] Also SOF/DCV (n=1), SOF/SIM (n=1).
[76] 8 weeks (n=1), 24 weeks (n=1); reasons not reported.
[77] All resolved.
[78] 4 studies. REML MA. I^2 12% (small heterogeneity).
[79] 3 studies. Pooled.

Regimen	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	Timing	Acute Rejection	Allograft Loss	Patient Death (≥12 Months)
DAAs, various[80]	Chevallier 2020 32636068 France (26)	12 wk (80%)[81]	Gt 1 68%[82] Gt 2 7% Gt 3 2% Gt 4 18% Gt 5 5% Unknown 0%	9% (TE)	Mean 951 d (SD 189)	NR	5/44 (11.4%)[83] OR 2.69 (0.30, 24.6)	0/44 (0%) OR~0.06 (<0.01, 1.27)
No DAA (Retrospective NRCS)			Gt 1 50% Gt 2 14% Gt 3 5% Gt 4 5% Gt 5 0% Unknown 27%	9% (TE)		NR	1/22 (4.5%)[84]	3/22 (13.6%)[85]
DAAs, various[86]	Gendia 2019 30580473 Italy (27)	NR	Gt 1a 15% Gt 1b 54% Gt 2 31% Gt 3 0% Gt 4 0%	NR	Mean 7.4 y (3.7)	NR	1/13 (7.7%) OR 0.38 (0.03, 4.81)	0/13 (0%) OR~0.06 (<0.01, 1.31)
No DAA (Retrospective NRCS)			Gt 1a 18% Gt 1b 55% Gt 2 9% Gt 3 9% Gt 4 9%	NR	Mean 4.9 y (4.1)	NR	2/11 (18.2%)	4/11 (36%)

[80] SOF/LDV±RBV (n=23), SOF/DCV (n=14), GZR/EBR (n=4), SOF/SIM (n=1) , SOF/RBV (n=1), PrO (n=1).
[81] 4 weeks (n=2), 8 weeks (n=1), 24 weeks (n=6), reasons not reported.
[82] P values vs. no DAA: genotype 1 P=0.07, genotype 2 P=0.18, genotype 3 P=0.30, genotype 4 P=0.06, genotype 5 P=0.15, Unknown P=0.0001.
[83] 4 of 5 had pre-DAA GFR <30; 5th GFR 50.
[84] Baseline GFR not reported.
[85] 1 hepatocellular carcinoma, 2 septic shock.
[86] SOF/RBV (n=6), SOF/LDV (n=4), SOF/DCV (n=2), SOF/VEL (n=1).

Regimen	Study, PMID Country (Reference)	Treatment Duration	Genotype	Liver Cirrhosis (Definition)	Timing	Acute Rejection	Allograft Loss	Patient Death (≥12 Months)
DAAs, various[87]	Goetsch 2020 33089036 US (28)	NR	NR	NR	>2 y	NR	2/29 (6.9%) OR 0.17 (0.03, 0.89)	2/29 (6.9%) OR 0.30 (0.05, 1.61)
							Allograft loss or death adjHR 0.32 (0.08, 1.21)	
No DAA (Retrospective NRCS)			NR	NR	>2 y	NR	9/30 (30%)	6/30 (20%)
DAA vs. no DAA Summary							**Unadj OR 0.50 (0.09, 2.73)**[88]	**Unadj OR 0.16 (0.04, 0.61)**[89]
DAAs, various[90]	Weigert 2018 29661427 Portugal (19)	12 (61%) or 24 (39%) wk[91]	Gt 1a 22% Gt 1b 56% Gt 3 13% Gt 4 9%	13% (TE)	1 yr	NR	NR	2/23 (9%)

Abbreviations: adjHR: adjusted hazard ratio, APRI: aspartate aminotransferase to platelet ratio index, DAA: direct-acting antiviral, DCV: daclatasvir, EBR: elbasvir, EOT: end of treatment, GFR: glomerular filtration rate, GLE: glecaprevir, Gt: genotype, GZR: grazoprevir, LDV: ledipasvir, mo: months, NR: not reported, NRCS: nonrandomized comparative study, OR: odds ratio (unadjusted), PIB: pibrentasvir, PMID: PubMed identifier, PrOD: paritaprevir/ritonavir/ombitasvir/dasabuvir, RBV: ribavirin, REML MA: restricted maximum likelihood meta-analysis, SIM: simeprevir, SOF: sofosbuvir, SVR12: sustained virologic response at 12 weeks post-treatment, TE: transient elastography, U/S: ultrasound, Unadj: unadjusted, VOX: voxipaprevir, wk: weeks.

[87] SOF/LDV+-RBV (n=22), SOF/SIM (n=2), SOF/RBV (n=1), SOF/VEL (n=1), SOF/VEL/VOX (n=1), GLE/PIB (n=1).
[88] 3 studies. REML MA. I^2 48% (moderate heterogeneity).
[89] 3 studies. REML MA. I^2 0% (no heterogeneity).
[90] SOF/LDV (n=15), SOF/RBV (n=4), SOF/DCV (n=2), GZR/ELB (n=2).
[91] 24 weeks for F4 fibrosis.

Table S15. Summary Table: DAAs in kidney transplant recipients, part 3 (eGFR and proteinuria)

Regimen	Study, PMID Country (Reference)	Duration	Genotype	Liver Cirrhosis (Definition)	Follow-up	eGFR Chg (95% CI), P value [Baseline]	Proteinuria Chg (95% CI), P value [Baseline]
GZR/EBR	Eisenberger 2019 30656217 Germany (2)	12 wk (most)[92]	Gt 1a 18% Gt 1b 73% Gt 4 9%	NR	SVR12	ΔMedian −2, NS [29]	ΔMedian ~−240 mg/g Cr, NS [~740]
PrOD	Iliescu 2020 31948406 Romania (5)	12 wk	Gt 1b 100%	0% (TE)	Mean 2.6 yr	NR	Decrease (NR), P=0.013 [NR]
SOF/RBV (400)	Hanif 2017 28260436 Pakistan (6)	24 wk	Gt 1 57% Gt 3 30% Gt 4 3% Mixed 11%	11% (Decompensated)	EOT (24 wk)	Mean Δ 17.0 (11.7, 22.5), P=0.04 [63.7]	NR
SOF/RBV (400)[93]	Taneja 2018 28703905 India (8)	12 wk (most)	Gt 1 79% Gt 3 14% Gt 4 7%	14% (TE, LSM)	EOT	ΔMedians −1.0, P=0.14 [72.8]	NR
SOF/DCV (400)	Duerr 2019 30717681 Germany (13)	12 wk	Gt 1a 6% Gt 1b 94%	0% (biopsy)	SVR12	NR	−165 mg/g Cr (−692, 363) P=0.75 [606]
SOF/DCV (400)	Huang 2019 30369001 China (16)	12 wk (89%)[94]	Gt 1 84% Others 16%	16% (FIB4 >3.25)	SVR12	NR P=0.07 [57.2]	−0.6 g/g Cr(−1.1, 0) P=0.048 [0.95] 8/12 >10% improvement 2/12 >10% worsening
SOF/DCV (400)	Michels 2020 32294735 Brazil[95] (29)	NR	NR	48% (TE)[96]	12 or 24 wk[97]	Mean Δ −1.4 (−6.0, 3.2), P=0.55 [51.4]	NR

[92] 16 weeks (n=1 patient with high HCV mRNA at baseline).
[93] In n=1, SOF dose 200 mg daily.
[94] 24 weeks for cirrhosis or prior treatment (n=2).
[95] This study did not report DAA or kidney outcomes specifically for the transplant population.
[96] Of whole population (n=241) including hemodialysis and other CKD.
[97] Unclear.

Regimen	Study, PMID Country (Reference)	Duration	Genotype	Liver Cirrhosis (Definition)	Follow-up	eGFR Chg (95% CI), P value [Baseline]	Proteinuria Chg (95% CI), P value [Baseline]
SOF/DCV (400)	D'Ambrosio 2019 30904073 Italy (12)	12 wk (86%)[98]	Gt 2 71% Gt 3 29%	14% (not defined)	48 wk	ΔMedians 3, P=0.57 [54]	NR
SOF/DCV±RBV (400)	Elmowafy 2020 33111161 Egypt (15)	24 wk	Gt 1 10% Gt 4 90%	2% (U/S), 0% (TE)	SVR12	Mean Δ −2.0 (NR), P NR [61.5]	NR
SOF/DCV±RBV (400)[99]	Taneja 2018 28703905 India (8)	12 wk (most)[100]	Gt 1 25% Gt 2 8% Gt 3 67% Gt 4 0%	17% (TE, LSM)	EOT	ΔMedians −2.0, P=0.51 [65.5]	NR
SOF/LDV (400)	Colombo 2017 27842383[101] Italy, France, Austria, Germany (17)	12 wk	Gt 1a 12% Gt 1b 74% Gt 1, no subtype 4% Gt 4 11%	14% (multiple)[102]	SVR12	ΔMedians 0, P NR [50]	NR
SOF/LDV (400)		24 wk	Gt 1a 18% Gt 1b 75% Gt 4 7%	16% (multiple)[103]	EOT	ΔMedians −5, P NR [60]	NR
SOF/LDV (400)	Eisenberger 2017 27495770 Germany (18)	12 wk (80%)[104]	Gt 1a 27% Gt 1b 67% Gt 4 7%	NR	EOT	ΔMedians −9, P NR [78]	NR
SOF/LDV±RBV (400)	Akin 2018 29899985 Turkey (21)	12 or 24 wk[105]	Gt 1a 25% Gt 1b 75%	17% (not defined)	SVR12	Mean Δ 4.1 (−2.1, 10.3), P=0.20 [68.6]	NR

[98] 24 weeks (n=2), reason not reported.
[99] In n=3, SOF dose 200 mg daily.
[100] 24 weeks if cirrhosis or detectable RNA at week 4 (n not reported).
[101] Randomized controlled trial of 12 vs. 24 weeks.
[102] TE, Ishak score >= 5 by biopsy, FibroScan >12.5 kPa, or FibroTest score >0.75 plus APRI >2.
[103] TE, Ishak score >= 5 by biopsy, FibroScan >12.5 kPa, or FibroTest score >0.75 plus APRI >2.
[104] 8 weeks (n=3), unclear reasons.
[105] Numbers not reported. Implied 24 weeks if cirrhosis, previous treatment failure and for specific genotypes.

Regimen	Study, PMID Country (Reference)	Duration	Genotype	Liver Cirrhosis (Definition)	Follow-up	eGFR Chg (95% CI), P value [Baseline]	Proteinuria Chg (95% CI), P value [Baseline]
SOF/LDV+RBV (400)	Taneja 2018 28703905 India (8)	12 wk (most)[106]	Gt 1 86% Gt 4 14%	5% (TE, LSM)	EOT	ΔMedians −12.2, P=0.30 [65.9]	NR
SOF±LDV+RBV (400)	Musialik 2019 30972854 Poland (20)	12 wk (most)[107]	Gt 1b 90% Gt 4 10%	5% (TE)[108]	EOT	Mean Δ −2 (−4.9, 0.9), P=0.17 [55]	NR
SOF/VEL (400)	Greco 2019 SP786[109] Italy (25)	12 wk	Gt 1 60% Gt 2 40%	0% (not defined)	EOT	Mean Δ 3 (NR), P NR [54][110]	Mean Δ −20 mg/24 hr (NR), P NR [430][111]
SOF-based regimens[112] (400)	Prasad 2018 29962673 India (9)	12 wk	Gt 1 27% Gt 2 4.5% Gt 3 64% Gt 4 4.5%	45% (TE)	EOT	Mean Δ −0.5 (−7.0, 6.0), NS [60.6]	NR
DAAs, various[113]	El Maghrabi 2019 31531807 Egypt (14)	24 wk	Gt 1 13% Gt 4 87%	3% (U/S), 0% (TE)	SVR12	Mean Δ −1.9 (−10.7, 6.9), P=0.67 [64.3]	NR
DAAs, various[114]	Chevallier 2020 32636068 France (26)	12 wk (most)[115]	Gt 1 68% Gt 2 7% Gt 3 2% Gt 4 18% Gt 5 5%	9% (TE)	EOT	Mean Δ 11 (−1, 23), P=0.075 [47]	NR

Abbreviations: APRI: aspartate aminotransferase to platelet ratio index, Chg: change, CI: confidence interval, DAA: direct-acting antiviral, DCV: daclatasvir, EBR: elbasvir, eGFR: estimated glomerular filtration rate, EOT: end of treatment, Gt: genotype, GZR: grazoprevir, ILDV: ledipasvir, LSM: liver stiffness measurement,

[106] 24 weeks if cirrhosis or detectable RNA at week 4 (n not reported).
[107] 24 weeks if genotype 2 or 3 and first several treated for genotype 1b or 4 (numbers not reported for this regimen).
[108] Including n=9 who received other regimens.
[109] European Renal Association – European Dialysis and Transplant Association (ERA-EDTA) 2019 conference abstract.
[110] Reported as P<0.01, but this is likely one-sided P for "not worsening."
[111] Reported as P<0.01, but this is likely one-sided P for "not worsening."
[112] SOF/RBV (n=14), SOF/DCV/RBV (n=5), SOF/LDV (n=3).
[113] SOF/DCV+-RBV (n=105), SOF/RBV (n=4), PrO/RBV (n=5).
[114] SOF/LDV±RBV (n=23), SOF/DCV (n=14), GZR/EBR (n=4), SOF/SIM (n=1) , SOF/RBV (n=1), PrO (n=1).
[115] 4 weeks (n=2), 8 weeks (n=1), 24 weeks (n=6), reasons not reported.

Mean Δ: mean change in value (final minus pre-treatment), NR: not reported, NS, nonsignificant, P: P value, PMID: PubMed identifier, PrOD: paritaprevir/ritonavir/ombitasvir/dasabuvir, RBV: ribavirin, SIM: simeprevir, SOF: sofosbuvir, SVR12: sustained virologic response at 12 weeks post-treatment, TE: transient elastography, U/S: ultrasound, VEL: velpatasvir, wk: weeks, yr: years, ΔMedian: difference between final and pre-treatment median values (not median change).

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Table S16: Summary Table
Isolation of HCV patients receiving hemodialysis
Categorical outcomes

PMID	Author	Year	Country	Type of article	Study design	Sample size	CKD stage	Blood transfusions	HD vintage	Outcome	Definition	Arm
19147995	**Agarwal SK., Dash SC., Gupta S., Pandey RM.**	2009	India	Peer-reviewed publication	Pre-post design	1285	CKD HD	6.6 (3.9)	nd	**HCV seroconversion**	**nd**	**Isolation for HCV+, 2003-2006**
												No isolation for HCV+, 1995-1998
7753458	**Blumberg A., Zehnder C., Burckhardt JJ.**	1995	Switzerland	Peer-reviewed publication	Pre-post design	131	CKD HD	nd	2 weeks-284 months	**HCV seroconversion**	**nd**	**Isolation for HCV+**
												No isolation for HCV+
16903625	**Karkar A., Abdelrahman M., Ghacha R., Malik TQ.**	2006	Saudi Arabia	Peer-reviewed publication	Pre-post design	265	CKD HD	nd	4-21 years	**HCV seroconversion**	**nd**	**Isolation for HCV+**
												No isolation for HCV+
										HCV (antibody) positivity	**nd**	**Isolation for HCV+**
												No isolation for HCV+
										HCV seroconversion	**nd**	
16685138	**Gallego E., López A., Pérez J., Llamas F., Lorenzo I., López E., Illescas ML., Andrés E., Olivas E., Gómez-Roldan C.**	2006	Spain	Peer-reviewed publication	Pre-post design	282	CKD HD	nd	nd			**Strict/complete isolation for HCV+**
												Isolation for HCV+
										HCV (antibody) positivity	**nd**	**Strict/complete isolation for HCV+**
												Isolation for HCV+
17382087	**Shebeb AM., Kotkat AM., Abd El Reheim SM., Farghaly AG., Fetohy EM.**	2006	Egypt	Peer-reviewed publication	nRCS, unclear	101	CKD HD	nd	2.3 (1.4)	**HCV seroconversion**	**nd**	Intervention program + no isolation
												No intervention program + isolation
												No intervention program + no isolation
												Intervention program + no isolation
												No intervention program + isolation
												No intervention program + no isolation
												Intervention program + no isolation
												No intervention program + isolation
												No intervention program + no isolation
15469615	**Shamshirsaz AA., Kamgar M., Bekheirnia MR., Ayazi F., Hashemi SR., Bouzari N., Habibzadeh MR., Pourzahedgilani N., Broumand V., Shamshirsaz AH., Moradi M., Borghei M., Haghighi NN., Broumand B.**	2004	Iran	Peer-reviewed publication	RCT	442	CKD HD	24%	21.6 month	**HCV (RNA) positivity**	**incidence of HC**	**dedicated HD machine**
												nondedicated HD machine
												dedicated HD machine
												nondedicated HD machine

Table S16: Summary Table
Isolation of HCV patients receiving hemodialysis
Categorical outcomes

PMID	Author	Year	Definition	Timepoint	Frequency/Rate	Relative effect	p-value	Quality
19147995	Agarwal SK., Dash SC., Gupta S., Pandey RM.	2009	separate room, separate machine	3 years	2.7%	nd	<0.001	Fair
			shared room, shared machine		36.2%			
7753458	Blumberg A., Zehnder C., Burckhardt JJ.	1995	separate machine, same room	1992+	7/<131 (unclear)	nd	nd	Poor
			nd	nd	0/nd (unclear)			
16903625	Karkar A., Abdelrahman M., Ghacha R., Malik TQ.	2006	separate rooms, separate machines	2005	0/265 (0%)	nd	nd	Poor
				2004	0/265 (0%)	nd	nd	
				2003	nd/265 (0.2%)	nd	nd	
				2002	1/265 (0.3%)	nd	nd	
			nd	2001	7/265 (2.6%)	nd	nd	
				2000	6/265 (2.3%)	nd	nd	
				1999	6/265 (2.1%)	nd	nd	
				1998	7/265 (2.7%)	nd	nd	
			separate rooms, separate machines	2005	77/265 (29%)	nd	nd	
				2004	85/265 (32%)	nd	nd	
				2003	90/265 (34%)	nd	nd	
				2002	117/265 (44%)	nd	nd	
			nd	2001	151/265 (57%)	nd	nd	
				2000	130/265 (49%)	nd	nd	
				1999	148/265 (56%)	nd	nd	
				1998	154/265 (58%)	nd	nd	
				1996-2003	0/171 (0%)	nd	nd	
16685138	Gallego E., López A., Pérez J., Llamas F., Lorenzo I., López E., Illescas ML., Andrés E., Olivas E., Gómez-Roldan C.	2006	separate room, separate staff, separate machine					Fair
			separate machines, shared room	1993	2/144 (1.4%)	nd	nd	
			separate room, separate staff, separate machine	2003	10/161 (6.2%)	nd	nd	
				2003	14/175 (8.0%)	nd	nd	
				2001	13/183 (7.1%)	nd	nd	
				2000	17/172 (10%)	nd	nd	
				1999	18/150 (12%)	nd	nd	
				1998	23/161 (14.3%)	nd	nd	
				1997	26/169 (15.4%)	nd	nd	
				1996	31/171 (18.1%)	nd	nd	
			separate machines, shared room	1995	34/157 (21.7%)	nd	nd	
				1994	38/146 (26.0%)	nd	nd	
				1993	37/144 (25.7%)	nd	nd	
				1992	37/121 (30.6%)	nd	nd	
17382087	Shebeb AM., Kotkat AM., Abd El Reheim SM., Farghaly AG., Fetohy EM.	2006	Health Education Program (HEP)	6 month	1/10 (10%)	nd	nd	Poor
			isolation of seropositive patients		11/45 (24%)	nd	nd	
			nd		2/19 (11%)	nd	nd	
			Health Education Program (HEP)	6-12 month	1/18	nd	nd	
			isolation of seropositive patients		3/32	nd	nd	
			nd		5/24	nd	nd	
			Health Education Program (HEP)	6-18 months	0/15	nd	nd	
			isolation of seropositive patients		3/30	OR 0.65 (0.23, 1.80)	nd	
			nd		2/12	nd	nd	
15469615	Shamshirsaz AA., Kamgar M., Bekheirnia MR., Ayazi F., Hashemi SR., Bouzari N., Habibzadeh MR., Pourzahedgilani N., Broumand V., Shamshirsaz AH., Moradi M., Borghei M., Haghighi NN., Broumand B.	2004	The only difference between two groups of HD cente	9 months	4/254 (1.6%)	nd	0.05	Good
					9/192 (4.7%)	nd		
			The only difference between two groups of HD cente	18 months	2/160 (1.3%)	nd	<0.05	
					7/121 (5.8%)	nd		

Table S17: Evidence profile: Isolation of HCV patients receiving hemodialysis

Outcome	# of Studies and Study Design	Total N of Patients	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence, including Applicability	Other Considerations	Summary of Findings: Quality of Evidence for Outcome	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
HCV seroconversion	4 Pre-post design, 1 nRCS	2064	Important Limitations* (-2)	No Important Inconsistencies† (0)	Some uncertainty about directness of evidence‡ (-1)	None	Very Low	Large reductions in rates of seroconversion in most studies, to 0% in 2 studies	High
HCV positivity	1 RCT 2 Pre-post design 1 cohort	1072	RCT: none Observational: Important Limitations* (-2)	No Important Inconsistencies§ (0)	Some uncertainty about directness of evidence‡ (-1)	None	Low	Significantly lower event rate with dedicated HD machines (1.3% vs. 5.8%) 1 study with reduction post-isolation, but rate was already falling annually without isolation 2 studies with rate falling by about half	Moderate
Total N		2543							

Balance of Potential Benefits and Harms:	Quality of Overall Evidence:
Introduction of isolation associated with reduction in HCV seroconversion and HCV positivity; however, unclear if this was the primary cause of the improvement.	Very Low

Studies included in EP: 19147995,7753458, 16903625, 16685138, 17382087, 15469615

Abbreviations: HCV = hepatitis C virus, HD = hemodialysis, nRCS = nonrandomized comparative study, RCT = randomized controlled trial.

Annotations:

***** Some studies did not incorporate or report the comparability of the baseline characteristics and co-interventions between groups in the study design and/or data analysis. Some studies did not report the compliance with the interventions. Mostly based on pre-post studies, where other factors may have changed also.

† HCV seroconversion rate was higher in the non-isolation group in most studies, while was slightly higher in the NRCS. There was great heterogeneity in measures for isolation.

‡ Mostly limited to developing countries (Iran, Egypt, India, Macedonia, Saudi Arabia, Turkey; also Spain and Switzerland).

§ HCV (antibody) positive rate was consistently higher in the non-isolation group, although it might reflect the decreased prevalence in the general population in the pre-post design.

Table S18: Summary table
Transplantation vs. waitlist among patients with HCV infection
Categorical outcomes

PMID	Author	Year	Type of article	Study design	Database	Sample size	CKD stage	Kidney function	HCV+	HCV genotype	Outcome	Definition	Arm
21546575	**Roth D, Gaynor JJ, Reddy R, Ciancio G, Sageshima J, Kupin W, Guerra G, Chen L, Burke GW**	2011	Peer-reviewed publication	nRCS	United Network Organ Sharing	175	nd	nd	nd	nd	**Patient survival**	**relative risk of the impact of kidney transplantation on the HR of death**	**KTx**
													No KTx (candidates)
													KTx
													No KTx (candidates)
													KTx
													No KTx (candidates)
												relative risk of the impact of kidney transplantation on the HR of death from infection; relative	**KTx**
													No KTx (candidates)
													KTx
													No KTx (candidates)
												relative risk of the impact of kidney transplantation on the HR od death from CV event	**KTx**
													No KTx (candidates)
25340605	**Scalea JR, Barth RN, Munivenkatappa R, Philosophe B, Cooper M, Whitlow V, LaMattina JC**	2015	Peer-reviewed publication	nRCS	nd	1679	nd	nd	nd	nd	**Graft survival**	**hazards ratio of multivariate analysis of variables associated with graft failure**	**Longer waitlist time**
													Shorter waitlist time
15636622	**Bloom RD, Sayer G, Kosunarty F, Constantinescu S, Abt, Reddy KR**	2005	Peer-reviewed publication	nRCS	nd	315	nd	nd	100%	nd	**Patient survival**	**Kaplan Meier**	**KTx**
													No KTx (candidates)
9100052	**Knoll GA, Tankersley MR, Lee JY, Julian BA, Curtis JJ**	1997	Peer-reviewed publication	nRCS	United Network for Organ Sharing	58	nd	nd	100%	nd	**Patient survival**	**Kaplan Meier**	**KTx**
													No KTx (candidates)
9573555	**Pereira BJ, Natov SN, Bouthot BA, Murthy BV, Ruthazer R, Schmid CH, Levey AS**	1998	Peer-reviewed publication	nRCS	New England Organ Bank	287	nd	nd	100%	nd	**Patient survival**	**relative risk of death**	**KTx**
													No KTx (candidates)
													KTx
													No KTx (candidates)
													KTx
													No KTx (candidates)
													KTx
													No KTx (candidates)

Table S18: Summary table
Transplantation vs. waitlist among patients with HCV infection
Categorical outcomes

PMID	Author	Year	Timepoint	Frequency/Rate	Relative effect	p-value	Adjusted for
21546575	Roth D, Gaynor JJ, Reddy R, Ciancio G, Sageshima J, Kupin W, Guerra G, Chen L, Burke GW	2011	0-6 mo post-txp	nd	2.51 (1.12, 5.66)	0.03	simultaneous panceas-kidney transplant listing, age55 years, and white race
			>6-≤84 mo post-txp	nd	0.32 (0.17, 0.62)	0.0007	simultaneous panceas-kidney transplant listing, age55 years, and white race
			>84 mo post txp	nd	0.74 (0.19, 2.92)	0.66	simultaneous panceas-kidney transplant listing, age55 years, and white race
			0-6 mo post-txp	nd	26.62 (5.01, 141.3)	0.0001	simultaneous panceas-kidney transplant listing
			>6 months post -txp	nd	1.69 (0.31, 9.14)	0.54	simultaneous panceas-kidney transplant listing
			overall post-txp effect	nd	0.20 (0.08, 0.47)	0.0002	simultaneous panceas-kidney transplant listing, age55 years, and white race
25340605	Scalea JR, Barth RN, Munivenkatappa R, Philosophe B, Cooper M, Whitlow V, LaMattina JC	2015	7.8 years	nd	1.05 (1.01, 1.10)	0.02	DGF, Donor age, R-D-, R+D+, R+D-, Retransplant, Pretransplant, History of receipient DM, Recipient race
15636622	Bloom RD, Sayer G, Kosunarty F, Constantinescu S, Abt, Reddy KR	2005	96 months	~80% ~50%	nd	0.003	nd
9100052	Knoll GA, Tankersley MR, Lee JY, Julian BA, Curtis JJ	1997	48 months	~85% ~70%	nd	0.043	nd
9573555	Pereira BJ, Natov SN, Bouthot BA, Murthy BV, Ruthazer R, Schmid CH, Levey AS	1998	0-3 months post-tx	nd nd	4.75 (2.76, 8.17)	nd	nd
			4-6 months post-tx	nd nd	1.76 (0.75, 4.13)	nd	nd
			7-47 months post-tx	nd nd	0.31 (0.18, 0.54)	nd	nd
			>48 months post-tx	nd nd	0.84 (0.51, 1.37)	nd	nd

Table S19: Evidence profile: Transplantation vs. waitlist among patients with HCV infection

Outcome	# of Studies and Study Design	Total N of Patients	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence, including Applicability	Other Considerations	Summary of Findings: Quality of Evidence for Outcome	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	4 retrospective	835	Serious limitations (-1)*	No important inconsistencies (0)	Direct (0)	None (0)	Moderate	Better long-term survival with kidney transplant	Critical
Graft loss	1 retrospective	1679	No limitations (0)	N/A	Indirect (-1)	Sparse (-1)	Low	Better graft survival with short waitlist time than long waitlist time.	Critical
Total N		2514							

Balance of Potential Benefits and Harms:
Kidney transplantation results in improved patient and graft survival than remaining on the waitlist.

Quality of Overall Evidence:
Moderate

Studies included in EP: 21546575, 25340605, 15636622, 9100052, 9573555

Annotations:
* Mostly unadjusted analyses.
† Study comparison was of short vs. long waitlist time.

Table S20: Summary table HCV infection as predictor of death among kidney transplant recipients

PMID	Author	Year	Type of article	Country	Study design	Registry Name	Population	Sample Size, Total	CKD stage, baseline	Kidney function, baseline	HCV genotypes	Follow-up Time	Predictor	Predictor Definition	Predictor units	# w/Predictor or mean (95%)	Outcome
	Fontaine	2013	Abstract	France	Retrospective Case Control	Cristal database	KTx	25,830	CKD 5T			15 y	HCV+	nd		983	Mortality
													HBV+ and HCV+	coinfection		76	
23400903	**Soliman**	2013	Peer-reviewed	Egypt	Retrospective cohort study	nd	KTx, HCV genotype 4	411	CKD 5T	nd	genotype 4 100%	8 years	HCV +	HCV Ab + one year after transplantation		137	Mortality
													HCV -	HCV Ab - one year after transplantation		274	
25098499	**Xia**	2014	Peer-reviewed	US	Retrospective cohort study	Donor kidney mate analysis	KTx	7036	CKD 5T			10 years max	HCV + (D-/R+)			3518	Mortality
													HCV - (D-/R-)			3518	
	Heo	2015	Abstract	US	Retrospective cohort study	OPTN	KTx	35,557	CKD 5T			10 years max	HCV +			2024	Mortality
													HCV -			33533	
22994610	**Ridruejo**	2010	Peer-reviewed	Argentina	Retrospective cohort study	none	KTx	542	CKD 5T			Long-term (unclear)	HCV +			180	Mortality
													HCV -			362	

Table S20: Summary table HCV infection as predictor of death among kidney transplant recipients

PMID	Author	Year	Outcome Definition	Cases (# w/Outcome)	Rate	Adjustments (reject if none)	Metric (HR, OR, RR, beta, change)	Estimate (95% CI)	Comparison	P_value	Risk of Bias
	Fontaine	2013		64	nd	"confounding factors"	HR	1.8 (1.6, 2.2)	vs. HCV-	<0.0001	LOW
				59	nd		HR	3.2 (1.9, 5.2)	vs. not coinfected	<0.0001	
23400903	**Soliman**	2013	nd	43	31%	Univariate. Excluded from multivariate model (with age, diabetes, sex, HD time, HTN, CMV, proteinuria)	HR	1.01 (0.97, 1.06)	vs. HCV -	0.5	HIGH
				101	37%			Univariate; Excluded from multivariate model.			
25098499	**Xia**	2014		nd	nd	recipient risk factors	HR	1.24 (1.06, 1.45)	vs HCV- (matched to donor kidney mate)	nd	LOW
				nd	nd						
	Heo	2015		nd	nd	not reported	HR	1.60 (1.40, 1.82)	vs. HCV -	nd	LOW
22994610	**Ridruejo**	2010		46	26%	age >49 y, HBsAg, cadaveric donor, pancreas transplant, IS triple therapy	HR	1.64 (1.05, 2.55)	vs. HCV -	0.027	LOW
				48	13%						

Table S20: Summary Table
HCV infection as predictor of graft loss among kidney transplant recipients

PMID	Author	Year	Type of article	Country	Study design	Registry Name	Population	Sample Size, Total	CKD stage, baseline	Kidney function, baseline	HCV genotypes	Follow-up Time	Predictor	Predictor Definition	Predictor units
	Fontaine	2013	Abstract	France	Retrospective Case Control	Cristal database	KTx	25,830				15 y	HCV+		
													HBV+ and HCV+		
16244496	**Mitwalli**	2006	Peer-reviewed	Saudi Arabia	Retrospective cohort study	nd	KTx	448	CKD 5T	nd	nd	5.85 (2.7)	HCV +	2nd and 3rd generation ELISA tests for HCV Ab, Ampiclor HCV amplification kit for HCV RNA	
													HCV -		
26900309	**Molmenti**	2015	Peer-reviewed publication	US	Retrospective cohort study	UNOS	KTx	88,284	CKD 5T	nd, all received KTx	nd	nd	HCV +		
													HCV -		
													HCV +		
													HCV -		
25098499	**Xia**	2014	Peer reviewed	US	Retrospective cohort study	Donor kidney mate analysis	KTx	7036	CKD 5T			10 years max	HCV + (D-/R+)		
													HCV - (D-/R-)		
	Heo	2015	Abstract	US	Retrospective cohort study	OPTN	KTx	35,557	CKD 5T			10 years max	HCV +		
													HCV -		
	Kassis	2014	Abstract	US	Retrospective cohort study	UNOS/OPTN	KTx	61,775	CKD 5T			14 years max	HCV +		
													HCV -		
22994610	**Ridruejo**	2010	Peer-reviewed	Argentina	Retrospective cohort study	none	KTx	542	CKD 5T			6 months	HCV +		
													HCV -		

Table S20: Summary Table
HCV infection as predictor of graft loss among kidney transplant recipients

PMID	Author	Year	# w/Predictor or mean (95%)	Outcome	Outcome Definition	Cases (# w/Outcome)	Rate	Adjustments (reject if none)	Metric (HR, OR, RR, beta, change)	Estimate (95% CI)	Comparison	P_value	Risk of Bias
	Fontaine	2013	983	Graft loss		53	nd	"confounding factors"	HR	1.3 (1.2, 1.6)	vs. HCV-	<0.0001	LOW
			76			54	nd		HR	1.8 (1.2, 2.7)	vs. not coinfected	<0.01	
16244496	Mitwalli	2006	286	Graft failure	nd	119	41.6%	age, sex, type of donor, hypertension	HR	4.37 (1.81, 4.77)	vs. HCV -	nd	LOW
			162			106	65.4%						
26900309	Molmenti	2015	799	graft failure (live donor recipient)	nd	nd	nd	donor age, donor BMI, donor gender, donor ethnicity, HLA mismatch level, donor preoperative creatinine, and donor hepatitis C antibodyrecipient drug-treated hypertension and Hep B SAg	HR	1.56 (1.33, 1.83)	vs. HCV-	<0.05	LOW
			23,544			nd	nd						
			4076	graft failure (deceased donor recipient)	nd	nd	nd	donor age, donor BMI, donor gender, donor ethnicity, HLA mismatch level, donor terminal laboratory creatinine, donor hepatitis C antibody, donor history of diabetes, donor history of hypertension, donor cause of death, and donor cardiac arrest postbrain death, recipient drug-treated hypertension and hepatitis B surface antigen	HR	1.44 (1.34, 1.54)	vs. HCV-	<0.05	
			49241			nd	nd						
25098499	Xia	2014	3518	Graft loss		nd	nd	recipient risk factors	HR	1.24 (1.04, 1.47)	vs HCV- (matched to donor kidney mate)	nd	LOW
			3518			nd	nd						
	Heo	2015	2024	Graft failure		nd	nd	not reported	HR	1.32 (1.15, 1.52	vs. HCV -	nd	LOW
			33533										
	Kassis	2014	3334	Graft failure		nd	nd	not reported	HR	1.34 (1.31, 1.37)	vs. HCV-	<0.001	
			58441										
22994610	Ridruejo	2010	180	Graft failure		36	20%	age >49 y, cadaveric donor, IS triple therapy	HR	2.65 (1.19, 3.23)	vs. HCV -	0.008	LOW
			362			31	9%						

Table S21: Evidence profile: HCV infection as predictor of death and graft loss among kidney transplant recipients

Outcome	# of Studies and Study Design	Total N of Participants	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence, including Applicability	Other Considerations	Summary of Findings: Quality of Evidence for Outcome	Summary of Findings: Description of Findings	Summary of Findings: Importance of Outcome
Death	5 retrospective cohorts	Cohorts: 411-35,557	No limitations (0)	No important inconsistencies (0)	Direct (0)	None	High	4 of 5 studies found significant association between HCV+ status and mortality among KTx recipients (HR 1.01-1.8). The study with a nonsignificant finding was restricted to HCV genotype 4 (in Egypt). 1 study found that HCV/HBV coinfection was associated with a higher death rate (HR 3.2).	Critical
Graft Loss	7 retrospective cohorts	Cohorts: 448-61,775	No limitations (0)	No important inconsistencies (0)	Direct (0)	None	High	All 7 studies found statistically significant associations between HCV+ status and increased risk of graft loss (ES 1.3-4.37). 1 study found similarly increased risk in patients with live or deceased donor kidneys. 1 study found that HCV/HBV coinfection was associated with a higher rate of graft loss (HR 1.8).	Critical
Total N	8 retrospective cohorts	Cohorts: 411-61,775							

Association of HCV+ status and CKD:	Quality of Overall Evidence:
Kidney transplant recipients with HCV infection are at higher risk of death and graft loss than similar patients without HCV.	High

Studies included in EP: 23400903, 16244496, 26900309, 25098499, 22994610, AASLD-850 2013, ATC Abstracts 2015, Kassis 2014

Abbreviations: CKD: chronic kidney disease; DAA: direct-acting antivirals; GFR: glomerular filtration rate; RCT: randomized controlled trial; SVR: sustained virological response

Table S22. Summary Table: DAA treatment in HCV-infected donor to HCV-uninfected recipient kidney transplantation

Study Year PMID[1] (Reference)	Design (Txp Years)	N Kidney: + → – [– → –]	F/up Time	Waitlist Time[2]	Time to DAA, Days	DAA Regimen (N) [Duration]	SVR12	DAA-Related SAE	Hepatic Events	Allograft Survival	Allograft GFR, mL/min/1.73 m²	Death	DGF	Acute Rejection	QoL
Prospective studies (with protocol)															
Chen R 2021 34489392 China (1)	Prosp (2019-21)	26 [0]	24 wk		0[3]	SOF/VEL (n=26) [12 wk]	26/26 (100%) ITT	0/26 (0%)	0/26 (0%)[4]	26/26 (100%)	Md 74.1 (IQR 37.0-102.4)	0/26 (0%)	6/26 (23%)	3/26 (12%)	NR
Concepcion 2021 34514116 (2)	Prosp (2018-19)	50 [50[5]]	180 d	NR	Md 26 (IQR 21, 37)	GLE/PIB (n=30) SOF/LDV (n=13) SOF/VEL (n=7) [12 wk]	49/50 (98%[6])	0/50 (0%)	0/50 (0%) [vs. 0/50 (0%)][7]	49/50 (98%)[8] [vs. 50/50 (100%)]	Mn 58.9 (16.5) [vs. 61.9 (23.5)], P=0.47	0/50 (0%) [vs. 1/50 (2%)]	2/50 (4%) [vs. 6/49 (12%)], P=0.27	5/50 (10%) [vs. 4/49 (8%)], P=1.0	NR
Daloul 2021 34169833 (3)	Prosp (2019-20)	30 [0]	Md 17 mo (IQR 12, 21)		Md 26 (IQR 20, 34)	SOF/VEL (n=24) GLE/PIB (n=5) SOF/LDV (n=1) [12 wk]	28/30 (93.3%[9])	0/30 (0%)	NR	30/30 (100%)	Md 46 (IQR 40, 63)	0/30 (0%)	4/30 (13%)	4/30 (13%)	NR
Durand 2018 29507971 (4)	Prosp (2015~17)	10 [0]	24 wk		0[10]	GZR/EBR (n=7), GZR/EBR/SOF[11] (n=3) [12 wk]	10/10 (100%) ITT	0/10 (0%)	Acute hepatitis 1/10 (10%)	10/10 (100%)	Md 63.5 (IQR 47.8-69.9)	0/10 (0%[12])	4/10 (40%)	0/10 (0%)	NR
Durand 2021 32894697 (5)	Prosp (2018-19)	10 [0]	12 mo		0[13]	GLE/PIB (n=10) [4 wk]	10/10 (100%) ITT	0/10 (0%)	NR	9/10 (90%)[14]	Md 54.5 (IQR 30-79)	0/10 (0%)	NR	1/10 (10%)	NR

[1] All conducted in the United States, except as indicated.
[2] For comparative studies only (**+ → – vs. – → –)**
[3] 2 hour prior to transplantation surgery, per protocol.
[4] Liver function stable in all.
[5] Matched by age, gender, race, presence of diabetes, kidney donor profile index, and calculated panel-reactive antibody.
[6] One SVR12 failure (sofosbuvir/ledipasvir) retreated with 12 wk sofosbuvir/velpatasvir/voxilaprevir with SVR12.
[7] No cases of fibrosing cholestatic hepatitis.
[8] One allograft loss at day 151 due to acute rejection.
[9] One SVR12 failure (sofosbuvir/velpatisvir) who was noncompliant retreated with 12 wk sofosbuvir/velpatasvir/voxilaprevir with SVR12. One SVR12 failure (sofosbuvir/velpatisvir) with 12 wk sofosbuvir/velpatasvir/voxilaprevir/ribavirin and was found to have NS5A-Y93H mutation and is continuing treatment with ribavirin.
[10] On call to operating room, per protocol.
[11] Added for genotypes 2 or 3.
[12] Implied.
[13] Before organ perfusion.
[14] One allograft loss at day 261 unrelated to HCV or DAA treatment.

Study Year PMID[1] (Reference)	Design (Txp Years)	N Kidney: + → − [− → −]	F/up Time	Waitlist Time[2]	Time to DAA, Days	DAA Regimen (N) [Duration]	SVR12	DAA-Related SAE	Hepatic Events	Allograft Survival	Allograft GFR, mL/min/1.73 m^2	Death	DGF	Acute Rejection	QoL
Feld 2020 32389183 Canada (6)	Prosp (2019)	10 [0]	36 wk		0[15]	GLE/PIB/ ezetimibe (n=10) [8 d]	10/10 (100%) ITT	1/10 (10%)[16]	0/10 (0%)[17]	10/10 (100%)	Md 74 (IQR 65-92)	0/10 (0%)	NR	0/10 (0%)[18]	NR
Gupta 2020/ 2021 34212479 / 31652392 (7, 8) vs. D-/R-[19]	Prosp (2017-20)	102 [102[20]]	4.5 yr [1.3 yr]	1.8 y [vs. 4.4 y] P<0.001	0[21]	SOF/VEL ± Ez[22]		NR	0/102 (0%)[23]	99/102 (97%) [vs. 69/71 (97%)][24]	Mn 57 (SD 23) [vs. 56 (22)][25]	2/102 (2%) [vs. 4/71 (6%)][26]	44/102 (43%) [vs. 54/102 (53%)][27]	8/102 (8%) [vs. 5/71 (7%)][28]	NR
						[2 d (n=10); 12 wk if viremia]	No viremia 7/10 (70%) After full course 8/10 (80%)[29]								
						[4 d (n=42); 12 wk if viremia]	No viremia 38/42 (90%) After full course 41/42 (98%)[30]								
						[7 d (n=50); 12 wk if viremia]	No viremia 48/50 (96%) After full course 50/50 (100%)								
Reese 2018 / Mazur 2019 30083748 / 30779252 (9, 10)	Prosp (2016-17)	20 [100[31]]	12 mo	NR	3±1[32]	GZR/EBR (n=17) [12 wk], GZR/EBR/RBV (n=3, NS5A) [16 wk]	20/20 (100%) ITT	0/20 (0%)	0/40 (0%)[33] [vs. NR]	20/20 (100%) [vs. NR]	Md 72.8 (IQR 58.6-74.4, N=10) [Md 57.7 (IQR 46.0-68.6) Median difference 13.6 (IQR 7.9-19.2), N=50, P<0.001]	0/20 (0%) [vs. NR]	5/20 (25%) [vs. 45/100 (45%), P=0.076]	0/20 (0%) [vs. NR]	Stable[34]

[15] 6–12 hours before transplant.
[16] Prolongation of hospital admission for transiently elevated liver enzymes (peak ALT 650 U/L on postoperative day 11 (possibly related to the therapy). No discontinuation due to adverse events.
[17] Acute liver failure or fibrosing cholestatic hepatitis.
[18] Implied.
[19] Gupta 2021 34212479 reported the comparison of D+/R- vs. D-/R- combining patients reported in this study (7 day regimen) with patients reported in Gupta 2020 31652392 (2-4 day regimen), plus two patients (and matched controls) who had a 4 day regimen.
[20] Matched by multiple factors.
[21] 6 hours prior to transplant.
[22] Patient requiring full 12-week course had different DAA regimens.
[23] No abnormal liver function except transient transaminitis that resolved with DAA therapy.
[24] 1 year. 31 D-/R- did not yet have 1 year follow-up. P=0.71 between groups.
[25] 6 months. P=0.84 between regimens.
[26] 1 year. 31 D-/R- did not yet have 1 year follow-up. P=0.38 between groups.
[27] P=0.21 between regimens.
[28] 1 year. 31 D-/R- did not yet have 1 year follow-up. P=0.92 between groups.
[29] 2 patients required a second 12-week course, with cure.
[30] 1 patient required a second 12-week course, with cure.
[31] Matched by Kidney Donor Profile Index (KDPI) score.

Study Year PMID[1] (Reference)	Design (Txp Years)	N Kidney: + → − [− → −]	F/up Time	Waitlist Time[2]	Time to DAA, Days	DAA Regimen (N) [Duration]	SVR12	DAA-Related SAE	Hepatic Events	Allograft Survival	Allograft GFR, mL/min/1.73 m^2	Death	DGF	Acute Rejection	QoL
Sise 2020/2021 32843477 / 35155863 (11, 12)	Prosp (2019)	30 [0]	12 mo		2-5 [35]	GLE/PIB (n=30) [8 wk]	30/30 (100%) ITT	0/30 (0%)	0/30 (0%)[36]	30/30 (100%)	Md 60 (IQR 51-81)	2/30 (6.7%)[37]	7/30 (23%)	3/30 (10%)	NR
Terrault 2020 32926749 (13)	Prosp (2018-19)	11 [0]	36 wk		Md 16.5 (IQR 9.8, 24.5)	SOF/VEL (n=10)[38] [12 wk]	11/11 (100%) ITT	0/10 (0%)	0/10 (0%)[39]	10/10 (100%)[40]	NR	0/11 (0%)[41]	2/10 (20%)[42]	0/10 (0%)[43]	NR
	Retrospective studies														
Graham 2020 32072689 (14)	Retro (2018-19)	30 [30[44]]	6 mo	1355 d (1230) [vs. 1162 d (1027)] P=0.51	Md 9 (Rn 5-41)	GLE/PIB (n=29)[45] [12 wk]	30/30 (100%) ITT	0/30 (0%)	0/30 (0%)[46] [vs. NR]	30/30 (100%) [vs. 29/20 (97%)]	Mn 55.5 (SD 18) [vs. 48.6 (SD 15), P=0.12][47]	0/30 (0%) [vs. 0/30 (0%)]	8/30 (27%) [vs. 18/30 (60%), P=0.01]	2/30 (6.6%) [vs. 2/30 (6.6)%]	NR
Hudson 2021 34132438 (15)	Retro (2019-20)	22 [0]	12 mo		Md 41 (SD 13)	GLE/PIB (n=15), SOF/VEL (n=7) [12 wk]	22/22 (100%) ITT	0/22 (0%)	0/22 (0%)[48]	NR	NR	0/22 (0%)	NR	0/22 (0%)	NR
Jandovitz 2020 33259125 (16)	Retro (2018)	25 [0]	12 mo		Md >20[49]	SOF/LDV (n=14), SOF/VEL (n=8), GLE/PIB (n=3) [12 wk]	24/25 (96%[50]) ITT	NR	NR	24/25 (96%)[51]	NR	NR	16/25 (64%)	NR	NR

[32] Per protocol.
[33] 5 with transient minor elevated transaminase. Unclear description of the additional 20 patients.
[34] SF-36 stabilized higher than baseline by post-op week 16.
[35] Per protocol.
[36] No “clinically significant liver disease”.
[37] Neither related to HCV or DAA treatment.
[38] 1 patient did not have viremia, not treated with DAA.
[39] 3 with elevated transaminases, 2 unrelated to HCV or DAA treatment; 3rd with mild increased transaminase that normalized by SVR12. 1 patient who was not treated with DAA was omitted from analysis.
[40] Implied. 1 patient who was not treated with DAA was omitted from analysis.
[41] Implied.
[42] 1 patient who was not treated with DAA was omitted from analysis.
[43] 1 patient who was not treated with DAA was omitted from analysis.
[44] Age-matched.
[45] Also n=1 SOF/VEL.
[46] No major liver function adverse events.
[47] Difference somewhat greater at 1 and 3 months, favoring HCV-donor.
[48] Abnormal liver function or fibrosing cholestatic hepatitis.
[49] After viremia detected (mean 7 days post-transplant) and genotype established, median 13 days (range 8-22).
[50] The 1 patient who did not achieve SVR12 was successfully retreated with sofosbuvir/velpatasvir/voxilaprevir.
[51] 1 allograft loss due to renal artery aneurysm 55 days post-transplant.

Study Year PMID[1] (Reference)	Design (Txp Years)	N Kidney: + → − [− → −]	F/up Time	Waitlist Time[2]	Time to DAA, Days	DAA Regimen (N) [Duration]	SVR12	DAA-Related SAE	Hepatic Events	Allograft Survival	Allograft GFR, mL/min/1.73 m^2	Death	DGF	Acute Rejection	QoL
Kapila 2020 31659775 (17)	Retro (2018)	64 [0]	≤55 wk		Md 72 (Rn 9-198)	GLE/PIB (n=33), SOF/LDV (n=24), No DAA (n=4)[52] [12 wk[53]]	44/45 (97.8%)[54] Incomplete	NR	2/64 (3.1%) FCH[55]	NR	NR	1/64 (1.5%)[56]	NR	NR	NR
Molnar 2019/2021 31306549 / 33333148 (18, 19)	Retro (2018)	65 [59][57]	12 mo	Dialysis vintage Median (IQR) 49 mo (33-69) [vs. 71 (50-98)] P=0.006	Md 76 (IQR 68-88)	GLE/PIB (n=59), SOF/VEL (n=5)[58] [12 wk[59]]	65/65 (100%) ITT	NR	1/65 (1.5%) [vs. NR] FCH[60]	65/65 (100%) [vs. 56/59 (95%)][61]	Mn 64 (SD 16) [Adjusted difference: −5.2 (−13.8, 3.5); P=0.2][62]	1/65 (2%) [vs. 3/59 (5%), P=0.3][63]	5/65 (7.7%) [vs. 7/59 (11.9%), P=0.4][64]	3/54 (6%) [vs. 3/42 (7%), P=0.8][65]	NR
Torabi 2021 32810315 (20)	Retro (2018-20)	52 [0]	12 mo		Md 11 (IQR 7-17)	GLE/PIB (n=48), SOF/VEL (n=3)[66] [12 wk]	39/39 (100%)[67] Incomplete	NR	NR	50/52 (96.2%)	Mn 62.3 (SD ~23)	0/52 (0%)	13/52 (25%)	3/52 (5.8%)	NR

[52] 4 (6%) did not receive DAA (3 no viremia, 1 had poor post-transplantation course); 1 treated with SOF/VEL. 2 were not accounted for.
[53] 1 treated for 16 weeks.
[54] Excluding 17 who had not yet reached SVR12. Including 1 who died with viremia who was not treated with DAA.
[55] With complete resolution.
[56] Died day 77 after complicated post-transplantation course. Did not receive DAA despite viremia. Cause of death unknown.
[57] Outcomes of 65 HCV-uninfected recipients of a kidney from an HCV-infected donor were compared with 59 HCV-uninfected recipients of a kidney from an HCV-uninfected kidney during the same time period (2018) .
[58] Also n=1 SOF/LDV.
[59] 5 patients treated for 16 weeks.
[60] With complete resolution.
[61] Article reported allograft loss including death: 1/65 vs. 6/59, unadjusted P value 0.04.
[62] Nominally favor D−/R−. Adjusted for recipients' age, sex, race, body mass index, calculated panel reactive antibody, comorbidities (diabetes, hypertension, peripheral vascular disease, coronary artery disease), donor optimal kidney donor profile index.
[63] Crude comparison (unadjusted).
[64] Crude comparison (unadjusted).
[65] Crude comparison (unadjusted). 28/124 (23%) missing (no biopsy).
[66] Also n=1 SOF/VEL/VOX.
[67] Excluding 13 who did not reach SVR12 follow-up time.

SUMMARY:

N=485 HCV + → − kidney allografts

Summary SVR12[68]:				
	Prospective	293/299 (10 studies)	97.2% (95.1, 98.7)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	Retrospective	224/226 (6 studies)	98.4% (96.3, 99.6)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	Total	517/525(16 studies)	97.7% (96.3, 98.8)	[meta-analysis. I^2 = 0% (no heterogeneity)]
Summary DAA SAE:				
	Prospective	1/196 (9 studies)	0.5% (0.1, 3.5)	[simple pooling] (likely an overestimate, SAE not reported for 100 patients)
	Retrospective	0/52 (2 studies)	0% (0, 13.7)	[simple pooling] (likely an overestimate, SAE not reported for 226 patients)
	Total	1/248 (11 studies)	0.4% (0.1, 2.8)	[simple pooling] (likely an overestimate, SAE not reported for 326 patients)
Summary FCH (assumes studies reporting any hepatic event included FCH):				
	Prospective	0/276 (8 studies)	0% (0, 2.8)	[simple pooling]
	Retrospective	3/181 (4 studies)	2.1% (0.4, 4.7)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	Total	3/457 (12 studies)	0.7% (0.2, 2.1)	[simple pooling]
Summary allograft survival ≥~1 year[69]:				
	Prospective	188/192 (5 studies)	97.3% (94.6, 99.1)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	Retrospective	139/142 (3 studies)	97.8% (94.7, 99.6)	[meta-analysis. I^2 = 6% (no heterogeneity)]
	Total	327/334 (8 studies)	97.6% (95.7, 98.9)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	vs. D-/R-	167 vs. 130 (2 studies)	OR 1.98 (0.27, 14.4)	[meta-analysis. I^2 not applicable]
Summary death ≥~1 year[70]:				
	Prospective	4/192 (5 studies)	2.9% (1.0, 5.9)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	Retrospective	4/303 (5 studies)	1.6% (0.5, 3.3)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	Total	8/495 (10 studies)	2.1% (1.0, 3.5)	[meta-analysis. I^2 = 0% (no heterogeneity)]]
	vs. D-/R-	167 vs. 130 (2 studies)	OR 0.32 (0.08, 1.26)	[meta-analysis. I^2 not applicable]
Summary DGF:				
	Prospective	74/278 (8 studies)	22.2% (12.5, 33.7), range 4.0-43%	[meta-analysis. I^2 = 75% (large heterogeneity)]
	Retrospective	42/172 (4 studies)	28.3% (8.9,53.5), range 7.7-64%	[meta-analysis. I^2 = 91% (large heterogeneity)]
	Total	116/450 (12 studies)	24.2% (15.1, 34.6), range 4.0-64%	[meta-analysis. I^2 = 82% (large heterogeneity)]
	vs. D-/R-	267 vs. 340 (5 studies)	OR 0.49 (0.31, 0.78)	[meta-analysis. I^2 = 12% (small heterogeneity)]
Summary acute rejection:				
	Prospective	24/298 (10 studies)	8.3% (5.5, 11.7)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	Retrospective	8/158 (4 studies)	5.2% (2.3, 9.2)	[meta-analysis. I^2 = 0% (no heterogeneity)]
	Total	32/456 (14 studies)	7.2% (5.0, 9.2)	[meta-analysis. I^2 = 0% (no heterogeneity)]]
	vs. D-/R-	267 vs. 340 (5 studies)	OR 1.06 (0.51, 2.19)	[meta-analysis. I^2 = 0% (no heterogeneity)]

Abbreviations: +/-: HCV infected/uninfected individuals, ASN: American Society of Nephrology (conference), CI: confidence interval, D: donor, DAA: direct acting antiviral, DGF: delayed graft function, EBR: elbasvir, Ez: ezetimibe, F/up: follow-up, FCH: fibrosing cholestatic hepatitis, GFR: glomerular filtration rate, GLE: glecaprevir, GZR: grazoprevir, HCV: hepatitis C virus, IQR: interquartile range, LDV: ledipasvir, Md: median, Mn: mean, NR: not reported, NS5A: nonstructural protein 5A resistance-associated substitutions (mutation), PIB: pibrentasvir, PMID = PubMed identifier, Prosp: prospective study design, QoL: quality of life, R: recipient, RBV: ribavirin, Retro: retrospective study design, Rn: range (full), SAE: serious adverse events (related to DAAs), SD: standard deviation, SF-36: Short Form 36 item questionnaire, SOF: sofosbuvir, SVR12: sustained virologic response at 12 weeks post-treatment, Txp = transplantation, VEL: velpatasvir, VOX: voxipaprevir.

[68] Patients treated with short course DAA who became viremic and were treated with full-course DAA assessed at SVR12 after full course.
[69] Median 10 months to 4.5 years; thus, omitting Chen 2021, Concepcion 2021, Durand 2018, Feld 2020, Graham 2020, Terrault 2020.
[70] Median 10 months to 4.5 years; thus, omitting Chen 2021, Concepcion 2021, Durand 2018, Feld 2020, Graham 2020, Terrault 2020.

Table S23. Evidence Profile: DAA treatment for HCV donor-infected to HCV-uninfected recipients

Outcome	# of Studies[71]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings: Certainty of Evidence for Outcome	Summary of Findings: Description of Findings		Summary of Findings: Importance of Outcome
Death, ≥~12 mo	10	495	Some limitations[72]	Consistent	Indirect[73]	None	Low	2.1% (1.0, 3.5)	High likelihood of survival to 1 year.	Critical
	2 vs. D−/R−	167 vs. 130	Some limitations[74]	Consistent	Direct	Imprecise	Very Low	OR 0.32 (0.08, 1.26)		
Allograft survival, ≥~12 mo	8	334	Some limitations[75]	Consistent	Indirect[76]	None	Low	97.6% (95.7, 98.9)	High likelihood of allograft survival to 1 year.	Critical
	2 vs. D−/R−	167 vs. 130	Some limitations[77]	Some inconsistency	Direct	Imprecise	Very Low	OR 1.98 (0.27, 14.4)		
SVR12	16	525	Some limitations[78]	Consistent	Direct	None	Moderate	97.7% (96.3, 98.8)	Very high SVR12	High
Serious adverse events	11	248	Some limitations[79]	Consistent	Direct	None	Moderate	0.4% (0.1, 2.8)	Low reported rate of SAE due to DAAs; likely is an overestimate	High
Fibrosing cholestatic hepatitis	12	447	Some limitations[80]	Consistent	Direct	None	Moderate	0.7% (0.2, 2.1)	Low reported rate of hepatic complications	High
Allograft GFR	12	435	No limitations	Consistent	Indirect[81]	None	Moderate	Range mean/median 46-74 mL/min	Achieved eGFR mostly in CKD G2-G3a range. Difference between D+/R−/ and D−/R− unclear.	High
	5 vs. D−/R−	267 vs. 341	Some limitations[82]	Inconsistent	Direct	None	Low	Inconsistent, but mostly no significant difference		
Delayed graft function	12	450	Some limitations[83]	Inconsistent	Indirect[84]	None	Low	24.2% (15.1, 34.6) Range 7.7-64%	Estimate unclear. ~1/2 risk compared with D−/R−	Moderate
	5 vs. D−/R−	267 vs. 340	Some limitations[85]	Consistent	Direct	None	Moderate	OR 0.49 (0.31, 0.78)		
Acute rejection	1514	456	No limitations	Consistent	Indirect[86]	None	Moderate	7.2% (5.0, 9.7)	Low reported rate	Moderate

[71] Single group studies, except as noted.
[72] Incomplete or unclear reporting.
[73] No comparator.
[74] Incomplete or unclear reporting.
[75] Incomplete or unclear reporting.
[76] No comparator.
[77] Unadjusted analysis.
[78] Some incomplete analyses (mostly preliminary analyses before all had reached SVR12, some missing data).
[79] Incomplete or unclear reporting.
[80] Incomplete or unclear reporting.
[81] No comparator.
[82] No or incomplete adjustment.
[83] Incomplete or unclear reporting.
[84] No comparator.
[85] No or incomplete adjustment.

Outcome	# of Studies[71]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings		
							Certainty of Evidence for Outcome	Description of Findings	Importance of Outcome
	5 vs. D−/R−	236 vs. 192	Some limitations[87]	Consistent	Direct	Imprecise	Very Low	OR 1.06 (0.51, 2.19) of acute rejection. Difference between D+/R−/ and D−/R− unclear.	

Balance of Potential Benefits and Harms:	Certainty of Overall Evidence:
Patient and allograft survival rates good with D+/R- transplantation but insufficient evidence for comparison with D−/R−. SVR rates high with DAA with low rates of serious adverse event rates. Moderate rates of delayed graft function and acute rejection and allograft GFR good, but insufficient evidence for comparison with D−/R−.	Low

Abbreviations: CKD: chronic kidney disease (category), D+/R−: donor HCV infected/recipient HCV uninfected, D−/R−: donor HCV uninfected/recipient HCV uninfected, DAA: direct-acting antiviral, GFR: glomerular filtration rate, HCV: hepatitis C virus, OR: odds ratio, SAE: serious adverse events, SVR12: sustained virologic response at 12 weeks post-treatment.

[86] No comparator.
[87] No or incomplete adjustment.

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Table S24. Summary Table: HCV-associated GN DAA treatment, categorical outcomes

Study, Country (Reference)	Population	Follow-Up	Treatment	Genotype	SVR12	Death	SCr Double or ESRD	Cryoglobulinemia
Affatato 2020 ERA-EDTA P0283 Italy (1)	Active cryoglobulinemic kidney disease NR by biopsy	6-48 mo	DAA, various[1]	1 83% 3 17%	NR	1/11 (9%)[2]#	NR	CR 10/12 (83%) PR 1/12 (8%) Relapse after CR 3/10 (30%)
Cacoub 2019, 29857143 France, Italy, Egypt (2)	HCV-related cryoglobulinemic vasculitis (n=148), of which GN (n=25) NR by biopsy	15.3 mo	SOF/LDV n=23 SOF/DCV n=53 SOF/SIM n=18 SOF/RBV n=51 NR n=3	1 53% 2 13% 3 11% 4 19% 5 3% 6 0.7%	141/145 (97%)[3] Not ITT	4/148 (3%)[4]	NR	All DAA (SVR12)[5]: CR 106/146 (73%) PR 33/146 (23%) No response 7/146 (5%) All DAA, GN (SVR12): CR NR/25 PR NR/25 No response 3/25 (12%) CR (15.3 mo), by DAA[6] SOF/LDV 87% adjOR 4.09 (1.19, 19.0) SOF/DCV 88% adjOR 2.28 (0.96, 5.63) SOF/SIM 72% adjOR 1.23 (0.40, 4.02) SOF/RBV 70% reference
Elnadry 2021 EASL 2021 Egypt (3)	HCV-related mixed cryoglobulinemia (n=80), of whom biopsy-proven kidney involvement (n=20)	30-36 mo	SOF/DCV±RBV 12 wk	NR	80/80 (100%) ITT	NR	NR	CR 43/80 (54%) PR 27/80 (34%)
Fayed 2022 35103020 Egypt, France, Italy (4)	HCV-related cryoglobulinemic vasculitis (n=913), of whom had kidney involvement (n=332) NR by biopsy	35 mo	DAA, various[7]	1 10% 2 3% 3 2% 4 85% 5 1% 6 0.1%	911/913 (99.8%)	20/913 (2.2%)[8]	NR	CR 814/888 (92%) PR 37/888 (4%) No response 31/888 (3%) Worsened 6/888 (0.7%)
Iliescu 2020 31948406 Romania (5)	HCV-related cryoglobulinemia with CKD G2 (n=13), G3 (n=39), G4 (n=3) NR by biopsy	SVR12	PrOD 12 wk	1b 100%	56/56 (100%) ITT	NR	0/56 ("significant worsening of kidney function during treatment")	CR 47/56 (84%)[9] PR 5/56 (9%) No response 4/56 (7%)

Study, Country (Reference)	Population	Follow-Up	Treatment	Genotype	SVR12	Death	SCr Double or ESRD	Cryoglobulinemia
Kondili 2021 34919289 Italy (6)	HCV-related cryoglobulinemic vasculitis (n=423[10]), of whom had kidney involvement by blood/urine measures (n=64)	3 yr	DAA, unspecified	NR	NR	NR	NR	End of treatment: FCR[11] 51/423 (12%) CR[12] 72/423 (17%) PR[13] 118/423 (28%) 12 mo post-SVR12: No response[14] 102/423 (24%) Across 3 yr[15] FCR 164/423 (39%) Relapse 18/137[16] (13%)
Pérez de José 2021 33623683 Spain (7)	HCV-related MC[17] 65/138 (47%) by biopsy DAA 37% IFN+RBV 83% No treatment 53%	138 mo (median)	DAA, unspecified	1b 78%	98/100 (98%) ITT	4/100 (4%) HR 0.12 (0.04, 0.40)	6/100 (6%) HR 0.10 (0.04, 0.33)	NR
			IFN+RBV	1b 71%	10/23[18] (43.5%)	9/24 (38%)	6/24 (25%)	NR
			No treatment	1b 60%	0/15 (0%)	10/15 (67%)	3/15 (20%)	NR

Abbreviations: adjOR: adjusted odds ratio, CKD G2/3/4: chronic kidney disease category G2/3/4, CR: complete remission, DAA: direct acting antiviral, DCV: daclatasvir, eGFR: estimated glomerular filtration rate, ERA-EDTA: European Renal Association – European Dialysis and Transplant Association, ESRD: end-stage renal disease, HCV: hepatitis C, HR: hazard ratio, IFN+RBV: interferon and ribavirin, LDV: ledipasvir, MC: mixed cryoglobulinemia, NR: not reported, PR: partial remission, PrOD: paritaprevir 150 mg qD/ritonavir 100 mg qD/ombitasvir 25 mg qD/dasabuvir 250 mg BID, RBV: ribavirin, SCr: serum creatinine, SIM: simeprevir, SOF: sofosbuvir, SVR12: sustained viral response 12 weeks after end of therapy (or this timepoint), wk: weeks.

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[1] PrOD (n=5), SOF/DCV (n=3), SOF/SIM (n=3), No treatment (n=1; unable to omit patient from results). Durations not reported.

[2] 1 died at 9 months after the end of DAA of acute disease reactivation. Omits untreated patient omitted who did not die during follow-up.

[3] 3 unexplained missing.

[4] Median follow-up 15.3 months: hepatocellular carcinoma (n=2), alveolar pulmonary hemorrhage (n=1), acute respiratory failure (n=1).

[5] 2 unexplained missing.

[6] Adjusted for severity of vasculitis and type of mixed cryoglobulin.

[7] Sofosbuvir 95%, Daclatasvir 50%, Simeprevir 31%, Ledipasvir 21%, Ribavirin 8%. 12 weeks 92%, 24 weeks 7%.

[8] 11/115 (9.6%; mostly due to immunosuppression-related infections) of relapsers; 9/798 (1.1%; unrelated to vasculitis or immunosuppresion) of non-relapsers.

[9] Complete response included disappearance of purpuric lesions. All with partial response , prior to treatment had cutaneous lesions, nephrotic syndrome, paresthesia and (in 4/5) ulcerative lesions. All with no response had nephrotic syndrome; renal function improved in all with no response.

[10] Of 523 treated with DAA for cryoglobulinemic vasculitis who both achieved SVR12 and had available follow-up data.

[11] Full clinical response: all symptoms disappeared.

[12] Complete clinical response: all baseline clinical manifestations had improved.

[13] Partial clinical response: improvement in at least half of the baseline symptoms.

[14] No clinical response: improvement in less than half of the baseline symptoms.

[15] At any time point during 3 years of follow-up.

[16] 137 of 164 who achieved FCR had continuing follow-up. Relapse was transient in 6 of 9 who had subsequent follow-up and 3 of 9 had no response status after relapse.

[17] Patients had HCV-related MC followed in a nephrology clinic. Implied, but not clearly stated, that all had kidney involvement, "defined by the presence of cryoglobulinaemic glomerulonephritis with a membranoproliferative pattern of injury on histological examination of kidney biopsy or by the presence of proteinuria or haematuria or an eGFR <60 mL/min/1.73 m^2 without an alternative cause for CKD in patients without kidney biopsy". Across all patients, mean eGFR = 56 mL/min and mean proteinuria = 2.1 g/day.

[18] Denominator based on reported percentage. One participant apparently missing.

Table S25. Summary Table: HCV-associated GN DAA treatment, continuous outcomes

Study, Country	Treatment	N	eGFR, mL/min, Mean (SD) [P Value]	Proteinuria, g/d, Mean [P Value Vs. T0]	Cryocrit, %, Mean (SD) [P Value Vs. T0]	c3, mg/dL, Mean (SD) [P Value Vs. T0]	c4, mg/dL, Mean (SD) [P Value Vs. T0]
Fayed 2022 35103020 Egypt, France, Italy (4)	DAA	913	NR	NR	NR	NR	Median [IQR] T0 8 [7-10] EOT 22 [16-26] [P NR]
Iliescu 2020 31948406 Romania (5)	PrOD	56	NR	"Decreased" (P 0.013)	NR	NR	NR
Pérez de José 2021 33623683 Spain (7)	DAA, unspecified	100	NS change	T0 2.4 EOT 1.3 [P 0.003] 138 mo* 0.9 [P <0.001]	T0 5.0 (4.4) EOT 1.7 (2.0) 138 mo* 0.7 (1.8) [P 0.001]	T0 32 (41) EOT 101 (30) [P <0.001] 138 mo* 100 (24) [P <0.001]	T0 5 (6) EOT 14 (18) [P <0.001] 138 mo* 20 (24) [P <0.001]
	IFN+RBV	34	NS change (implied)	NS change	NR	NS change	NS change
	No treatment	15	NS change (implied)	NS change (implied)	NR	NR	NR

Abbreviations: c3/4: complement component 3/4, DAA: direct acting antiviral, eGFR: estimated glomerular filtration rate, EOT: end of therapy, IFN+RBV: interferon and ribavirin, IQR: interquartile range, NR: not reported, NS: statistically nonsignificant, PrOD: Paritaprevir 150 mg qD/Ritonavir 100 mg qD/Ombitasvir 25 mg qD/Dasabuvir 250 mg BID, SD: standard deviation, T0: baseline (time 0).

* Median follow-up duration across all arms (138 months)

Table S26. Evidence Profile: Management of HCV-associated glomerulonephritis

Outcome	Treatment	# of Studies[19]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings: Certainty of Evidence for Outcome	Summary of Findings: Description of Findings		Summary of Findings: Importance of Outcome
Death, ≥12 mo	DAA, various	4	1172 (GN in 506)	Some limitations[20]	Consistent	Indirect (no comparator, most without GN)	None	Low	2.4% (1.6, 3.4)	Low rate. Indication of possible large effect of DAA.	Critical
	vs. No Tx	1	100 vs 15	No limitation[21]	N/A	Direct	Sparse Very large effect	Very Low	OR 0.02 (0.005, 0.09)		
SCr double or ESRD	DAA, various	2	156	Some limitations[22]	Inconsistent	Indirect (no comparator)	None	Low	3.8% (1.7, 8.3)	Possible low rate. Indication of possible large effect of DAA.	Critical
	vs. No Tx	1	100 vs 15	No limitation[23]	N/A	Direct	Sparse Very large effect	Very Low	OR 0.06 (0.02, 0.12)		
Cryoglobulinemia, complete remission	DAA, various	6	1616 (GN in ~500)	Some limitations[24]	Serious inconsistency[25]	Indirect (no comparator, most without GN)	None	Very Low	70% (49, 88) Range 29-92%	High rate of complete remission	High
	DAA comparisons	1	146 (GN in 25)	Serious limitations[26]	N/A	Indirect (83% no GN)	Sparse	Very Low	adjOR (vs. SOF/RBV) SOF/LDV 4.1 (1.2, 19) SOF/DCV 2.3 (0.96, 5.6) SOF/SIM 1.2 (0.4, 4.0)		
	vs. No Tx	0							ND		
SVR12	DAA, various	5	1294 (GN in ~479)	Some limitations[27]	Consistent	Indirect (most without GN)	None	Low	99.0% (97.7, 99.8)	Very high SVR12	High
Serious adverse events	0								ND		High
eGFR	DAA vs. No Tx	1	100 vs. 15	Serious limitations[28]	N/A	Direct	Sparse, no analysis	Very Low	NS	NS change with DAA	Moderate
Proteinuria	DAA, various	2	156	Some limitations[29]	Consistent	Indirect (no comparator)	None		Significant decrease	Proteinuria reduced with DAA treatment	Moderate
	vs. No Tx	1	100 vs. 15	Some limitations[30]	N/A	Direct	Sparse, no analysis	Very Low	ND		
Cryocrit, c3, c4	DAA, various	2 (c4)	1013	Some limitations[31]	Consistent	Indirect (no comparator, most without GN)	None	Low	Significant improvements	Measures improved with DAA treatment	Moderate
	DAA vs. No Tx	1	100 vs. 15	No limitation	N/A	Direct	Sparse, no analysis	Very Low	Significant improvements		
Balance of Potential Benefits and Harms: Very high rate of SVR12 and complete remission, with indication of large reduction in risks of death and SCr doubling or ESRD, with DAA								**Certainty of Overall Evidence:** Low			

Outcome	Treatment	# of Studies[19]	Total N of Patients on Treatment	Methodological Quality of Studies	Consistency Across Studies	Directness of the Evidence	Other Considerations	Summary of Findings		
								Certainty of Evidence for Outcome	Description of Findings	Importance of Outcome
treatment. Unclear evidence regarding best DAA regimen. No direct evidence regarding adverse event rates with DAA treatment. Unclear evidence regarding intermediate markers of glomerulonephritis and cryoglobulinemia.										

Abbreviations: CKD: chronic kidney disease, CR: complete remission, DAA: direct acting antiviral, eGFR: estimated glomerular filtration rate, ERA-EDTA: ESRD: end-stage renal disease, HCV: hepatitis C, IFN+RBV: interferon and ribavirin, MC: mixed cryoglobulinemia, NR: not reported, PR: partial remission, PrOD: paritaprevir 150 mg qD/ritonavir 100 mg qD/ombitasvir 25 mg qD/dasabuvir 250 mg BID, RBV: ribavirin, SCr: serum creatinine, SVR12: sustained viral response 12 weeks after end of therapy, wk: weeks

[19] Single group studies, except as noted.
[20] Unclear, possibly incomplete reporting.
[21] Unadjusted, but very large effect.
[22] Unclear, possibly incomplete reporting.
[23] Unadjusted, but very large effect.
[24] Unclear, possibly incomplete reporting.
[25] $I^2 = 97\%$
[26] Unclear, possibly incomplete reporting; possibly insufficient adjustment for confounders.
[27] Unclear, possibly incomplete reporting.
[28] No actual data reported, unclear reporting.
[29] Unclear, possibly incomplete reporting.
[30] Unclear, possibly incomplete reporting.
[31] Unclear, possibly incomplete reporting.

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